

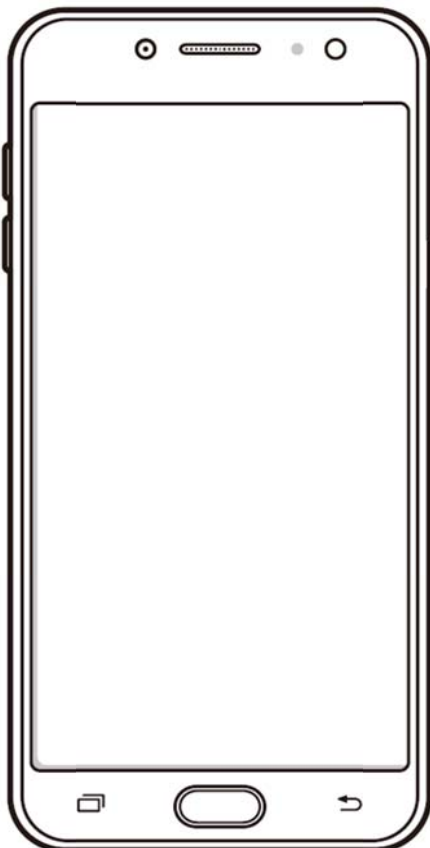
SAMSUNG

Mobile Device SM-C710F

SERVICE **Manual**

Mobile Device

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1. Safety Precautions

1-1. Repair Precaution

Before attempting any repair or detailed tuning, shield the device from RF noise or static electricity discharges.

Use only demagnetized tools that are specifically designed for small electronic repairs, as most electronic parts are sensitive to electromagnetic forces.

Use only high quality screwdrivers when servicing products. Low quality screwdrivers can easily damage the heads of screws.

Use only conductor wire of the properly gauge and insulation for low resistance, because of the low margin of error of most testing equipment.

We recommend 22-gauge twisted copper wire.

Hand-soldering is not recommended, because printed circuit boards (PCBs) can be easily damaged, even with relatively low heat. Never use a soldering iron with a power rating of more than 100 watts and use only lead-free solder with a melting point below 250°C (482°F).

Prior to disassembling the battery charger for repair, ensure that the AC power is disconnected. Always use the replacement parts that are registered in the SEC system. Third-party replacement parts may not function properly.

1. Safety Precautions

1-2. ESD(Electrostatically Sensitive Devices) Precaution

Many semiconductors and ESDs in electronic devices are particularly sensitive to static discharge and can be easily damaged by it. We recommend protecting these components with conductive anti-static bags when you store or transport them.

Always use an anti-static strap or wristband and remove electrostatic buildup or dissipate static electricity from your body before repairing ESDs.

Ensure that soldering irons have AC adapter with ground wires and that the ground wires are properly connected.

Use only desoldering tools with plastic tips to prevent static discharge.

Properly shield the work environment from accidental electrostatic discharge before opening packages containing ESDs.

The potential for static electricity discharge may be increased in low humidity environments, such as air-conditioned rooms. Increase the airflow to the working area to decrease the chance of accidental static electricity discharges.

2. Specification

2-1. CDMA & GPS General Specification

ITEM	CDMA	GPS
Tx Freq. range	824.025~848.985MHz	-
Rx Freq. range	869.025~893.985MHz	1575.42MHz
Channel Bandwidth	1.23MHz	2MHz
Channel Spacing	30KHz	Not Used
Number of Channel	832	1
Duplex Separation	45MHz	-
In/Output Impedance	50Ohm	50Ohm
Tx Local Frequency	$F_{Tx} * 1.6666$	-
Rx Local Frequency	$F_{Rx} * 2$	$F_{Rx} * 2$
TCXO Frequency	26MHz	26MHz
Freq. Stability	(FRX-45MHz)±300Hz	-
Operating Temperature	-30°C ~ +60°C	-30°C ~ +60°C

2. Specification

2-2. GSM General Specification

Item		GSM 850	EGSM 900	DCS1800	PCS1900
Freq. Band[MHz] Uplink/Downlink		824~849 869~894	880~915 925~960	1710~1785 1805~1880	1850~1910 1930~1990
ARFCN range		128~251	0~124 & 975~1023	512~885	512~810
Tx/Rx spacing		45MHz	45MHz	95MHz	80MHz
Mod. Bit rate/ Bit Period		270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us
Time Slot Period/ Frame Period		576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms
Modulation	GSM/ GPRS	0.3GMSK	0.3GMSK	0.3GMSK	Modulation
MS Power		33dBm ~5dBm	33dBm ~5dBm	30dBm ~0dBm	30dBm ~0dBm
Power Class		5pcl ~ 19pcl	5pcl ~ 19pcl	0pcl ~ 15pcl	0pcl ~ 15pcl
Sensitivity		-110dBm	-110dBm	-109dBm	-109dBm
TDMA Mux		8	8	8	8
Cell Radius		35Km	35Km	2Km	2Km

13008098

2. Specification

2-3. GSM Tx Power Class

TX Power control level	GSM850	TX Power control level	EGSM900	TX Power control level	DCS1800	TX Power control level	PCS1900
5	33±2 dBm	5	33±2 dBm	0	30±3 dBm	0	30±3 dBm
6	31±2 dBm	6	31±2 dBm	1	28±3 dBm	1	28±3 dBm
7	29±2 dBm	7	29±2 dBm	2	26±3 dBm	2	26±3 dBm
8	27±2 dBm	8	27±2 dBm	3	24±3 dBm	3	24±3 dBm
9	25±2 dBm	9	25±2 dBm	4	22±3 dBm	4	22±3 dBm
10	23±2 dBm	10	23±2 dBm	5	20±3 dBm	5	20±3 dBm
11	21±2 dBm	11	21±2 dBm	6	18±3 dBm	6	18±3 dBm
12	19±2 dBm	12	19±2 dBm	7	16±3 dBm	7	16±3 dBm
13	17±2 dBm	13	17±2 dBm	8	14±3 dBm	8	14±3 dBm
14	15±2 dBm	14	15±2 dBm	9	12±4 dBm	9	12±4 dBm
15	13±2 dBm	15	13±2 dBm	10	10±4 dBm	10	10±4 dBm
16	11±3 dBm	16	11±3 dBm	11	8±4 dBm	11	8±4 dBm
17	9±3 dBm	17	9±3 dBm	12	6±4 dBm	12	6±4 dBm
18	7±3 dBm	18	7±3 dBm	13	4±4 dBm	13	4±4 dBm
19	5±3 dBm	19	5±3 dBm	14	2±5 dBm	14	2±5 dBm
-	-	-	-	15	0±5 dBm	15	0±5 dBm

2. Specification

2-4. WCDMA General Specification

	WCDMA2100	WCDMA1900	WCDMA850	WCDMA900
Freq. Band[MHz] Uplink/Downlink	1922~1977 2112~2167	1852~1907 1932~1987	824~849 869~894	880~915 925~960
ARFCN range	UL: 9612~9888 DL: 10562~10838	UL: 9262~9538 DL: 9662~9938	UL: 4132~4233 DL: 4357~4458	UL: 2712~2863 DL: 2937~3088
Tx/Rx spacing	190MHz	80MHz	45MHz	45MHz
Mod. Bit rate/ Bit Period	3.84 Mcps	3.84 Mcps	3.84 Mcps	3.84 Mcps
Time Slot Period/Frame Period	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms
Modulation	QPSK/HQPSK	QPSK/HQPSK	QPSK/HQPSK	QPSK/HQPSK
MS Power	24dBm~-50dBm	21dBm~-50dBm	24dBm~-50dBm	24dBm~-50dBm
Power Class	3(max+24dBm)	4(max+21dBm)	3(max+24dBm)	3(max+24dBm)
Sensitivity	-106.7dBm	-106.7dBm	-106.7dBm	-106.7dBm
TDMA Mux	8	8	8	8
Cell Radius	2Km	2Km	2Km	2Km

2. Specification

2-5. LTE General Specification

	Band1 (FDD)	Band3 (FDD)	Band5 (FDD)	Band7 (FDD)	Band8 (FDD)
Freq. Band[MHz] UL/DL	1920~1980 2110~2170	1710~1785 1805~1880	824~849 869~894	2500~2570 2620~2690	880~915 925~960
ARFCN range UL/DL	18000~18599 0~599	19200~19949 1200~1949	20400~20649 2400~2649	20750~20449 2750~3449	21450~21799 3450~3799
Tx/Rx spacing	190MHz	95MHz	45MHz	120MHz	45MHz
TCXO Frequency	26MHz	26MHz	26MHz	26MHz	26MHz
Modulation	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM
MS Power	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm
Power Class	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)
Sensitivity	-99.3(BW: 5 MHz) -96.3(BW: 10 MHz) -94.5(BW: 15MHz) -93.3(BW: 20MHz)	-101(BW: 1.4MHz) -98(BW: 3MHz) -96.3(BW: 5 MHz) -93.3(BW: 10 MHz) -91.5(BW: 15MHz) -90.3(BW: 20MHz)	-102.5(BW: 1.4MHz) -99.5(BW: 3MHz) -97.3(BW: 5 MHz) -94.3(BW: 10 MHz)	-97.3(BW: 5 MHz) -94.3(BW: 10 MHz) -92.5(BW: 15MHz) -91.3(BW: 20MHz)	- 101.5(BW: 1.4MHz) -98.5(BW: 3MHz) -96.3 (BW: 5 MHz) -93.3(BW: 10 MHz)
Cell Radius	>5Km	>5Km	>5Km	>5Km	>5Km
In/Output Impedance	50Ω	50Ω	50Ω	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃

2. Specification

	Band20(FDD)	Band28(FDD)	Band38 (TDD)	Band40 (TDD)	Band41 (TDD)
Freq. Band[MHz] Uplink/Downlink	704~716 734~746	703~748 758~803	2570~2620	2300~2400	2555~2655
ARFCN range	24150~24449 6150~6449	27210~27659 9210~9659	37750~38249	38650~39649	40240~41240
Tx/Rx spacing	41MHz	55MHz	-	/	-
TCXO Frequency	26MHz	26MHz	26MHz	26MHz	26MHz
Modulation	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM
MS Power	-35~25.7 dBm	-35~25.7dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm
Power Class	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)
Sensitivity	101.5(BW: 1.4MHz) -98.5(BW: 3MHz) -96.3 (BW: 5 MHz) -93.3(BW: 10 MHz)	101.5(BW: 1.4MHz) -98.5(BW: 3MHz) -96.3 (BW: 5 MHz) -93.3(BW: 10 MHz)	-99.3(BW: 5 MHz) -96.3(BW: 10 MHz) -94.5(BW: 15MHz) -93.3(BW: 20MHz)	-99.3(BW: 5 MHz) -96.3(BW: 10 MHz) -94.5(BW: 15MHz) -93.3(BW: 20MHz)	-97.3(BW: 5 MHz) -94.3(BW: 10 MHz) -92.5(BW: 15MHz) -91.3(BW: 20MHz)
Cell Radius	>5Km	>5Km	>5Km	>5Km	>5Km
In/Output Impedance	50Ω	50Ω	50Ω	50Ω	50Ω
Operating Temperature	-	-30℃ ~ +60℃	-	-30℃ ~ +60℃	-30℃ ~ +60℃

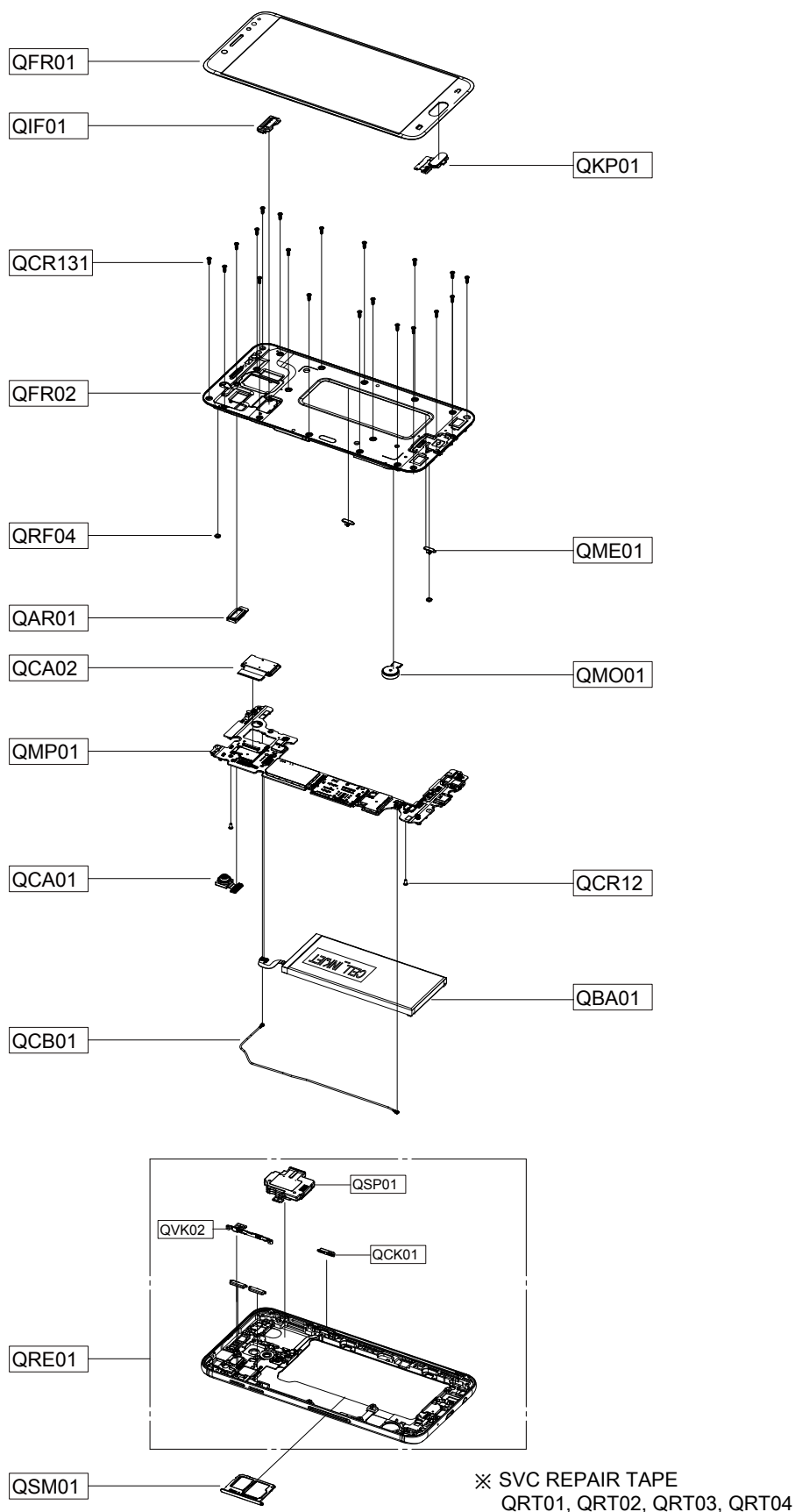
3. Operation Instruction and Installation

Main Function

Item	Description
OS	Android V7.1
RF	GSM850 /GSM900/ DCS1800/ PCS1900 CDMA:BC0 WCDMA: B1/ B2/ B5/ B8 LTE: B1/ B3/ B5/B7/ B8/ B20/28/B38/ B40/ B41
Battery	3000mAh
Base Band	Octa-Core 2.39Ghz+1.69GHz
Other RF	GPS, Glonass, BT4.2, Micro USB, WIFI 802.11 a/b/g/n 2.4 + 5GHz, FM
Camera	13MP+5MP Main CAM; 16MP Sub CAM
LCD	5.5." AMOLED, 1920*1080(FHD)
RAM	4GB
Sensor	Accelerometer, Gyro, Proximity, RGB light sensor, Fingerprint sensor, Geomagnetic sensor, Grip
Accessory	Charger: 5V/1.55A

4. Exploded View and Parts List

4-1. Cellular phone Exploded View



5. MAIN Electrical Parts List

Parts Code	Design Loc	Description
0401-001110	D5002	SDS511Q
0404-001250	D5000,D5001	RB520G-30
0406-001459	ZD6000,ZD6001,ZD6006	RD1301B
0406-001459	ZD6007	RD1301B
0406-001506	ZD5004,ZD5005,ZD5006	UCLAMP0541Z
0406-001506	ZD5007,ZD7000,ZD7001	UCLAMP0541Z
0406-001506	ZD7002,ZD7003,ZD7004	UCLAMP0541Z
0406-001506	ZD7005,ZD7006,ZD7008	UCLAMP0541Z
0406-001506	ZD7009,ZD7010,ZD7011	UCLAMP0541Z
0406-001592	ZD6009,ZD6010,ZD6011	PESD5V0V1BDSF
0406-001592	ZD6012,ZD6013,ZD6014	PESD5V0V1BDSF
0406-001592	ZD6016,ZD7007	PESD5V0V1BDSF
0406-001633	D5004	RDL3081
0406-001640	ZD5002,ZD5003	RCLAMP3521P.TCT
0406-001648	ZD6020	AOZ8831ADI-05
0406-001709	ZD5010,ZD6017,ZD6021	RDL1151B
0406-001709	ZD6022	RDL1151B
0406-001728	ZD5000,ZD5001,ZD5009	ES45D3HP
0406-001740	ZD6002,ZD6005	UCLAMP1211P
0406-001743	D6000,D6001,D6006	RCLAMP3331ZA
0406-001743	D6007,ZD6003,ZD6004	RCLAMP3331ZA
0406-001763	D6008,D6009,D6010	VS3V3BC1HS
0406-001763	D6011,D6012,D6013	VS3V3BC1HS
0406-001763	D6014	VS3V3BC1HS
0406-001781	ZD6019,ZD6024,ZD6031	D3V3H1B2LP
0406-001781	ZD6032	D3V3H1B2LP
0505-002088	Q5000,Q5002	2SK3541
0505-002341	Q5001	SI1013X-T1-GE3
0505-002388	Q6001	RUE003N02
0601-003338	LED5002	SCF0165G
0601-003569	LED5001	SPFCW02201AR
0601-003611	LED5000	SPFCW13331CRS1D4T0
1001-001650	U5001,U6004	DG2750DN-T1-E4
1001-001911	U1004	BGSA14GN10-1
1001-001956	U2010,U2017	BGS15MA12
1001-001969	U2006,U2008,U2009	MXD8631
1001-001970	U2014,U2015	MXD8621
1001-001992	U1002,U1003,U1006	RF1646

5. MAIN Electrical Parts List

1001-001999	U2002	MXD8643
1001-002022	U1007	QM13114
1001-002030	U1001	MXD8631SF
1108-000643	UME5000	KMRX60014M-B614
1201-003786	U2003	BGU6009
1201-003915	PAM1000	SKY77621-31
1201-003975	U1011	SFMLB7RB002
1201-003978	U1016	SFMLB3RB002
1201-003979	U1017	SFMLB8RB002
1201-003985	U2011	BGS8L2
1201-004100	PAM1001	WIPS33232-01
1201-004122	U1008,U1009,U1013	MXD8011HU
1201-004122	U2001,U2012,U2013	MXD8011HU
1201-004148	U6000	RT5509WSC
1202-001120	U6006	NCX2200GM
1203-008249	U7009	AP7341D-33FS4-7
1203-008250	U6002,U7012	AP7340D-18FS4-7
1203-008326	U7001,U7003	AP7340D-30FS4-7-71
1203-008603	U6001	AP7346
1203-008711	U5009	ET9530L
1203-008751	U7002	SM3004B
1203-008859	U7007	AP7341D-12FS4-7R
1203-008865	U5005,U7004,U7005	ET51528YB
1203-008867	U7006	ET51518YB
1203-008985	U5003	MT6355W
1203-008995	U5008	S2MU005X03-6030
1205-005217	U7008	FSA644BUCX
1205-005711	U2004	MT6625L
1205-005813	U1010	MT6176V
1205-005835	UCP3000	MT6757V-WD
1205-005845	U5004	A96T326AUN-OHF01-T
1209-002413	U7010	YAS539
1209-002450	U7013	LSM6DSL
1209-002510	U7011	CM36657VM3OH-A3
1404-001664	TH3000,TH3001	NCP15WF104F03RC
1404-001724	TH1000,TH5000	NCP03WF104F05RL
1405-001403	V1003,V1009,V2008	IMG0530C101JFR
1405-001404	VR2003	ESPV022R0V321
1405-001410	V1001,V2010	ESPC02015JS1

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5. MAIN Electrical Parts List

1405-001412	U7016,V1008,V2001	ESPC02039JS1
1405-001412	V2006,V2009,V2011	ESPC02039JS1
1405-001412	V6000,V6001,V6002	ESPC02039JS1
1405-001412	VR2004,VR2005	ESPC02039JS1
1405-001415	C2110,V1005,V2004	IMG0330C101JFR
1405-001420	V2003	IMG0330C390JFR
1405-001423	V6003	ESPC014R7JS1
1405-001424	V2005	ESPC021R5KS1
1405-001425	VR2000	HESP02100JTC
2007-003015	R5035,R5036	MCR01MZP5J2R2
2007-007100	R2006	RK73K1ETP106J
2007-007142	R6032,R7013	RC1005F103CS
2007-007199	R6033,R6046	TSR16GJ2R7V
2007-007318	R2005	MCR01MZP5F1001
2007-007528	R3020,R3021	RK73H1ETP1501F
2007-007538	R3026	RK73H1ETQTP5602F
2007-007698	R5012	RK73H1ETP5101F
2007-007741	R1009,R3043,R5007	M5534204B10E0R
2007-007741	R5013,R5029,R5042	M5534204B10E0R
2007-007741	R5044,R5049,R5054	M5534204B10E0R
2007-007741	R6013,R7015	M5534204B10E0R
2007-007766	R1011	RK73H1ETP2001F
2007-007942	R3023	RC1005F105CS
2007-007946	R6096	RC1005F474CS
2007-007981	R6094	ERJ2RKF1803X
2007-008045	R3051,R5075,R5076	ERJ1GEJ101C
2007-008137	R1010	RC1005F243CS
2007-008210	R1000,R6004	RMC1/20-180JTP
2007-008294	R3028	RC1005F330CS
2007-008304	R5027	RC1005F304CS
2007-008312	R1008,R3030,R3031	RC1005F394CS
2007-008531	R5037,R7001,R7008	MCR006MZPJ0100
2007-008531	R7022	MCR006MZPJ0100
2007-008582	R1002,R1003	RC0603J301CS
2007-008587	R5045,R5047	ERJ1GEJ561C
2007-008588	R7000	MCR006YZPJ222
2007-008788	R3006	RC0603J270CS
2007-008816	R6038	MCR006MZPJ822
2007-009111	R7014	RC0603J220CS

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5. MAIN Electrical Parts List

2007-009157	R5003,R5017,R5019	MCR006YZPF1003
2007-009157	R5020,R5024,R6034	MCR006YZPF1003
2007-009157	R6048	MCR006YZPF1003
2007-009158	R5004	MCR006MZPF1503
2007-009168	R5022,R6036	MCR006MZPJ563
2007-009171	R5023	MCR006YZPJ513
2007-009212	R5009,R5071,R6017	MCR006YZPF1001
2007-009315	R5000,R6001,R6002	RC0603J4R7CS
2007-009315	R6003,R6005,R6006	RC0603J4R7CS
2007-009315	R6007,R6008,R6009	RC0603J4R7CS
2007-009315	R6010,R6011,R6012	RC0603J4R7CS
2007-009352	R3022,R3027	MCR006YZPF4701
2007-009354	R6039	ERJ1GEF3301C
2007-009408	R3000,R3001,R3007	ERJ1GEF222C
2007-009408	R3008,R3009,R3010	ERJ1GEF222C
2007-009408	R3011,R3012,R3013	ERJ1GEF222C
2007-009408	R3014,R3015,R3016	ERJ1GEF222C
2007-009408	R5034	ERJ1GEF222C
2007-009801	R5016,R5018,R5056	MCR006YZPF1000
2007-009801	R6031	MCR006YZPF1000
2007-009805	R1001	RC0603F510CS
2007-009866	R5021	MCR006YZPF2003
2007-009879	R3047	ERJ1GEF1101C
2007-009920	R5058,R5060	RC0603F241CS
2007-009954	R5028	MCR006YZPF6802
2007-009969	R6024	MCR006YZPF1201
2007-010029	R5032	MCR006YZP5F5104
2007-010202	R5063,R5064,R6028	RC0603F473CS
2007-010685	R5080,R6000	RC0201FR-071ML
2007-011377	R6037	RC0603F433CS
2007-011532	R6091	RC0603F121CS
2007-011648	R1018,R5025,R5026	RC0603F1R0CS
2007-011648	R5031	RC0603F1R0CS
2007-012038	R5005	ESR03EZPF10R0
2007-012068	R5033	RU2012FR010CS
2203-000233	C7087	GRP1555C1H101J
2203-000359	C2500	GRP1555C1H151JD01E
2203-000995	C5104	GRP1555C1H470J
2203-001153	C5110	GRP1555C1H680J

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2203-005682	C1025,C2001,C2046	GRP0335C1E330JD01E
2203-005682	C2067,C2081,C5074	GRP0335C1E330JD01E
2203-005682	C5111,C5146,C5921	GRP0335C1E330JD01E
2203-005682	C6000,C6001,C6006	GRP0335C1E330JD01E
2203-005682	C6013,C6017,C7012	GRP0335C1E330JD01E
2203-005682	C7020,C7053,C7054	GRP0335C1E330JD01E
2203-005717	C1027	GRP0335C1E270JD01E
2203-005725	C1061,C5075	GRP0335C1E220JD01E
2203-005726	C1060,C5151	GRM0335C1E560J
2203-005727	C1135,C5136	CL03C120JA3GNNC
2203-005729	C5044,C6023,C6024	CL03C150JAGC
2203-005729	C6030,C7014,L1023	CL03C150JAGC
2203-005731	C6041,C6046	GRM0335C1E180J
2203-005732	C5105,C5900,C6018	GRP0335C1E680JD01E
2203-005732	C6020,C6029	GRP0335C1E680JD01E
2203-005734	C2095,C2107	GRM0335C1E820JD01D
2203-005736	C1002,C1003,C1007	GRP0335C1E101JD01E
2203-005736	C1008,C1009,C1011	GRP0335C1E101JD01E
2203-005736	C1013,C1015,C1021	GRP0335C1E101JD01E
2203-005736	C1022,C1023,C1029	GRP0335C1E101JD01E
2203-005736	C1033,C1034,C1040	GRP0335C1E101JD01E
2203-005736	C1053,C1057,C1064	GRP0335C1E101JD01E
2203-005736	C1068,C1070,C1072	GRP0335C1E101JD01E
2203-005736	C1076,C1092,C1110	GRP0335C1E101JD01E
2203-005736	C1125,C1126,C1127	GRP0335C1E101JD01E
2203-005736	C1128,C1129,C1130	GRP0335C1E101JD01E
2203-005736	C1131,C1134,C1136	GRP0335C1E101JD01E
2203-005736	C1137,C1139,C1141	GRP0335C1E101JD01E
2203-005736	C1142,C1149,C1151	GRP0335C1E101JD01E
2203-005736	C1153,C1154,C1155	GRP0335C1E101JD01E
2203-005736	C1157,C1158,C1159	GRP0335C1E101JD01E
2203-005736	C1163,C1165,C1166	GRP0335C1E101JD01E
2203-005736	C1167,C1168,C1169	GRP0335C1E101JD01E
2203-005736	C1170,C1186,C1190	GRP0335C1E101JD01E
2203-005736	C1198,C1203,C1205	GRP0335C1E101JD01E
2203-005736	C1207,C1210,C1216	GRP0335C1E101JD01E
2203-005736	C1218,C1229,C2002	GRP0335C1E101JD01E
2203-005736	C2003,C2004,C2006	GRP0335C1E101JD01E
2203-005736	C2014,C2015,C2018	GRP0335C1E101JD01E

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2203-005736	C2020,C2021,C2030	GRP0335C1E101JD01E
2203-005736	C2039,C2052,C2069	GRP0335C1E101JD01E
2203-005736	C2080,C2091,C2094	GRP0335C1E101JD01E
2203-005736	C2098,C2099,C2103	GRP0335C1E101JD01E
2203-005736	C2104,C6033,C7026	GRP0335C1E101JD01E
2203-005736	C7031,C7034,L1036	GRP0335C1E101JD01E
2203-005736	L1107	GRP0335C1E101JD01E
2203-005739	C1095	GRM0335C1E4ROCD01D
2203-005777	C1083,C1211	CL03C010CA3GNNC
2203-005789	C1012,C1016,C1042	GRM0335C1ER50B
2203-005789	C1046,C1176,C2025	GRM0335C1ER50B
2203-005789	C2115,L1050	GRM0335C1ER50B
2203-005792	C1032,C1041,C1050	CL03C1R5CA3GNNC
2203-005792	C1161,L1000,L1013	CL03C1R5CA3GNNC
2203-005806	C1140,C1147,C1164	GRP033R71C102KD01E
2203-005806	C2019,C2071	GRP033R71C102KD01E
2203-006120	C5039,C5041	GRP033R71C221KD01E
2203-006123	C1051	GRM0335C1E5R6CD01D
2203-006187	C1212,C2023	GRM0335C1E3R9B
2203-006194	C1175,C2017,C6027	GRP033R70J103KA01E
2203-006305	C1047,C2008	CM03CG100J25AH
2203-006318	C1031,C1066	GRP0335C1E1R2CD01
2203-006379	C2024	GRP033R70J472KA01
2203-006399	C5071	GRM155R60J105KE19D
2203-006400	C1063,C1152,C1173	GRM033R71E102K
2203-006400	C2033,C2049,C2059	GRM033R71E102K
2203-006400	C5000,C5030,C5149	GRM033R71E102K
2203-006400	C6037,R6052	GRM033R71E102K
2203-006410	C1124,C1191,C1208	CL03C1R8CA3GNNC
2203-006410	C6016	CL03C1R8CA3GNNC
2203-006426	C6039	GRM033R71C681KD01J
2203-006556	C1014,C1049,C1056	GRM0335C1E470JD01D
2203-006556	C1084,C1086,C1132	GRM0335C1E470JD01D
2203-006556	C1150,C5139,C5904	GRM0335C1E470JD01D
2203-006556	C5906	GRM0335C1E470JD01D
2203-006562	C1143,C5027,C5052	CV05X5R105K10AH
2203-006562	C5073,C5080,C5081	CV05X5R105K10AH
2203-006562	C5098,C5108,C5142	CV05X5R105K10AH
2203-006562	C5144,C5911,C5912	CV05X5R105K10AH

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2203-006562	C5914,C6051,C7001	CV05X5R105K10AH
2203-006562	C7008,C7023,C7035	CV05X5R105K10AH
2203-006562	C7037,C7039,C7041	CV05X5R105K10AH
2203-006562	C7043,C7046,C7050	CV05X5R105K10AH
2203-006604	C6015	GRM0335C1E2R4BD01D
2203-006642	C6008,C6009	GRM033R71E331KA01D
2203-006647	C5132	GRM033R60J473KE19D
2203-006665	C1019	GRM0335C1E2R2CD01D
2203-006707	C1099,C1100,C5006	GRM0335C1E4R7C
2203-006707	C5008,C5011	GRM0335C1E4R7C
2203-006841	C1079,C5002,C5072	CV05X5R105K16AH
2203-006841	C5097,C5913	CV05X5R105K16AH
2203-006846	C1111	ECJZEC1E2R0B
2203-006872	C4066,C5042,C5048	GRM155R60J225ME15D
2203-006872	C5920	GRM155R60J225ME15D
2203-006978	C5109	GMK107BJ105KA-T
2203-007165	C1101,C5009,C5019	CL05A475KQ5NRNC
2203-007165	C5083,C5117,C5129	CL05A475KQ5NRNC
2203-007165	C6010,C6045	CL05A475KQ5NRNC
2203-007194	C7061,L1007	CL03C6R8BA3GNNC
2203-007210	C5116,C5128	CV03X5R224K06AH
2203-007240	C5026,C5915	CL10A226MQ8NRNE
2203-007269	C7025,C7032,C7033	CL21A226MPCLRNC
2203-007271	C1080,C1091,C1146	CL05A225KP5NSNC
2203-007271	C5022,C5051,C5077	CL05A225KP5NSNC
2203-007271	C5084,C5085,C5086	CL05A225KP5NSNC
2203-007271	C5087,C5091,C5092	CL05A225KP5NSNC
2203-007271	C5101,C5106,C5107	CL05A225KP5NSNC
2203-007271	C5134,C5138,C5908	CL05A225KP5NSNC
2203-007317	C1093,C4006,C4007	CV05X5R475M06AH
2203-007317	C4028,C4033,C4041	CV05X5R475M06AH
2203-007317	C4055,C4092,C5018	CV05X5R475M06AH
2203-007317	C5053,C5070,C7015	CV05X5R475M06AH
2203-007391	C2048,C2056,C2064	CL03A474KQ3NNNH
2203-007391	C4001,C4002,C4003	CL03A474KQ3NNNH
2203-007391	C4004,C4030,C4031	CL03A474KQ3NNNH
2203-007391	C4034,C4035,C4036	CL03A474KQ3NNNH
2203-007391	C4067,C4068,C4069	CL03A474KQ3NNNH
2203-007391	C4070,C5121	CL03A474KQ3NNNH

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2203-007393	C2016,C5082,C5088	CL05A475KP5NRNC
2203-007393	C5089,C5090,C5093	CL05A475KP5NRNC
2203-007393	C5901,C5916,C7000	CL05A475KP5NRNC
2203-007393	C7022	CL05A475KP5NRNC
2203-007474	C7059	CL05A106MQ5NUNC
2203-007775	C5094,C7047	CV05X5R475M10AH
2203-007781	C5023,C5024,C5079	CL10A226MP8NUNE
2203-007796	C1088,C2031,C2032	CL03A105MP3NSNC
2203-007796	C2034,C3006,C4005	CL03A105MP3NSNC
2203-007796	C4009,C4010,C4011	CL03A105MP3NSNC
2203-007796	C4012,C4013,C4014	CL03A105MP3NSNC
2203-007796	C4015,C4016,C4017	CL03A105MP3NSNC
2203-007796	C4018,C4019,C4020	CL03A105MP3NSNC
2203-007796	C4022,C4023,C4024	CL03A105MP3NSNC
2203-007796	C4025,C4026,C4027	CL03A105MP3NSNC
2203-007796	C4038,C4042,C4043	CL03A105MP3NSNC
2203-007796	C4044,C4045,C4046	CL03A105MP3NSNC
2203-007796	C4047,C4052,C4057	CL03A105MP3NSNC
2203-007796	C4059,C4060,C4061	CL03A105MP3NSNC
2203-007796	C4065,C4086,C4097	CL03A105MP3NSNC
2203-007796	C5004,C5007,C5010	CL03A105MP3NSNC
2203-007796	C5012,C5013,C5014	CL03A105MP3NSNC
2203-007796	C5015,C5016,C5017	CL03A105MP3NSNC
2203-007796	C5025,C5034,C5035	CL03A105MP3NSNC
2203-007796	C5036,C5038,C5045	CL03A105MP3NSNC
2203-007796	C5046,C5047,C5054	CL03A105MP3NSNC
2203-007796	C5055,C5056,C5057	CL03A105MP3NSNC
2203-007796	C5058,C5059,C5060	CL03A105MP3NSNC
2203-007796	C5061,C5062,C5063	CL03A105MP3NSNC
2203-007796	C5065,C5066,C5067	CL03A105MP3NSNC
2203-007796	C5078,C5112,C5113	CL03A105MP3NSNC
2203-007796	C5114,C5118,C5119	CL03A105MP3NSNC
2203-007796	C5123,C5145,C5148	CL03A105MP3NSNC
2203-007796	C5155,C5156,C6012	CL03A105MP3NSNC
2203-007796	C6014,C6021,C6022	CL03A105MP3NSNC
2203-007796	C6032,C6035,C6038	CL03A105MP3NSNC
2203-007796	C6050,C6052,C7004	CL03A105MP3NSNC
2203-007796	C7006,C7007,C7009	CL03A105MP3NSNC
2203-007796	C7013,C7016,C7017	CL03A105MP3NSNC

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2203-007796	C7024,C7036,C7038	CL03A105MP3NSNC
2203-007796	C7040,C7042,C7044	CL03A105MP3NSNC
2203-007796	C7049,C7051,C7058	CL03A105MP3NSNC
2203-008097	C1078,C1081,C1085	CL03A104KA3NNNC
2203-008097	C1096,C1098,C1102	CL03A104KA3NNNC
2203-008097	C1106,C1109,C1112	CL03A104KA3NNNC
2203-008097	C1119,C1120,C1156	CL03A104KA3NNNC
2203-008097	C2005,C3000,C3002	CL03A104KA3NNNC
2203-008097	C3003,C3004,C3005	CL03A104KA3NNNC
2203-008097	C4049,C4050,C4051	CL03A104KA3NNNC
2203-008097	C4053,C4054,C4056	CL03A104KA3NNNC
2203-008097	C4058,C4062,C4063	CL03A104KA3NNNC
2203-008097	C4064,C4071,C4073	CL03A104KA3NNNC
2203-008097	C4074,C4075,C4076	CL03A104KA3NNNC
2203-008097	C4077,C4078,C4082	CL03A104KA3NNNC
2203-008097	C4083,C4084,C4085	CL03A104KA3NNNC
2203-008097	C5005,C5020,C5122	CL03A104KA3NNNC
2203-008097	C5127,C5130,C5131	CL03A104KA3NNNC
2203-008097	C5133,C5157,C5158	CL03A104KA3NNNC
2203-008097	C5903,C5926,C5927	CL03A104KA3NNNC
2203-008097	C5928,C6002,C6003	CL03A104KA3NNNC
2203-008097	C6011,C6028,C6049	CL03A104KA3NNNC
2203-008097	C7018,C7019,C7052	CL03A104KA3NNNC
2203-008097	C7055	CL03A104KA3NNNC
2203-008242	C4029,C4048	CL03A225MQ3CRNC
2203-008860	C5068,C5069,C5096	CL03A105MO3NRNC
2203-008860	C6019	CL03A105MO3NRNC
2203-009223	C2043,C2119	CL05C101JC5NNNC
2203-009328	C5115	CL05A226MQ5N6J8
2203-009537	C4000,C4008,C4021	GRM155R60J156ME47D
2203-009537	C4032,C4037,C4039	GRM155R60J156ME47D
2203-009537	C4040,C4079,C4080	GRM155R60J156ME47D
2203-009537	C4081,C4087,C4088	GRM155R60J156ME47D
2203-009537	C4089,C4090,C4091	GRM155R60J156ME47D
2203-009537	C4093,C4094,C4095	GRM155R60J156ME47D
2203-009537	C4096	GRM155R60J156ME47D
2203-009551	C1039,C1055,C6034	GRM0335C1H390JA01D
2203-009618	C5001,C5003,C6004	CL10A226MO7JZNC
2203-009618	C6005	CL10A226MO7JZNC

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2203-009665	C6042	GRM0335C1H151JA01D
2203-009733	C5037,C5099,C5100	CL05A106MP6NUN8
2203-009733	C5102,C5103,C5902	CL05A106MP6NUN8
2203-009733	C5905,C7010,C7011	CL05A106MP6NUN8
2203-009733	C7030,C7045,C7056	CL05A106MP6NUN8
2203-009733	C7057	CL05A106MP6NUN8
2203-009858	C1038,C1087,C6007	CL05A106MP9NUB8
2203-009858	C7027,C7081,C7082	CL05A106MP9NUB8
2203-009858	C7083,C7084,C7085	CL05A106MP9NUB8
2203-009858	C7086	CL05A106MP9NUB8
2703-001728	C2120	HK1005-1N5S-T
2703-002313	L6000,L6001	CIH05TR10JNC
2703-002649	L1046	LLV0603-F6N8J
2703-002842	L1108	LLV0603-F2N2S
2703-002858	L1062,L1112	HK0603-4N7S-T
2703-002901	L1019,L1029,L2037	LLV0603-F15NJ
2703-002907	L1095	LLV0603-F8N2J
2703-002951	L1048,L1111,L2002	CIH03T12NJNC
2703-002951	L2045	CIH03T12NJNC
2703-002958	L1011,L1012,L6037	CIH03T56NJNC
2703-002958	L6038	CIH03T56NJNC
2703-002959	L6015	MLG0603S68NJT
2703-002960	L2060,L6026,L6029	MLG0603S82NJT
2703-002960	L6031	MLG0603S82NJT
2703-002961	L2014,L5018,L5019	MLG0603SR10JT
2703-002999	L2003	CIH03T22NJNC
2703-003004	C1213,L1037	LLV0603-F47NJ
2703-003878	L6022	CIG10WR27MNC
2703-003904	L1002,L1027,L1038	CML03061N0BNH
2703-003904	L1080	CML03061N0BNH
2703-003921	L2031	CML0306-1N2-BNH
2703-004001	L1021,L1054,L6006	LQP03TN39NJ02D
2703-004012	C1214,L1047,L1053	LQP03TN2N2B02D
2703-004012	L1075,L1089,L1096	LQP03TN2N2B02D
2703-004013	L1004,L1070,L1092	LQP03TN1N8B02D
2703-004013	L1093,L2029,L2046	LQP03TN1N8B02D
2703-004013	L2052	LQP03TN1N8B02D
2703-004014	C2114	LQP03TN1N2B02D
2703-004018	C1018,C1071,L1040	LQP03TN5N6J02D

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2703-004018	L1058,L1066,L1084	LQP03TN5N6J02D
2703-004018	L2011,L2054,L2057	LQP03TN5N6J02D
2703-004033	L1006,L1091	LQP03TN2N4B02D
2703-004034	C1065,C1113,C2010	LQP03TN1N0B02D
2703-004034	L1074,L2000,L2017	LQP03TN1N0B02D
2703-004034	L2041,L2049,L2050	LQP03TN1N0B02D
2703-004035	L1015,L1017,L1065	LQP03TN3N3B02D
2703-004035	L2038	LQP03TN3N3B02D
2703-004036	C2011,L1032,L1056	LQP03TN12NJ02D
2703-004037	L1103,L2007,L2012	LQP03TN2N7B02D
2703-004037	L2028	LQP03TN2N7B02D
2703-004038	C2013,L1044,L1081	LQP03TN1N5B02D
2703-004230	L6018	LQG15HS68NJ02D
2703-004259	L1043	AML0603E9N1-T
2703-004286	L1064	CIH03S3N3BNC
2703-004289	L1003	CIH03S2N2BNC
2703-004300	L2022,L2030	CIH03S1N5BNC
2703-004328	L1028,L1042,L1049	LQP03TN4N7H02D
2703-004363	L1024	LQP03TG18NJ02D
2703-004366	C1177,L1030,L1055	LQP03TG6N8J02D
2703-004366	L2036	LQP03TG6N8J02D
2703-004367	C2074	LQP03TG8N2J02D
2703-004408	L1061	LQP03TN15NJ02
2703-004763	L2001,L2032,L2040	LQP03TN4N3J02D
2703-004763	L2053	LQP03TN4N3J02D
2703-004764	L1010,L1085,L1086	LQP03TN10NH02D
2703-004853	L2058,L2059	LQP03TNR12J02D
2703-004862	L1083,L2025	LQP03TN3N6B02D
2703-004911	L1113	LQP03TN5N1J02D
2703-005058	C2073,L1022	LQP03TN5N6J02D
2703-005085	C1044,L1069,L1079	LQP03HQ2N7B02D
2703-005085	L1109,L1110,L2026	LQP03HQ2N7B02D
2703-005086	L2015,L2051	LQP03HQ3N3B02D
2703-005087	L1063,L1072,L2044	LQP03HQ3N9B02D
2703-005088	L1076	LQP03HQ4N7J02D
2703-005089	L1039,L1073	LQP03HQ15NJ02D
2703-005109	L5025	CIGT252010LM1R0SNE
2703-005117	L7004,L7006	SDEM25201T-4R7MD
2703-005118	L7003	SDEM25201T-100MD

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2703-005136	L5012,L5013,L5016	CIGT252010LMR47SNE
2703-005136	L5017	CIGT252010LMR47SNE
2703-005295	L1014,L2034	LQP03HQ8N2H02D
2703-005296	L1077,L6005	LQP03HQ10NH02D
2703-005366	L1071	LQP03TN2N0B02D
2703-005438	L5029	CIH03T68NJNC
2703-005505	L5003,L5004,L5011	CIGT201610EH1R0SNE
2703-005505	L5014,L5015,L6002	CIGT201610EH1R0SNE
2801-005393	OSC5000	TFX-04
2809-001434	OSC1000	KT2016K26000ZAW23ZMS
2901-001673	F7006	ACFE1A2G900E
2901-001674	F7002,F7004	ACFE2B4G101E
2904-002134	F1004	SAFFB2G14AA0F0A
2904-002143	F2007	SAFFB2G65AA0F0A
2904-002257	F2010	SFHG52AA002
2904-002262	F2014	SFH881AA002
2904-002263	F2006	SFH942AA002
2904-002280	F2008	SFH806BA002
2904-002336	F2009	B8853
2904-002339	F2004	SFWG62DBB02
2904-002344	F2011	SFWG42CBB02
2904-002349	F2013	SFHG93AQ102
2904-002356	F2001	SFHG89XA402
2904-002365	F1007	SAFRH2G59MA0F3K
2904-002367	F1006	SFDG35AQ102
2910-000390	F1005	SAYEY2G53BD0F3K
2910-000402	F1003	LMDP2MGA-J81
2911-000448	F1002	LMSWKQGR-J80
3003-001236	MIC6000,MIC6001	SM0103B-DA381-S01
3301-001812	L6033,L6036	BLM15HD102SN1
3301-001885	L6007,L6008,L6010	BLM15HD182SN1D
3301-001885	L6012,L6014,L6016	BLM15HD182SN1D
3301-001885	L6019,L6020,L6025	BLM15HD182SN1D
3301-002223	L7000	BLM15PX601SN1D
3301-002228	L1041	BLM15PX330SN1
3301-002236	C6043,L5020,L5022	BLM03AX102SN1D
3301-002237	L1020,L1025,L2005	BLM03AX601SN1D
3301-002237	L2006,L2013	BLM03AX601SN1D
3301-002240	L5000,L5002,L5005	BLM03AX100SN1D

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5. MAIN Electrical Parts List

3301-002240	L5006,L5007,L5008	BLM03AX100SN1D
3301-002240	L5009,L5010,L6013	BLM03AX100SN1D
3301-002244	L6003,L6004,L6034	BLM15PX330SN1D
3301-002244	L6035	BLM15PX330SN1D
3301-002254	L5024	BLM18KG260TN1D09
3301-002279	L7001,L7008	BLM15PX331SN1
3301-002286	L6021	BKH0603LM102-T
3301-002305	L6017,L6023	NFZ15SG462SN10D
3301-002312	L5023,L7002,L7005	BLM15EX121SN1D
3301-002312	L7007	BLM15EX121SN1D
3301-002331	L5001	BLM18FD220SN1
3705-001708	RFS1002,RFS1004	MS-156C
3705-001937	RFS1000,RFS1001	20449-001E
3709-001891	SIM6000	DS254-C22B-C14
3711-007107	HDC7001	AXE630124AW1
3711-007173	HDC3000	AXE612124
3711-008182	HDC7000	503776-4051
3711-008847	HDC5000	505006-0811
3711-008969	HDC7002	BM30B-54DP-0.35V(46)
3711-009058	HDC6000	GB355-16P-H06-E15000
3712-001634	ANT2006,ANT2007	CS-074A
3712-001634	ANT2023,ANT5004	CS-074A
3712-001634	ANT5006	CS-074A
3712-001666	ANT6000,ANT6001	CS-090A
3712-001694	ANT2003,ANT2004	CS-070B
3712-001694	ANT2005,ANT2016	CS-070B
3712-001694	ANT5001,ANT5002	CS-070B
3712-001694	ANT5005,ANT5007	CS-070B
3712-001694	ANT5008,ANT6002	CS-070B
3712-001694	ANT6003	CS-070B
3712-001749	ANT1000,ANT1001	CS-111A
3712-001749	ANT1002,ANT1003	CS-111A
3712-001749	ANT1005,ANT2000	CS-111A
3712-001749	ANT2001,ANT2008	CS-111A
3712-001749	ANT2010,ANT2011	CS-111A
3712-001749	ANT2012,ANT2018	CS-111A
3712-001749	ANT2019,ANT2020	CS-111A
3712-001749	ANT2024,ANT2025	CS-111A
3712-001749	ANT2026,ANT2027	CS-111A

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5. MAIN Electrical Parts List

3712-001750	ANT5000,ANT5003	CS-112A
3722-003954	IFC5000	HY07-AB1790
3722-004081	EAR6000	7234-383000
4709-002196	F1008	DPX202700DT-4162A1
4709-002226	F1001	DPX202700DT-4062A1
4709-002412	F1000,F1010,F1011	HHM22137A3K
4709-002464	F1009	DPX252690DT-5225A1K
4709-002566	F2002	SFTG45AU002
GH60-00009A	FPC2001,FPC2002	PCC1011
GH60-00009A	FPC2003	PCC1011
GH60-00010A	FPC2004	PCC1007AW
GH61-07836A	FPC2005,FPC2006	PCC1006A
GH63-11154A	SC6000	SC-SMG928V-SUBPBA
GH63-14647A	SC6005	SC-SMC7100-BTWIFI
GH63-14648A	SC6001	SC-SMC7100-PMIC
GH63-14649A	SC6002	SC-SMC7100-TRV
GH63-14650A	SC6003	SC-SMC7100-AP
GH63-14659A	SC6006	SC-SMC7100-SPK
GH63-14674A	SC6004	SC-SMC7100-RF
GH63-14675A	SC6008	SC-SMC7100-FILTER
GH63-14676A	SC6007	SC-SMC7100-CHARGER
GH02-15424A	SC6003SP	SHIELD COVER AP

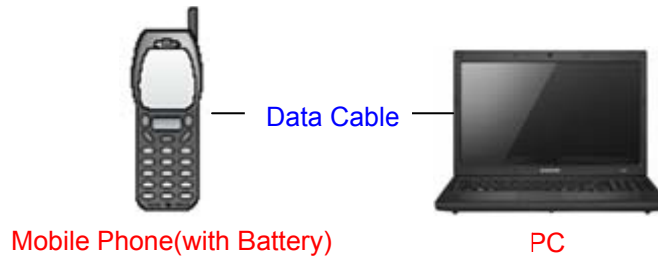
Please consult the GSPN website (Samsung Portal) for the most recent version of the product's part list.

6. Level 1 Repair

6-1. S/W Download

6-1-1. Prepare for S/W Downloading

- Diagram of connection



6-1-2. How to download S/W

1) Downloading Binary Files

- Binary file for downloading SM-C710F
 - [xxx.pit](#)
 - [BL_XXXX.tar.md5](#)
 - [AP_XXXX.tar.md5](#)
 - [CP_XXXX.tar.md5](#)
 - [CSC_xxxx.tar.md5](#)

(file size is about 3.8GB)

2) Prepare for Downloading

- Downloader Program ([Odin3 v3.12.3.exe](#))
- Mobile Phone
- Data Cable
- Binary file

3) Boot the mobile phone by pressing 'Home + Vol Down + Power key at the same time, If you do properly, you can see the following message on the main LCD as the following.



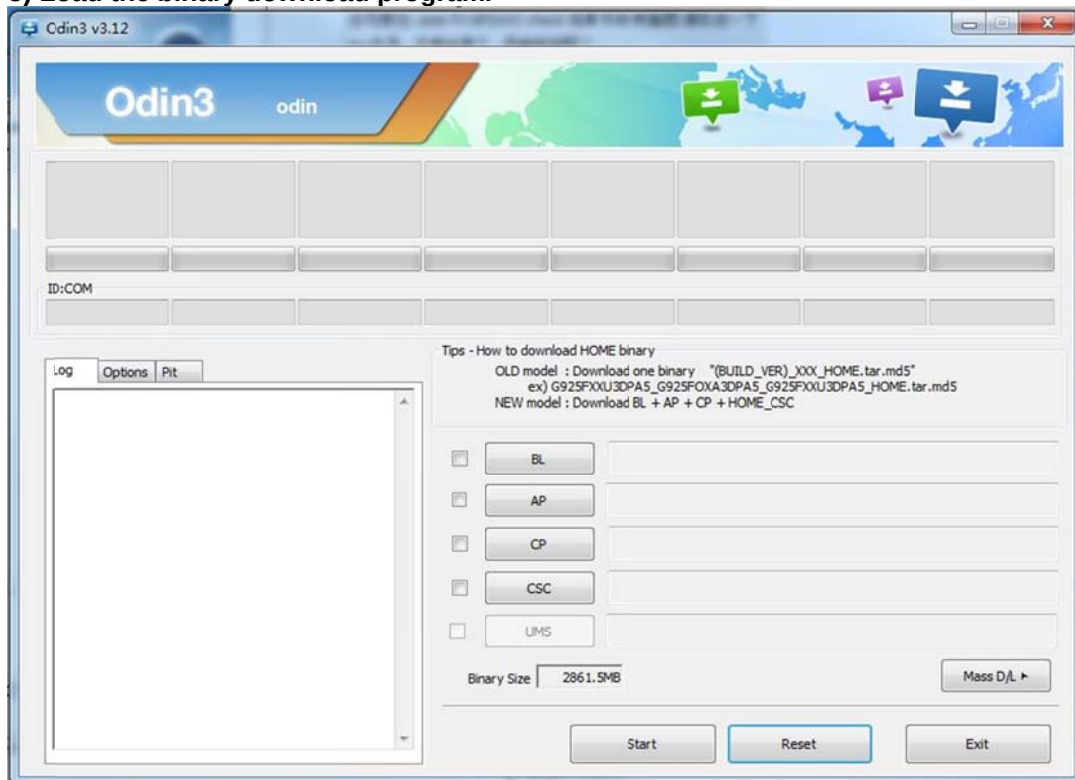
6. Level 1 Repair

4) Press the Vol Up Key again, and you will see below message on Main LCD.



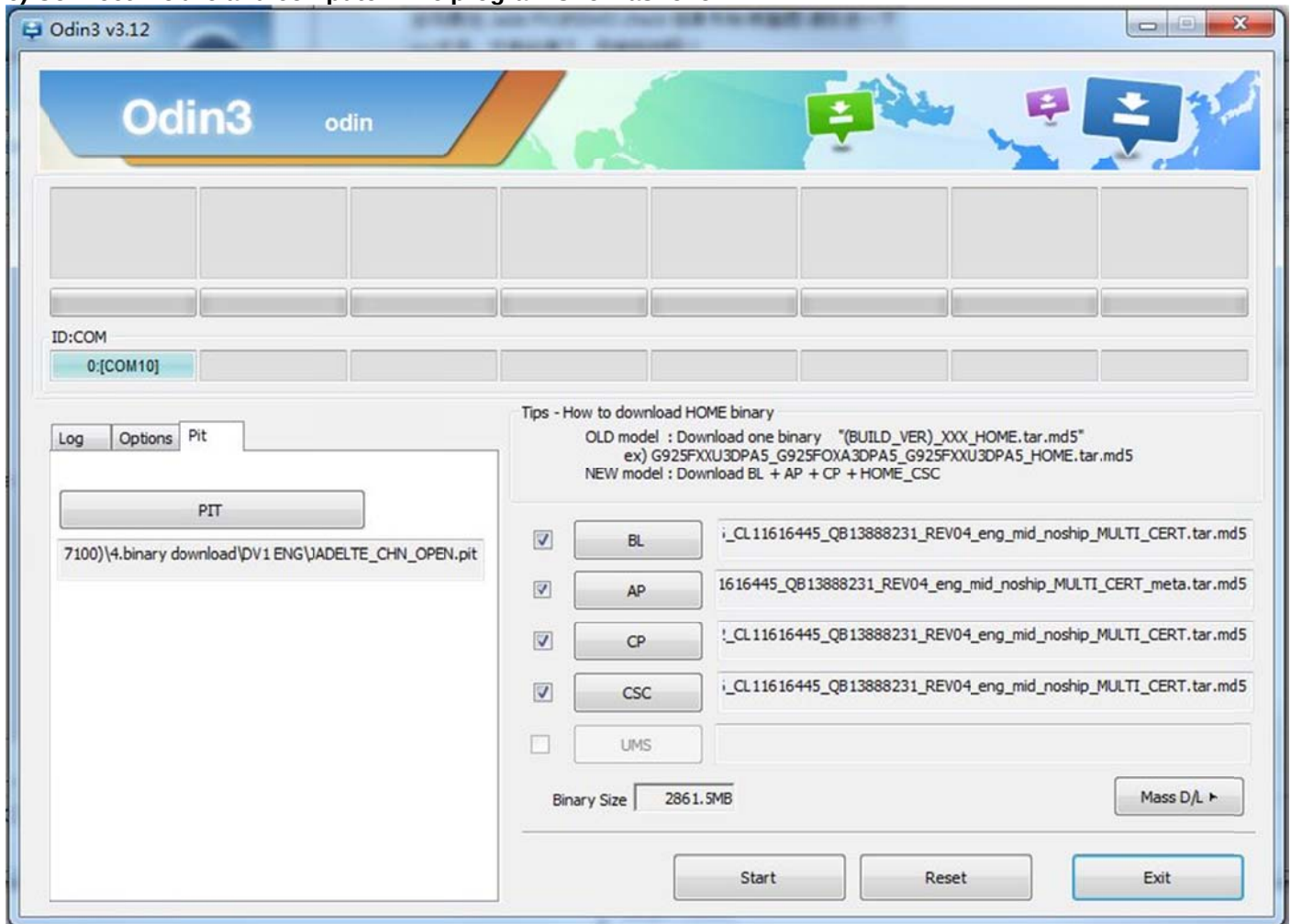
“ Downloading...”

5) Load the binary download program.



6. Level 1 Repair

6) Connect mobile and computer. The program show as follow.



Select the file as above:

- BL_XXXX.tar.md5
- AP_XXXX.tar.md5
- CP_XXXX.tar.md5
- CSC_xxxx.tar.md5

7) Now press the button "Start".

8) Now it's time to take a rest and finish the downloading.

9) After finished downloading of phone binary, the mobile phone will restart automatically.

※ **Caution. Never disconnect during the S/W downloading.**

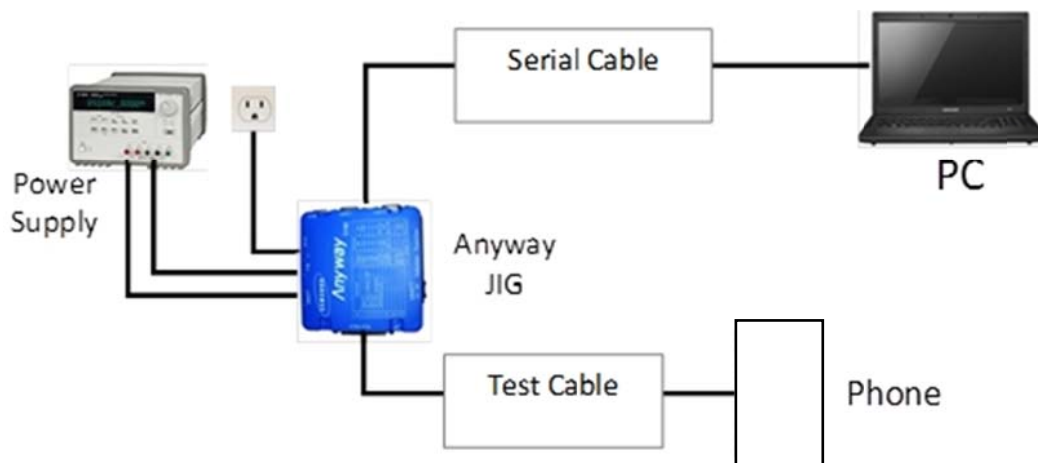
6. Level 1 Repair

6-2. IMEI writing

6-2-1. Preparation

- New IMEI writing Program has been released.
- Supported Model: Models which CAB files are uploaded on HHP svc INI File category, instead of ini file.
- Refer to below IMEI writing procedure.

- H/W



- S/W

① Library Install	To use Daseul, library files should be installed. Refer to SVC Bulletin “(11-82) Daseul (New IMEI writing Program) Library Install guide_rev1.0”
② Launcher	DASEUL_SVC_Launcher_v3_0_25 or higher -Uploaded on HHPsvc Notice
③ Runtime File	1. DASEUL_Runtime_Ver_3.1.139.0.CAB or higher -Uploaded on HHPsvc Notice 2. Make 'ModelName' folder at the same position with launcher & Runtime file.
④ Model File	Copy Model File under the 'Model Name' folder

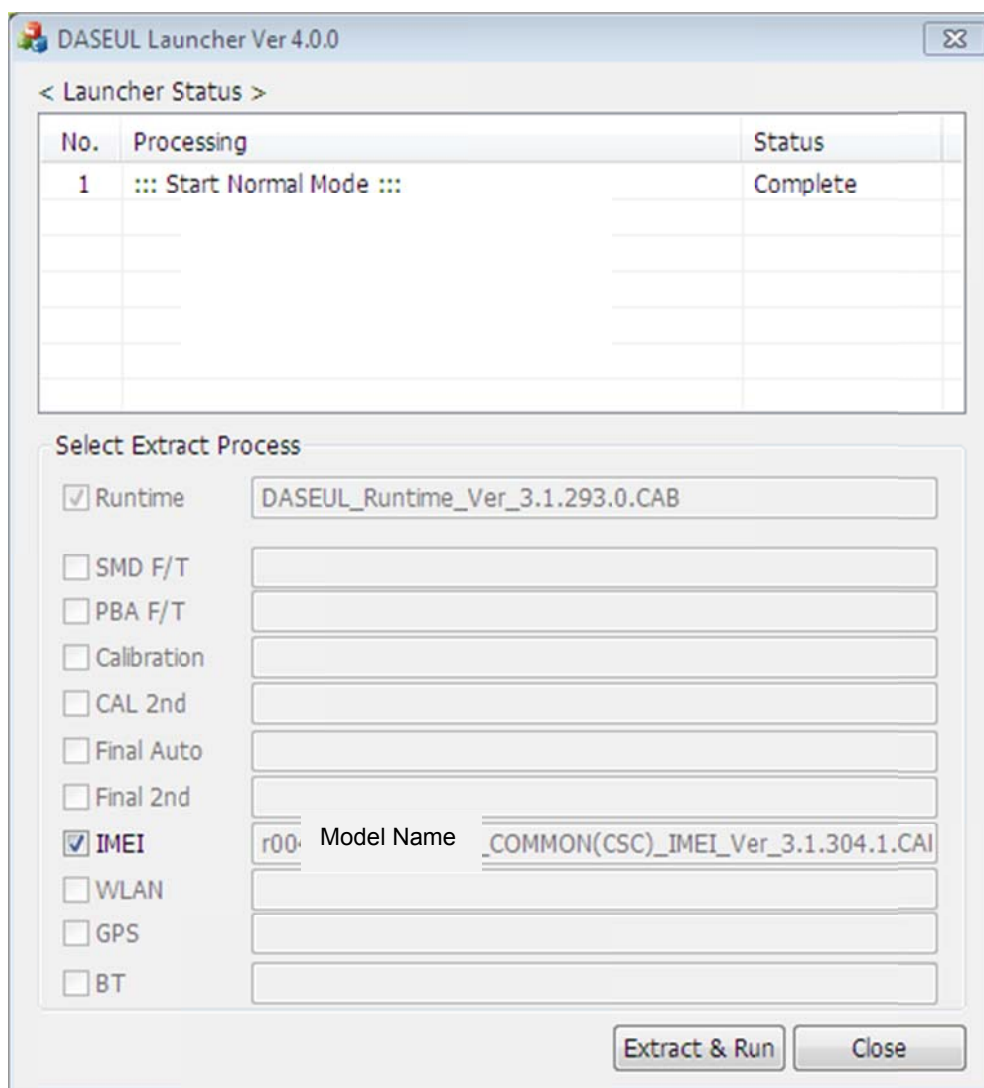
6. Level 1 Repair

6-2-2. IMEI writingProcess

1. 1. Run DASEUL_Launcher_v4.0.0

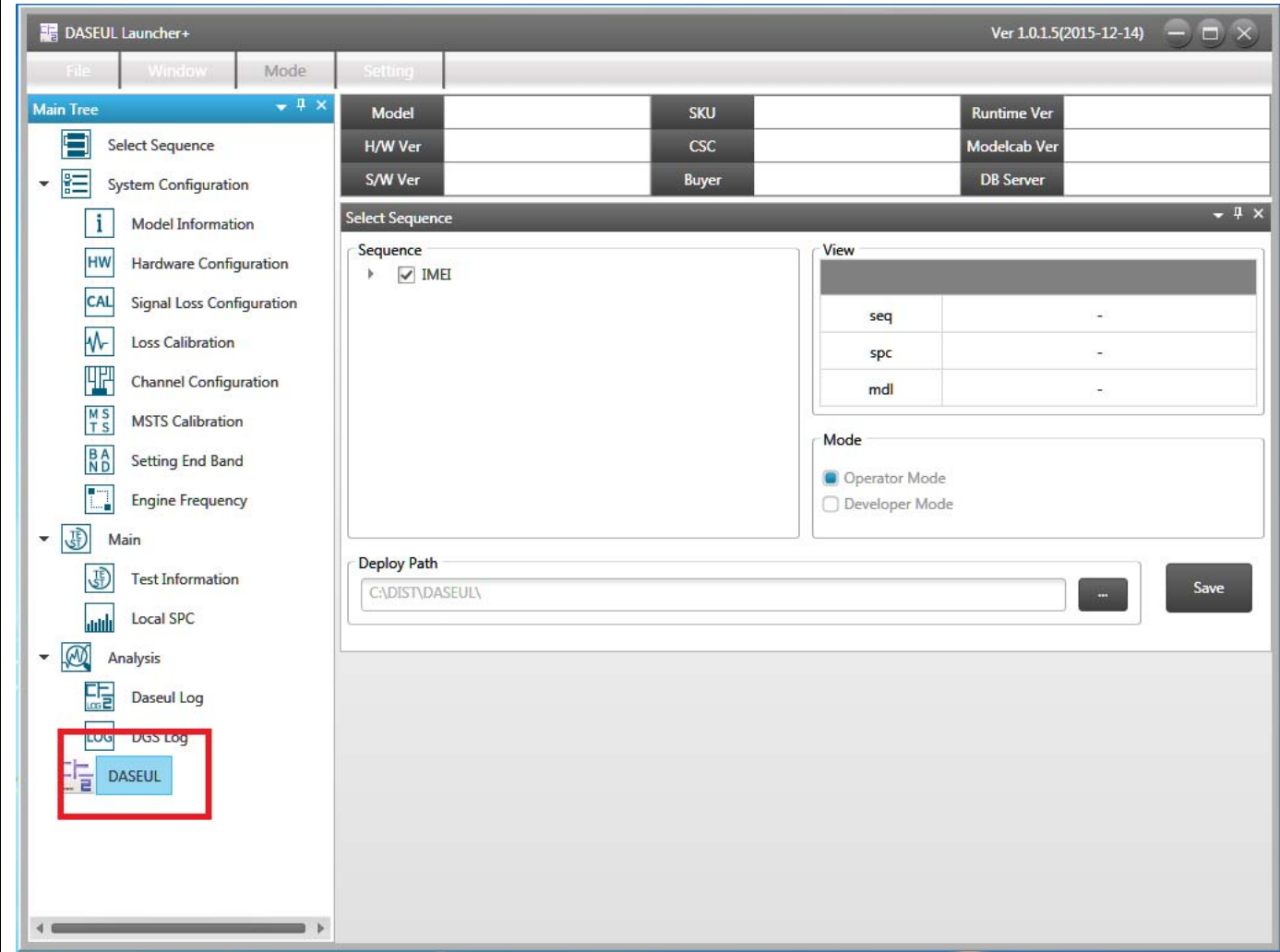
 DASEUL_IMEI_ALL_Component_r00425.CAB	2017/4/10 10:17	ALZip cab File	195,594 KB
 DASEUL_IMEI_ALL_Runtime_3.1.293.0_r00425.CAB	2017/4/11 9:16	ALZip cab File	231,261 KB
 DASEUL_Launcher_v4.0.0.exe	2016/10/21 15:11	Application	5,896 KB
 DASEUL_Runtime_Ver_3.1.293.0.CAB	2017/4/8 7:14	ALZip cab File	37,075 KB
 SM-C7100_COMMON(CSC)_IMEI_Ver_3.1.304.1.CAB	2017/6/8 10:25	ALZip cab File	42 KB

2. Select Service Mode



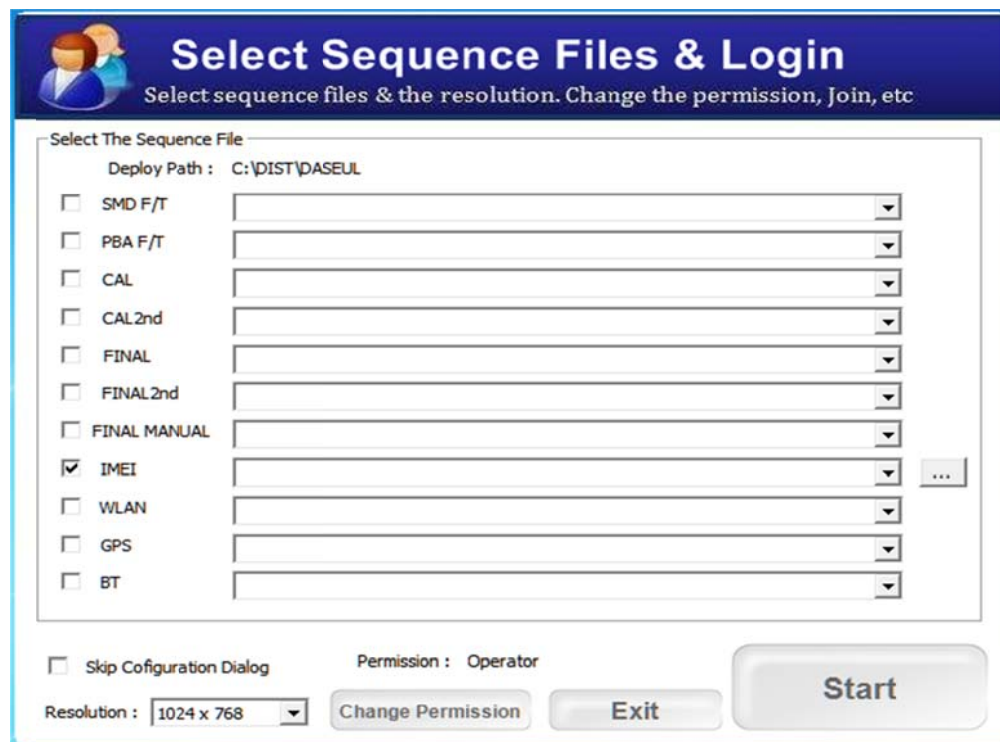
6. Level 1 Repair

3. Click DASEUL Select folder where the Launcher exists

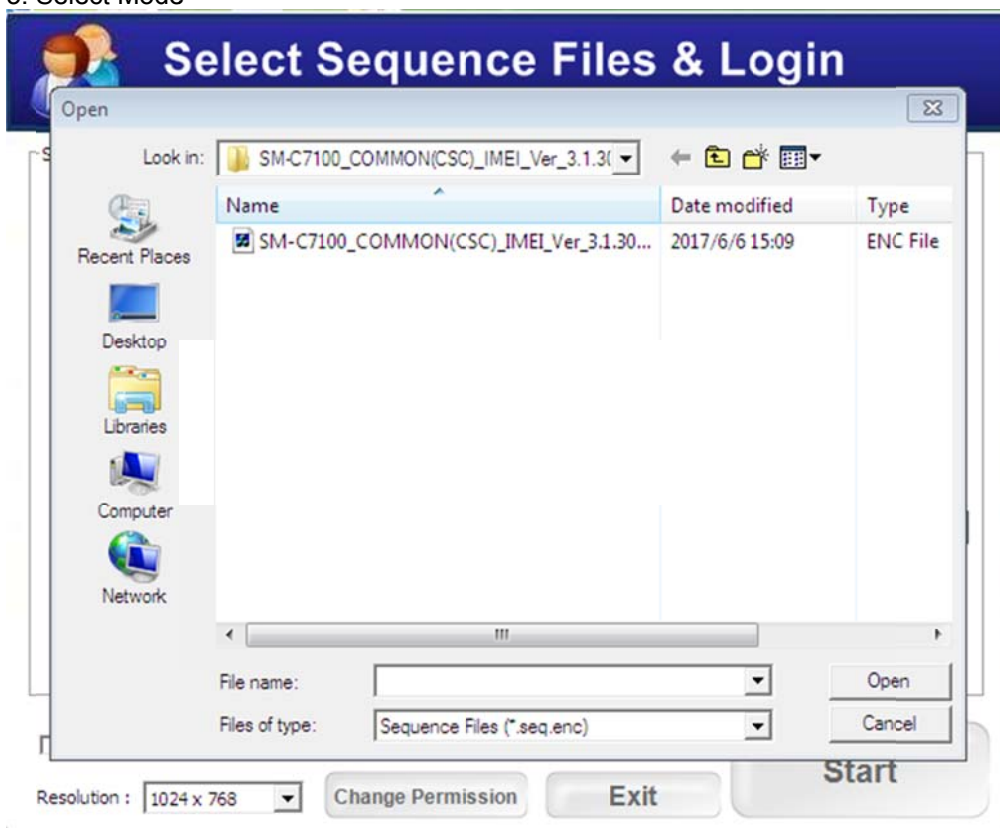


6. Level 1 Repair

4.  Select folder where the Launcher exists

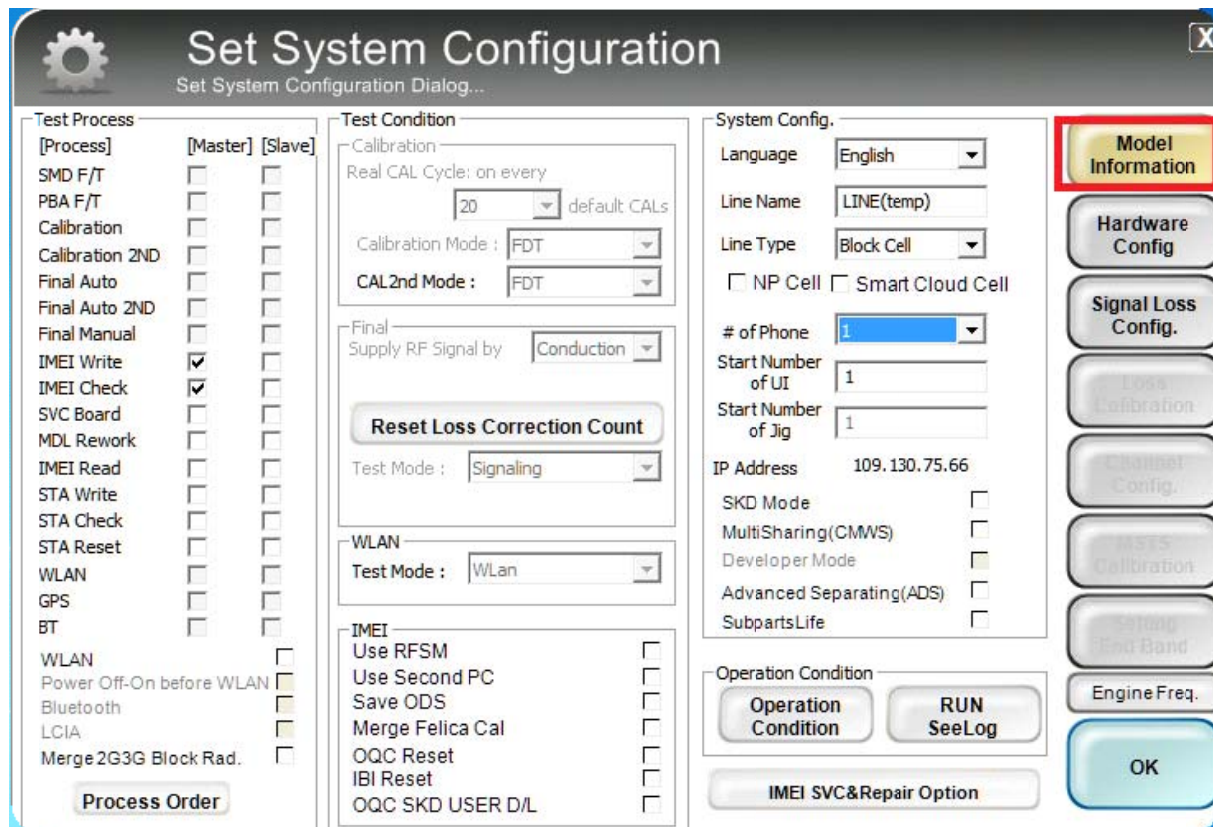


5. Select Mode



6. Level 1 Repair

6. Check 'Model information'



The screenshot shows the 'Set System Configuration' dialog box. The 'Test Process' tab is active, showing a list of test items with checkboxes for Master and Slave. The 'Test Condition' tab is also visible, showing settings for Calibration, Final Supply RF Signal, and WLAN. The 'System Config.' tab is highlighted with a red box, showing settings for Language, Line Name, Line Type, NP Cell, Smart Cloud Cell, # of Phone, Start Number of UI, Start Number of Jig, IP Address, SKD Mode, MultiSharing(CMWS), Developer Mode, Advanced Separating(ADS), and SubpartsLife. The 'Model Information' button is highlighted with a red box. The 'Operation Condition' tab is also visible, showing buttons for 'Operation Condition', 'RUN SeeLog', and 'IMEI SVC&Repair Option'.

Set System Configuration
Set System Configuration Dialog...

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☐	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☒	☐
IMEI Check	☒	☐
SVC Board	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

Calibration
Real CAL Cycle: on every default CALs
Calibration Mode:
CAL2nd Mode:
Final
Supply RF Signal by:
Reset Loss Correction Count
Test Mode:
WLAN
Test Mode:
IMEI
Use RFSM ☐
Use Second PC ☐
Save ODS ☐
Merge Felica Cal ☐
OQC Reset ☐
IBI Reset ☐
OQC SKD USER D/L ☐

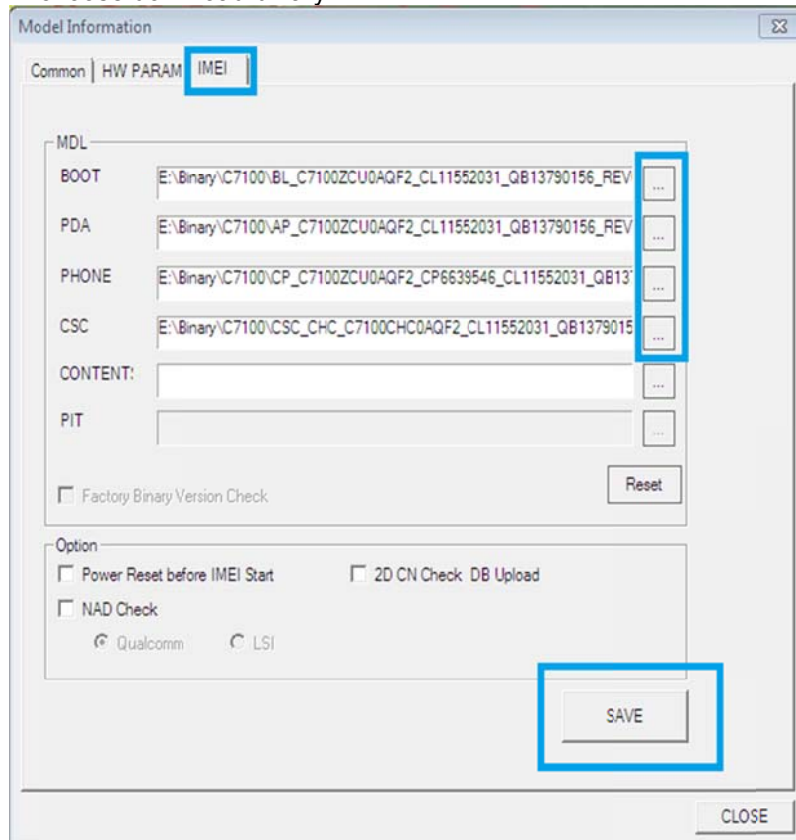
System Config.

Language:
Line Name:
Line Type:
☐ NP Cell ☐ Smart Cloud Cell
of Phone:
Start Number of UI:
Start Number of Jig:
IP Address: 109.130.75.66
SKD Mode ☐
MultiSharing(CMWS) ☐
Developer Mode ☐
Advanced Separating(ADS) ☐
SubpartsLife ☐

Operation Condition

Model Information

7. choose download binary



The screenshot shows the 'Model Information' dialog box. The 'IMEI' tab is active, showing a list of MDL files with checkboxes for BOOT, PDA, PHONE, CSC, CONTENT, and PIT. The 'Factory Binary Version Check' checkbox is checked. The 'Option' section has checkboxes for 'Power Reset before IMEI Start', '2D CN Check DB Upload', and 'NAD Check'. The 'Qualcomm' and 'LSI' radio buttons are selected. The 'SAVE' button is highlighted with a blue box.

Model Information

Common | HW PARAM | **IMEI**

MDL

BOOT	E:\Binary\C7100\BL_C7100ZCU0AQF2_CL11552031_QB13790156_REV	☐
PDA	E:\Binary\C7100\AP_C7100ZCU0AQF2_CL11552031_QB13790156_REV	☐
PHONE	E:\Binary\C7100\CP_C7100ZCU0AQF2_CP6639546_CL11552031_QB13790156_REV	☐
CSC	E:\Binary\C7100\CSC_CHC_C7100CHC0AQF2_CL11552031_QB13790156_REV	☐
CONTENT		☐
PIT		☐

☐ Factory Binary Version Check

Option

☐ Power Reset before IMEI Start ☐ 2D CN Check DB Upload
☐ NAD Check
☒ Qualcomm ☐ LSI

6. Level 1 Repair

8. Click IMEI, Then choose binary, click SAVE.

Model Information

Common | HW PARAM | **IMEI**

MDL

BOOT: E:\Binary\7100\BL_C7100ZCU0AQF2_CL11552031_QB13790156_REV ...

PDA: E:\Binary\7100\AP_C7100ZCU0AQF2_CL11552031_QB13790156_REV ...

PHONE: E:\Binary\7100\CP_C7100ZCU0AQF2_CP6639546_CL11552031_QB13 ...

CSC: E:\Binary\7100\CSC_CHC_C7100CHC0AQF2_CL11552031_QB1379015 ...

CONTENT: ...

PIT: ...

☐ Factory Binary Version Check

Reset

Option

☐ Power Reset before IMEI Start ☐ 2D CN Check DB Upload

☐ NAD Check

☒ Qualcomm ☐ LSI

SAVE

CLOSE

9. Click 'Hardware Config'

Set System Configuration

Set System Configuration Dialog...

Test Process

[Process] [Master] [Slave]

SMD F/T

PBA F/T

Calibration

Calibration 2ND

Final Auto

Final Auto 2ND

Final Manual

IMEI Write

IMEI Check

SVC Board

MDL Rework

IMEI Read

STA Write

STA Check

STA Reset

WLAN

GPS

BT

WLAN

Power Off-On before WLAN

Bluetooth

LCIA

Merge 2G3G Block Rad.

Process Order

Test Condition

Calibration

Real CAL Cycle: on every

20 default CALs

Calibration Mode: FDT

CAL2nd Mode: FDT

Final

Supply RF Signal by: Conduction

Reset Loss Correction Count

Test Mode: Signaling

WLAN

Test Mode: WLAN

IMEI

Use RFSM

Use Second PC

Save ODS

Merge Felica Cal

OQC Reset

IBI Reset

OQC SKD USER D/L

System Config.

Language: English

Line Name: LINE(temp)

Line Type: Block Cell

☐ Smart Cloud Cell

of Phone: 1

Start Number of UI: 1

Start Number of Jig: 1

IP Address: 109.131.10.178

SKD Mode

MultiSharing(CMWS)

Developer Mode

Advanced Separating(ADS)

Operation Condition

Operation Condition

RUN SeeLog

IMEI SVC&Repair Option

Model Information

Hardware Config

Signal Loss Config.

Block Calibration

Smart Config.

WLAN Calibration

Signal Band

Engine Freq.

OK

6. Level 1 Repair

10. Click 'Port Setting'

Hardware Component Configuration

Controller Type, IO Bus Type, Port Setting...

Phone
 Count: 1
 I/F - 1 Type: Serial COM
 I/F - 2 Type: Serial COM
 I/F 3g Type: AnyWay3g
 Multi 3g Cable Type: UART Line
☒ Use Portable ID Check JIG
 MSTS
 Count: 0
 I/F Type: GP2B
 Port Setting

MSTS Sharing Controller
 Count: 0
 Control Type: N/A
 I/F Type: Serial COM
 Terminal
 Port Setting
 Switch Box Port Setting
 Robot / ShieldBox
 Control Type: N/A
 I/F Type: Serial COM
 Port Setting
 Power Supply
 E3632A
 GP2B
 Port Setting
 Label Printer
 Control Type: None
 I/F Type: Serial COM
 Port Setting
 Offset X / Y

DEMS
 Server: HOME(GLINE)
 Type: Outside-Web5
 Barcode Reader
 Type: N/A
 I/F Type: Serial COM
 Port Setting
 LCIA ShieldBox
 I/F Type: Serial COM
 Port Setting
 MES PH Sender
 Type: N/A
 Port Setting
 Barometer
 I/O Type: Serial COM
 Port Setting

PBA F/T
 Function Test Jig
 NE-DAQ
 Port Setting
 Power Detector
 Port Setting
 HDMI Type
 HDMI JIG
 Port Setting
 SMD F/T
 Type: N/A
 I/d Address: 5
 Port Setting
 AP(2.4G) SSID
 AP(5G) SSID
 GPS Generator
 Port Setting
 WLAN
 Count: 0
 I/F Type: Serial COM
 Port Setting

SAVE **Cancel**

11. Select Port Number and SAVE

Hardware Component Configuration
Controller Type, IO Bus Type, Port Setting...

Set IO BUS Configuration

Phone IO Bus Setting

Common

BaudRate: 115200
Data Bit: 8
Parity: No
Stop Bit: 1

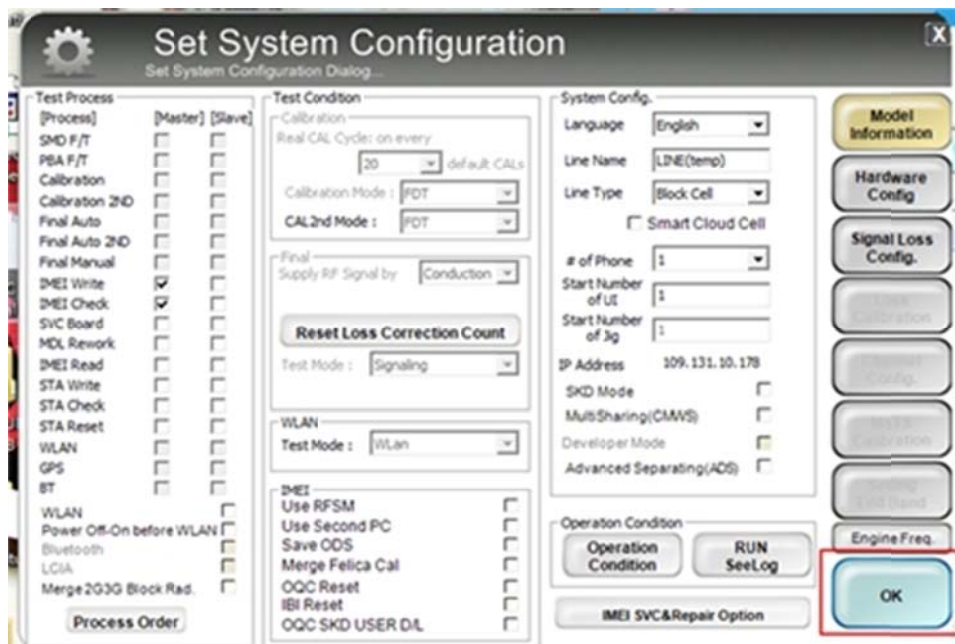
No.	Port #1	Port #2
1	4	13

serial port binary-DL-port

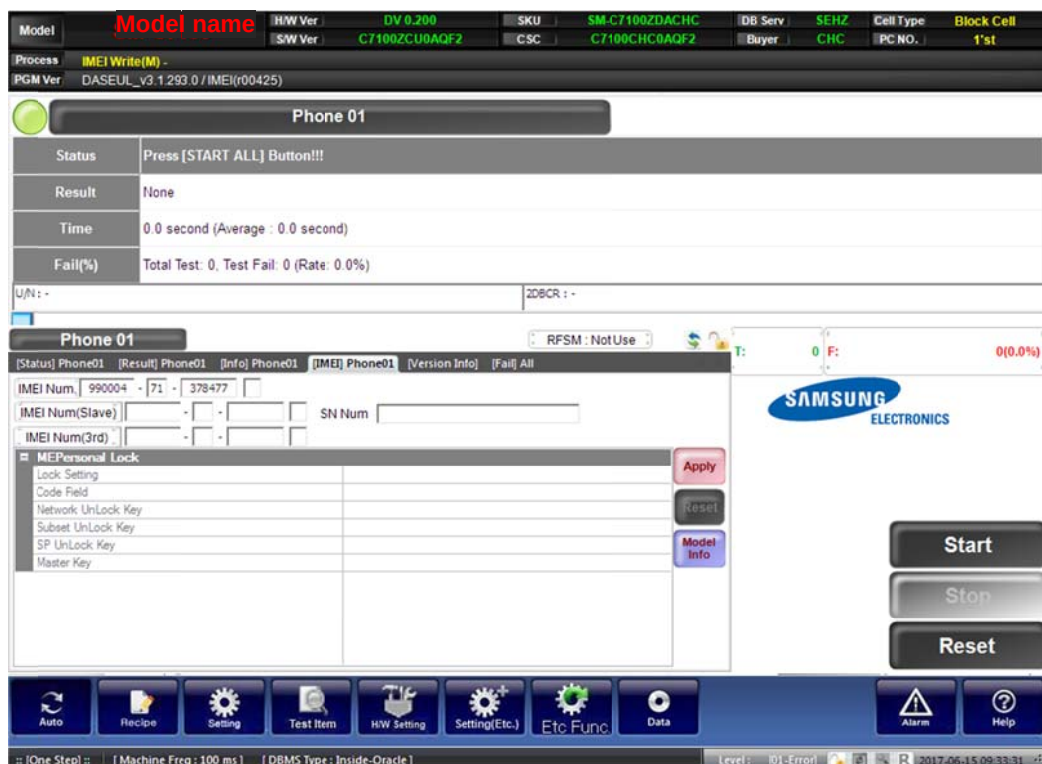
SAVE Cancel

6. Level 1 Repair

12. Click OK to proceed



13. Click Model Info and OK when pop-up shows

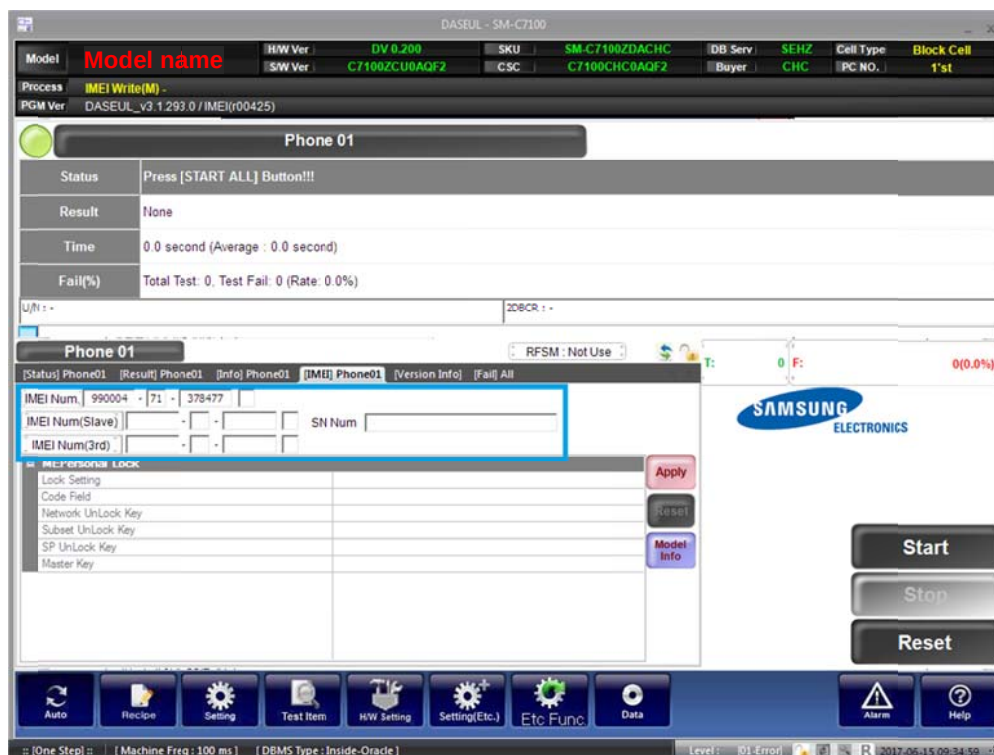


6. Level 1 Repair

14. Click OK

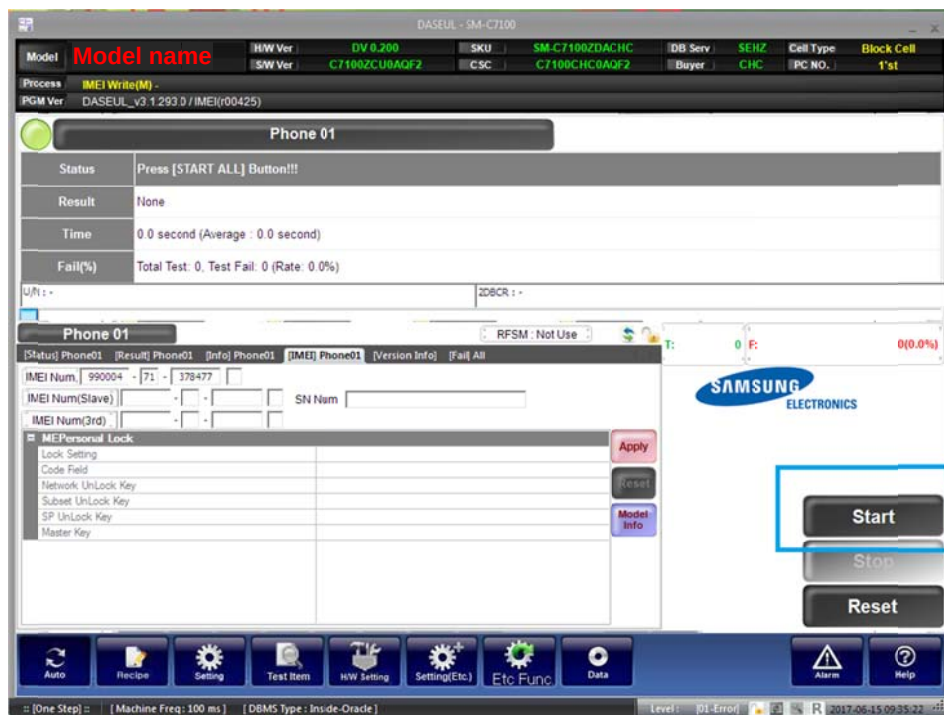


15. Input IMEI Number and click apply.



6. Level 1 Repair

16 Click Start

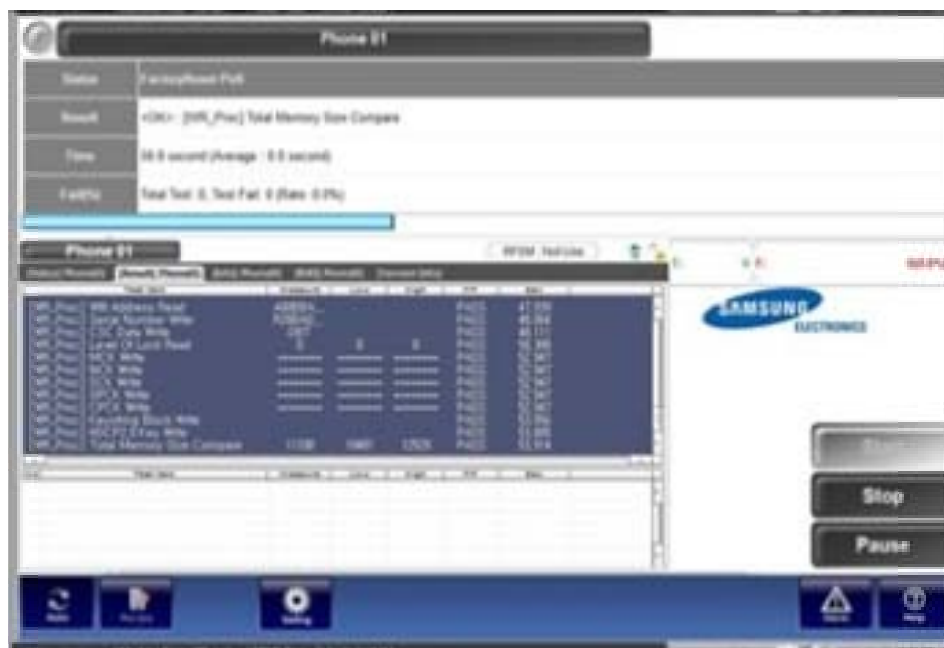


17. Connect the phone to Anyway JIG

※ When you connect the phone, the phone should be turned off.

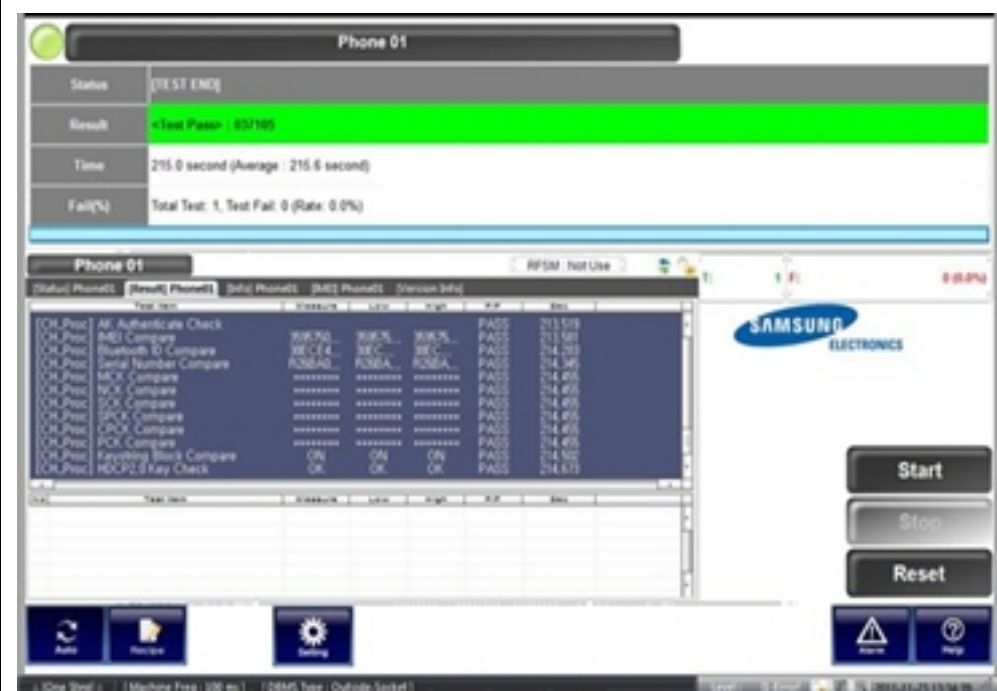
After connecting the phone, the phone will be booted automatically.

18. IMEI Writing Proceeding



6. Level 1 Repair

19. IMEI Writing Success



6. Level 1 Repair

6-3. RF Calibration

6-3-1. Required items in order to calibrate RF

- Installation program :RF Calibration Program
 - Daseul_Launcher_vx.x.xx.exe
 - Daseul_CAL_ALL_Runtime_x.x.xxx.x.CAB
 - Model File ([SM-C710F_OPEN_CALIBRATION_Ver_3.1.300.1.CAB](#))

※ It is required to use the latest program.

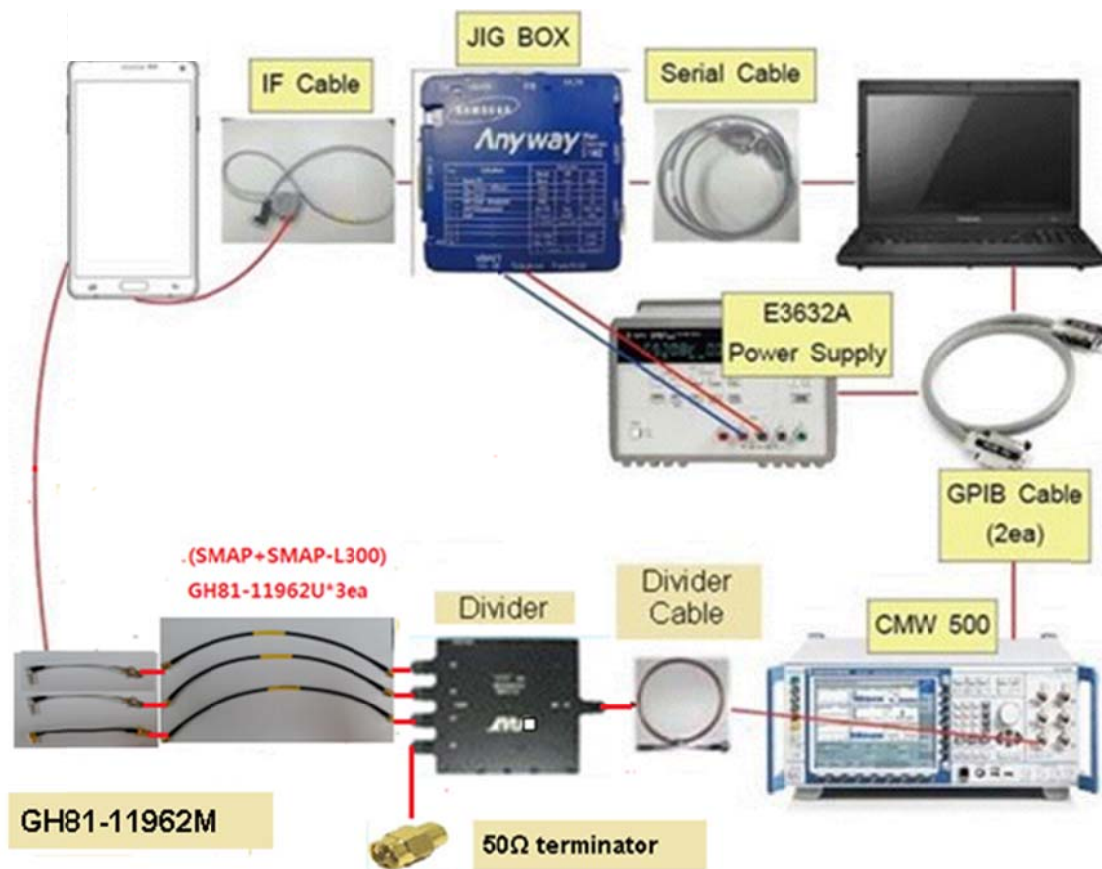
- Mobile Phone
- R&S CMW500
- E3632A Power Supply
- GPIB Cable(2ea)
- JIG BOX(Anyway S103)
- Type-7Pin IF Cable(GH81-10952A)
- Adapter (GH81-11888K)
- UART Serial Cable
- RF Cable(GH81-11962U) 3EA
- 1.2T RF Cable (GH81-11962M) 3EA
- 50Ω terminator(GH81-11962E 1EA)
- 4Port Divider(GH81-11962A 1EA)
- Divider Cable(GH81-11962B 1EA)

❖ Table of test cables

IF Cable	GH81-10952A		
	USB 2.0, Type-7Pin		
RF Cable (Manual)	GH81-11962M	GH81-11962U	
	1.2T ,102mm 3EA 	SMAP, 300mm, 3EA 	
4 Port Divider	GH81-11962A	GH81-11962B	GH81-11962E
	Divider 1EA 	Divider Cable 1EA 	50Ω terminator 1EA 

6. Level 1 Repair

❖ Setting



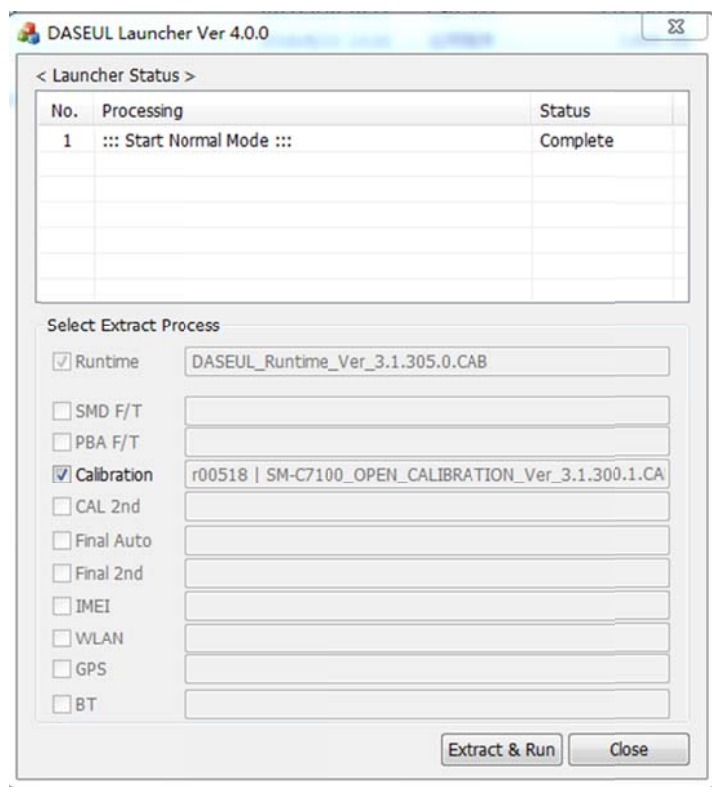
6. Level 1 Repair

6-3-2. RF Calibration Program

1. Run the RF Calibration Program Launcher, '[DASEUL_Launcher_vx.x.xx.exe](#)'.

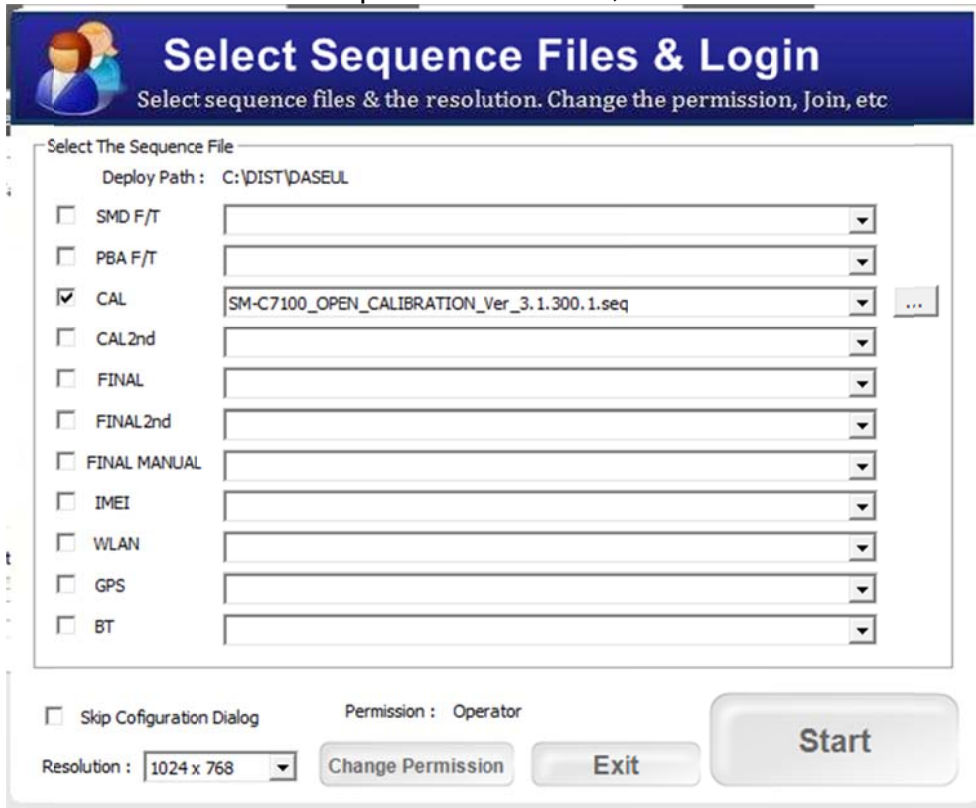
	DASEUL_CAL_ALL_Component_r00518.CAB	2017/5/24 16:44	Cab 文件	95,726 KB
	DASEUL_CAL_ALL_Runtime_3.1.305.0_r00518.CAB	2017/5/26 16:11	Cab 文件	131,230 KB
	DASEUL_Launcher_v4.0.0.exe	2016/8/23 13:16	应用程序	5,896 KB
	DASEUL_Runtime_Ver_3.1.305.0.CAB	2017/5/23 10:23	Cab 文件	37,514 KB
	SM-C7100_OPEN_CALIBRATION_Ver_3.1.300.1.CAB	2017/6/11 8:39	Cab 文件	264 KB

2. Check the '[Calibration](#)' menu, and select '[Extract & Run](#)'.



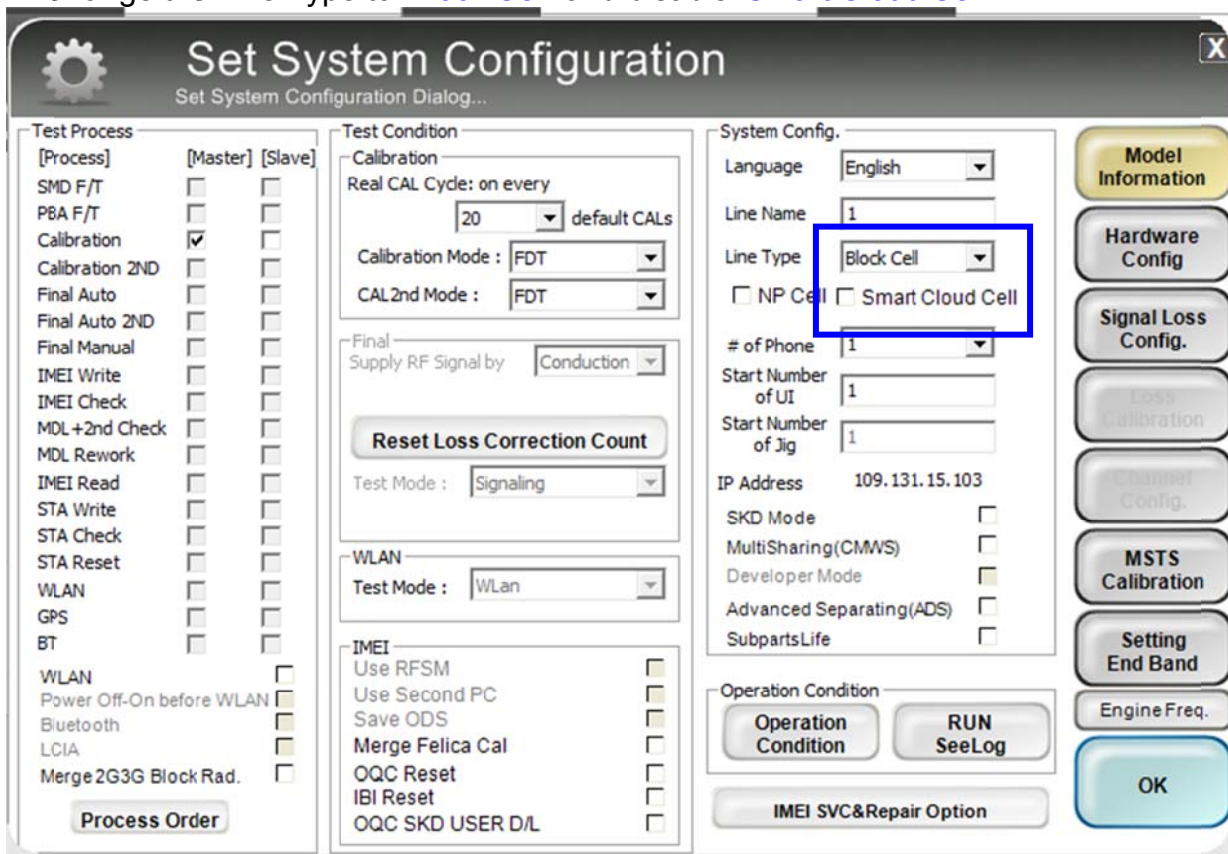
6. Level 1 Repair

3. Check the 'CAL' and open the [model file](#), then select 'Start' button.



The 'Select Sequence Files & Login' dialog box is shown. It has a title bar with a blue header and a subtitle 'Select sequence files & the resolution. Change the permission, Join, etc'. Below the header, there is a section 'Select The Sequence File' with a 'Deploy Path' of 'C:\DIST\DAEUL'. A list of sequence files is shown with checkboxes and dropdown menus. The 'CAL' checkbox is checked, and its dropdown menu is open, showing 'SM-C7100_OPEN_CALIBRATION_Ver_3.1.300.1.seq'. Other sequence files include SMD F/T, PBA F/T, CAL2nd, FINAL, FINAL2nd, FINAL MANUAL, IMEI, WLAN, GPS, and BT. At the bottom, there is a 'Skip Configuration Dialog' checkbox, a 'Permission' dropdown set to 'Operator', a 'Resolution' dropdown set to '1024 x 768', and buttons for 'Change Permission', 'Exit', and 'Start'.

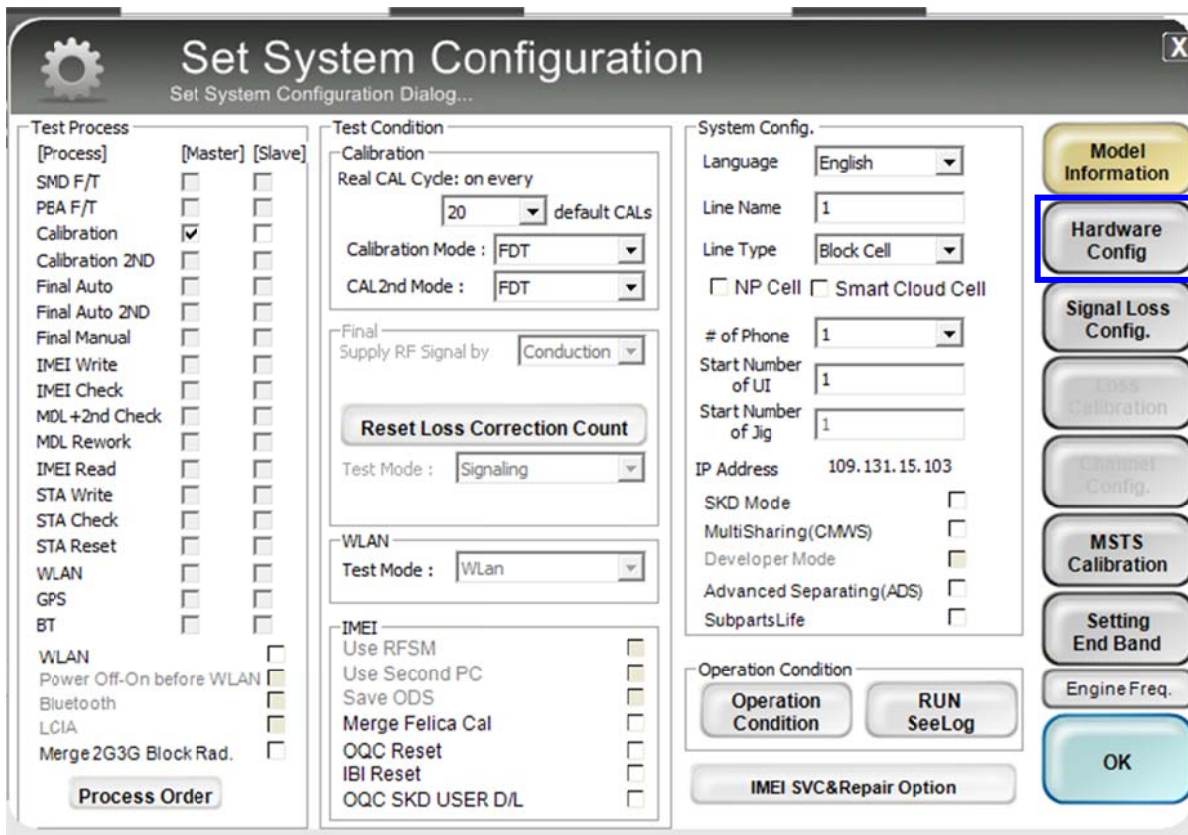
4. Change the Line Type to 'Block Cell' and disable 'Smart Cloud Cell'.



The 'Set System Configuration' dialog box is shown. It has a title bar with a gear icon and a subtitle 'Set System Configuration Dialog...'. The dialog is divided into several sections. On the left, there is a 'Test Process' section with a list of test items and checkboxes for 'Master' and 'Slave'. The 'Calibration' checkbox is checked. In the center, there is a 'Test Condition' section with 'Calibration' settings: 'Real CAL Cycle: on every 20 default CALs', 'Calibration Mode: FDT', and 'CAL2nd Mode: FDT'. Below this is a 'Final' section with 'Supply RF Signal by: Conduction' and a 'Reset Loss Correction Count' button. The 'Test Mode' is set to 'Signaling'. In the bottom left, there is a 'WLAN' section with 'Test Mode: WLAN' and an 'IMEI' section with various checkboxes. On the right, there is a 'System Config.' section with 'Language: English', 'Line Name: 1', 'Line Type: Block Cell' (highlighted with a blue box), and 'NP Cell' and 'Smart Cloud Cell' checkboxes. Below this are fields for '# of Phone', 'Start Number of UI', 'Start Number of Jig', 'IP Address', 'SKD Mode', 'MultiSharing(CMWS)', 'Developer Mode', 'Advanced Separating(ADS)', and 'SubpartsLife'. At the bottom right, there is an 'Operation Condition' section with 'Operation Condition' and 'RUN SeeLog' buttons, and an 'IMEI SVC&Repair Option' button. On the far right, there is a vertical stack of buttons: 'Model Information', 'Hardware Config', 'Signal Loss Config.', 'Loss Calibration', 'Channel Config.', 'MSTS Calibration', 'Setting End Band', 'Engine Freq.', and 'OK'.

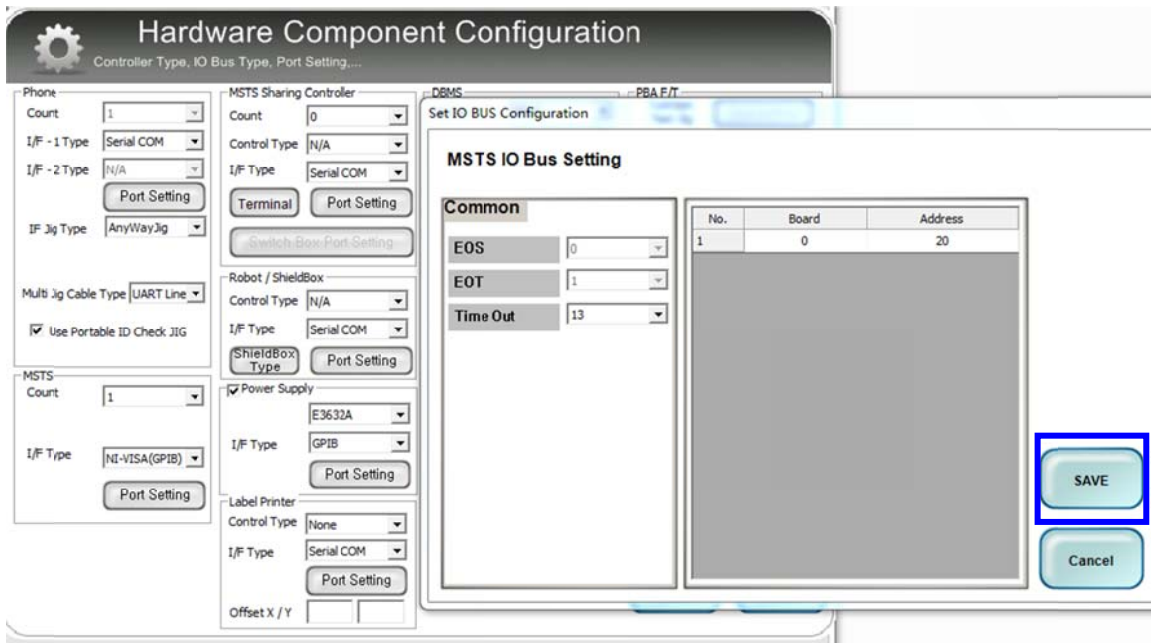
6. Level 1 Repair

5. Set the GPIB address of MST5(CMW500) and Power Supply(E3632A) to enter



The 'Set System Configuration' dialog box is shown. The 'Hardware Config' button on the right sidebar is highlighted with a blue box. The 'Test Process' section on the left has 'Calibration' checked under the 'Master' column. The 'Test Condition' section shows 'Real CAL Cycle: on every 20' and 'Calibration Mode: FDT'. The 'System Config.' section shows 'Language: English', 'Line Name: 1', 'Line Type: Block Cell', and 'IP Address: 109.131.15.103'. The 'Operation Condition' section has 'Operation Condition' and 'RUN SeeLog' buttons. The 'IMSI SVC&Repair Option' button is at the bottom.

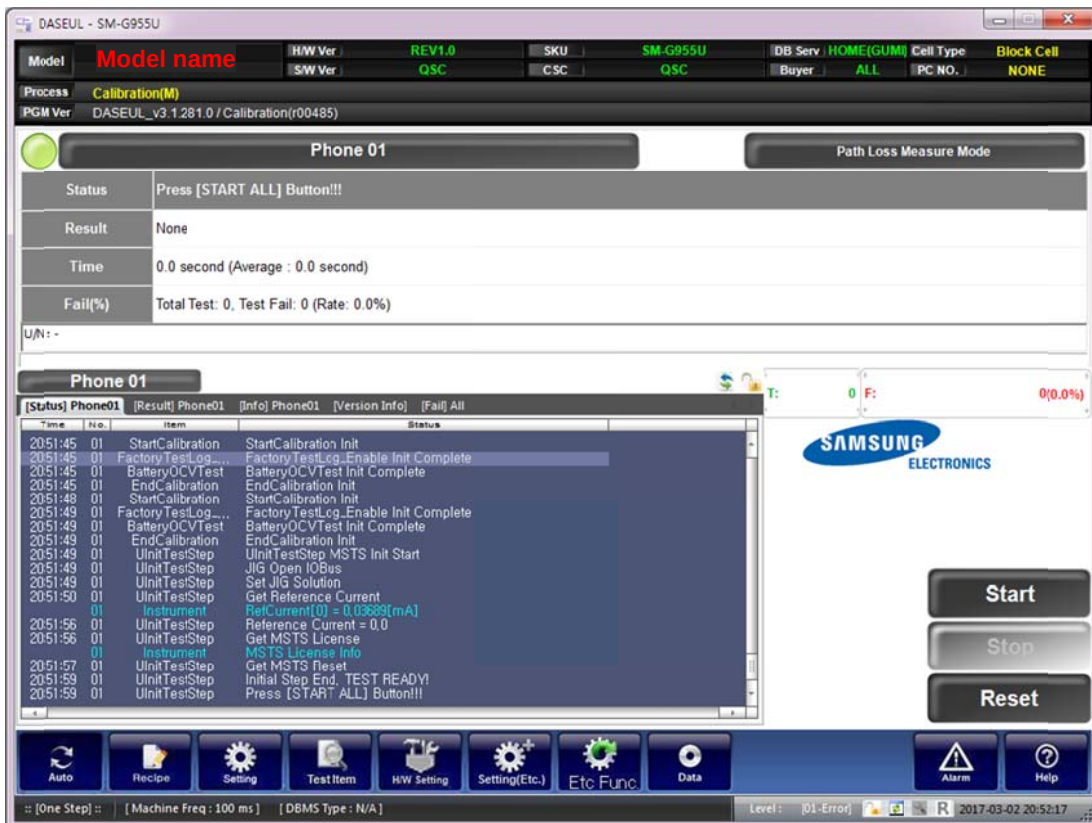
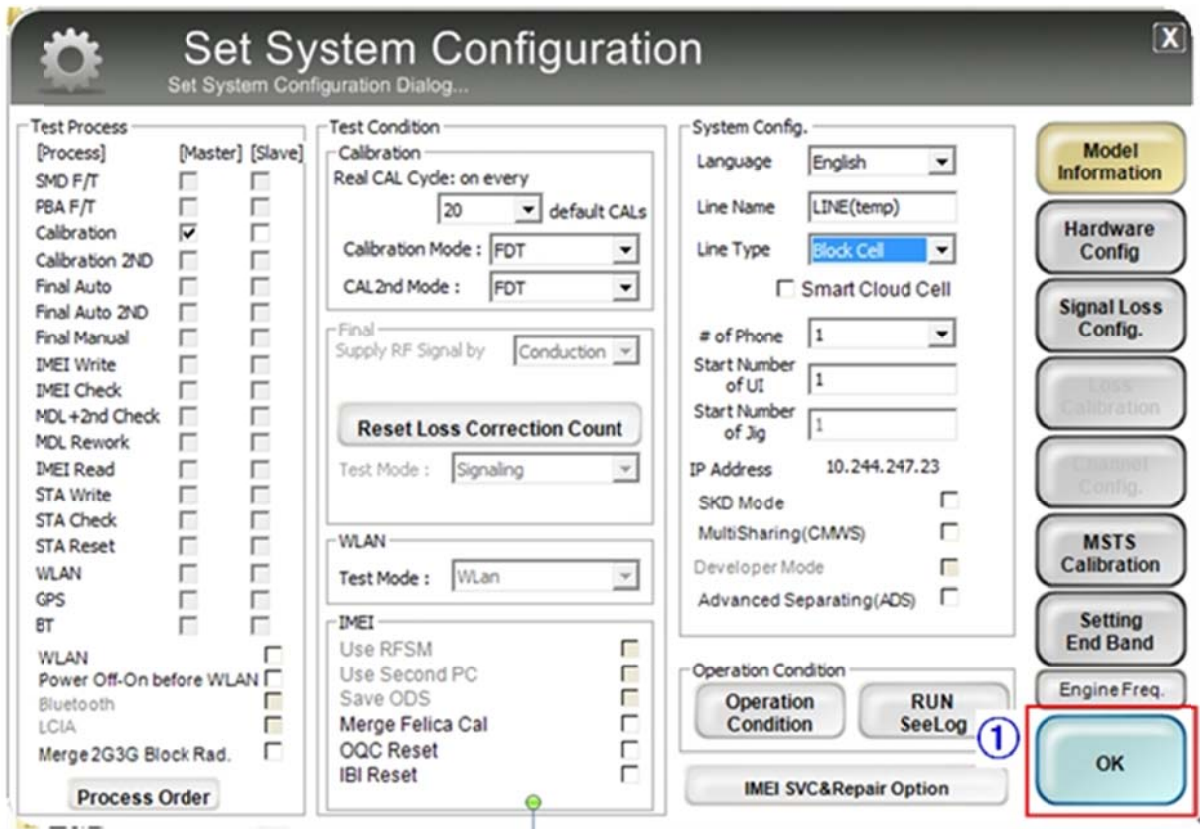
'Hardware Config' and 'Save'. (Check the GPIB address of equipments in advance)



The 'Hardware Component Configuration' dialog box is shown. The 'MST5 IO Bus Setting' sub-dialog is open, showing 'Common' settings for 'EOS', 'EOT', and 'Time Out'. The 'SAVE' button is highlighted with a blue box. The 'MST5' section on the left shows 'Count: 1' and 'I/F Type: NI-VISA(GPIB)'. The 'Power Supply' section shows 'E3632A' and 'GPIB'.

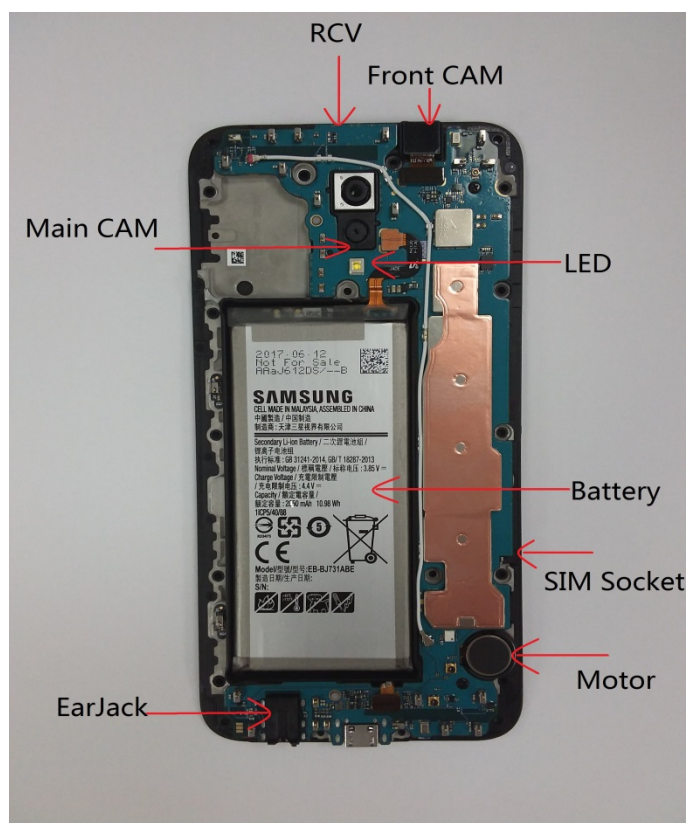
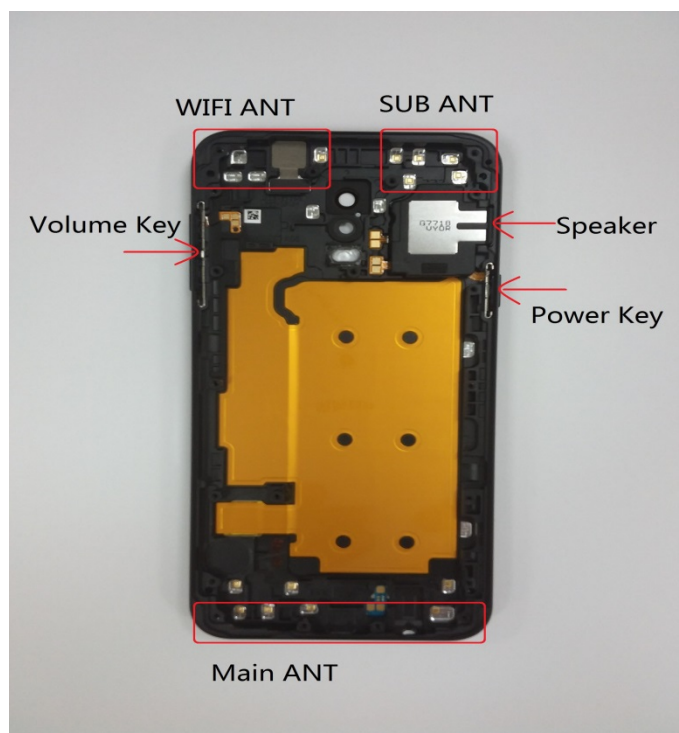
6. Level 1 Repair

6. Press 'OK' to start RF Calibration after completing all settings.



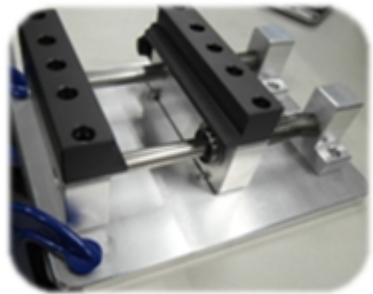
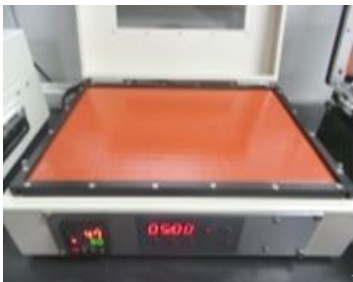
7. Level 2 Repair

7-1. Components on the Rear Case



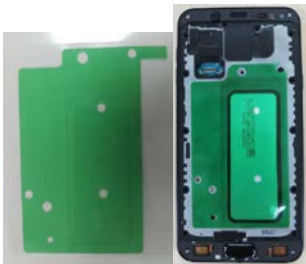
7. Level 2 Repair

7-2. Pre-requisite

	
Tweezers / Disass'y Stick / Screw Driver	Anti-static Gloves
	
OCTA Disassembly Holder	Anti-static Mat
	
Hot Plate	OCTA Disassembly Upper
	
Ethyl Alcohol	Cotton Swab

7. Level 2 Repair

7-3. Parts which must be changed after repair

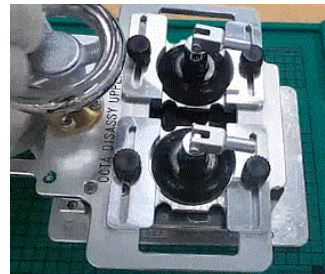
BOM Description	Picture	Remarks
A/S-TAPE_ MAIN_WINDOW_SVC [GH02-14924A] [GH02-14923A] Touch Key Top Tape GH81-14737A A/S-TAPE HOTMELT-HR SVC [GH02-15149A]	 [TOP_ GH02-14924A]  [BOT_ GH02-14923A]	Replace when repair Front part
		
		

7. Level 2 Repair

7-4. Disassembly

※ Be careful to put on antistatic wrist strap!

1. Put the device in the chamber or hot plate as following below heating condition
2. Temp : $70^{\circ}\pm 5$, Time : 15 Minutes
※ Please confirm the heating condition released lastly, and follow it.



Take out SET, when OCTA and Front ASS'Y Detach

[1] After heating for 15~30 minutes at $70\sim 80^{\circ}\text{C}$, take out the device from Mobile dryer.

[2] Detach the Back Cover from SET using Back Cover Disassembly Holder.



[3] Loosen 20 screws.

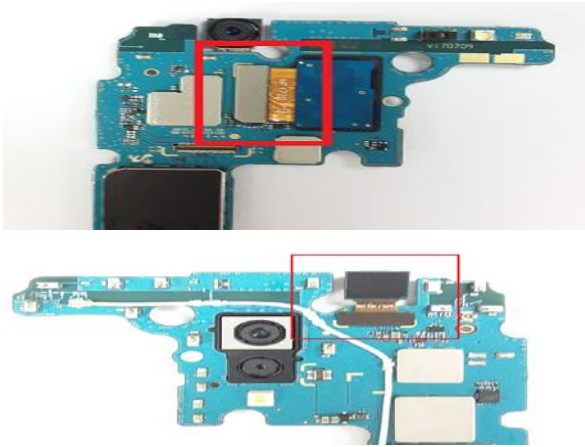
※ Caution
Take care of scratch and damage.



[4] Disassemble rear from bracket assy.(from top to bottom)

※ Caution
Be careful not to Damage

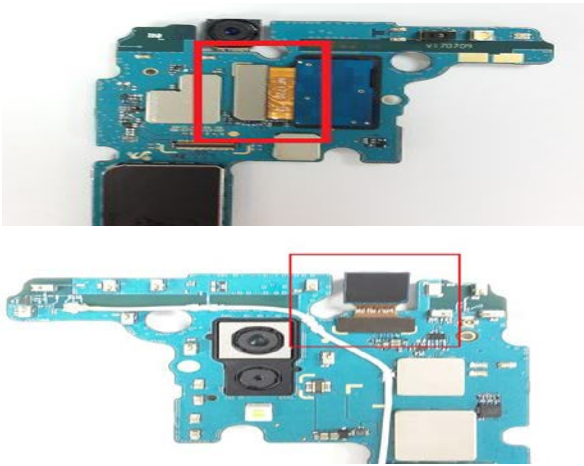
7. Level 2 Repair

	
[5] 1) Disassemble Home Key Connector/Battery Connector 2) Separate PBA from Brocket. Detach 2 screws point ※ Caution 1) Be careful not to damage. 2) Separate PBA after detach the PBA Conn.	[6] Disassemble Mega Cam and Front Cam on PBA. ※ Caution Be careful not to Damage
	
[7]Disassemble RCV, Motor	[8]Disassemble SPK from the REAR

7. Level 2 Repair

7-5. Assembly

※ Be careful to put on antistatic wrist strap!

	
[1] Assemble the Main CAM and VT CAM to the PBA	[2] Assemble Motor, RCV
	
[3] Assemble PBA to Bracket with 2 point screws ※ Caution Be careful not to Damage	[4] Assemble Homekey connector to the PBA

7. Level 2 Repair

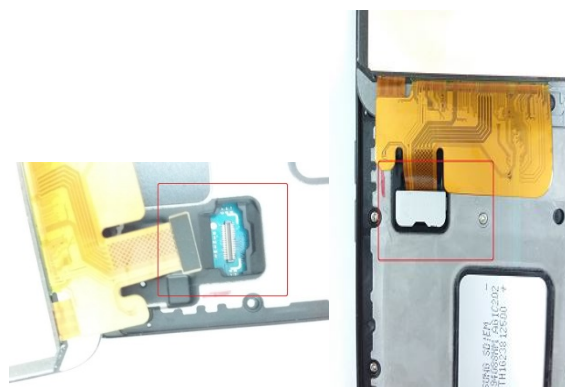


[5] Assemble Rear to Bracket assy (from bottom to Top)

※ **Caution**

Be careful not to Damage

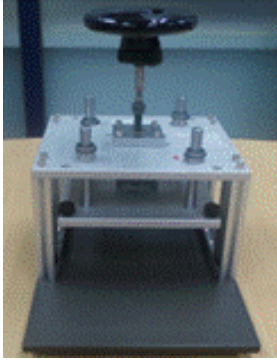
[6] Assemble Screw 20 points



[7] Assemble HR tape Touch key TOP tape to the Bracket and window tape to OCTA

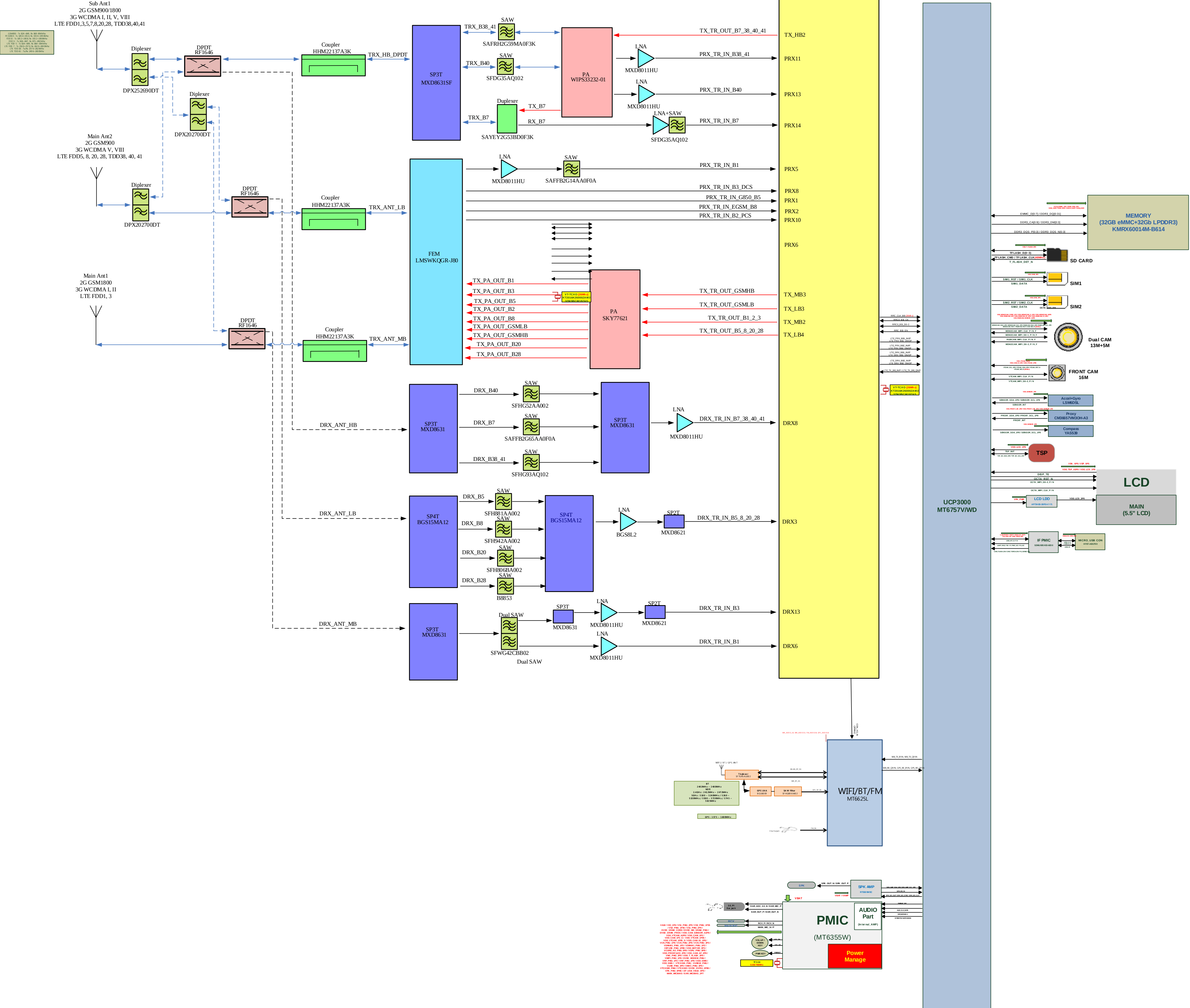
[8] Assemble OCTA connector & OCTA Conn CAP

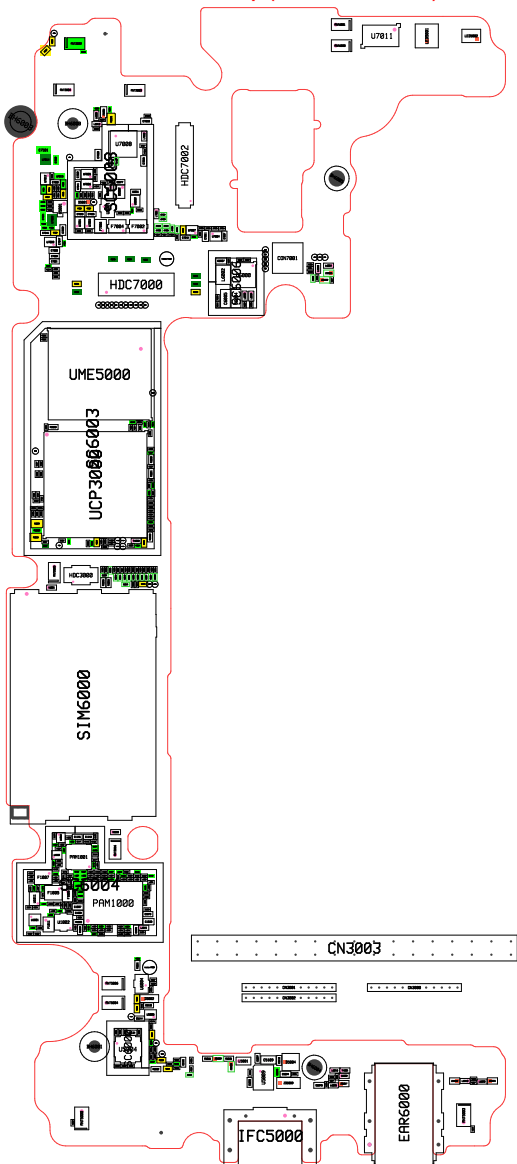
7. Level 2 Repair

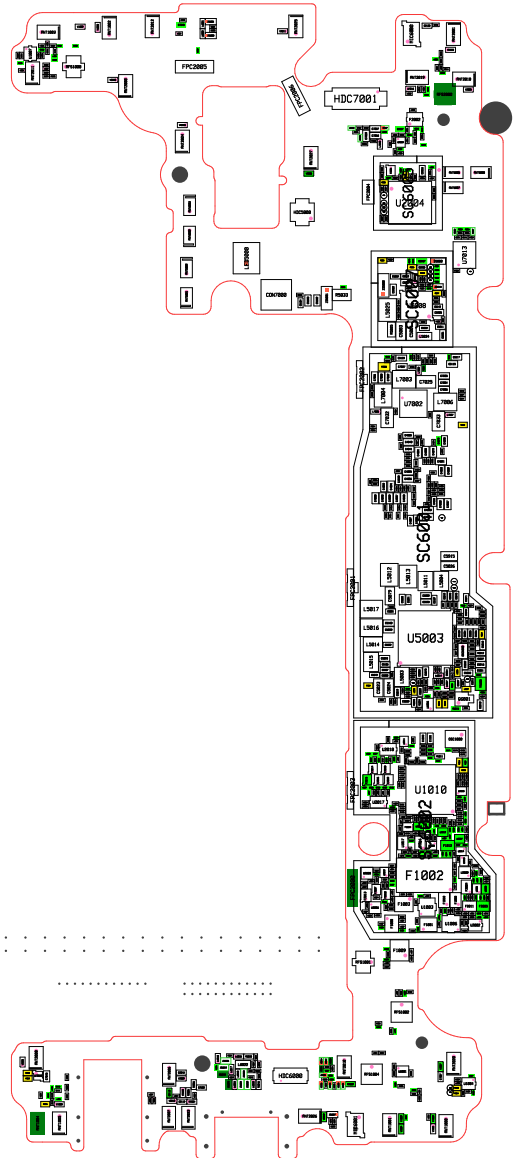
	1. Press the device using pressing jig - Force : 1N - Time : 1minute ※ Please confirm the pressing condition released lastly, and follow it. 
[9]Assemble OCTA to Bracket ※ Caution Be careful not to Damage the home key	[10]Press SET using OCTA PRESS Jig (1N for 1 minute)

SM-C710F/DS REV0.1 BLOCK DIAGRAM

Transceiver MT6176V

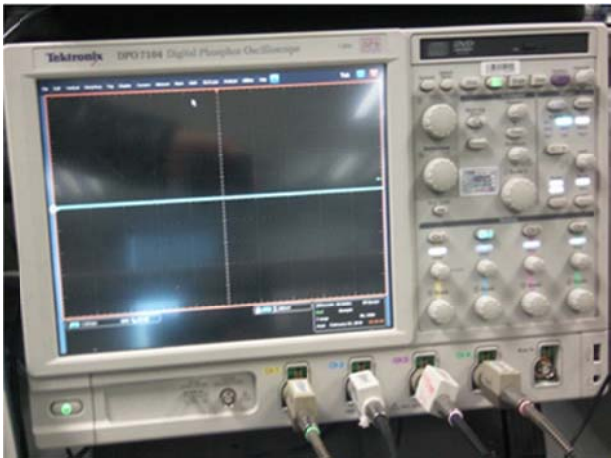
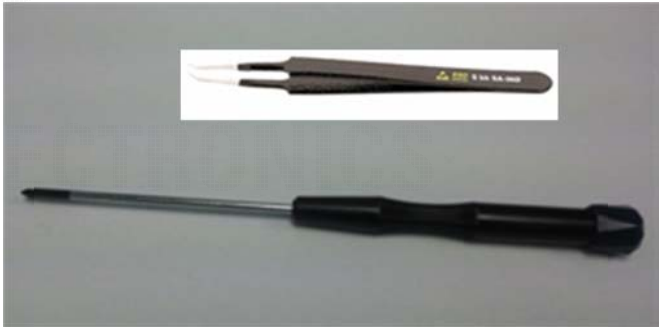
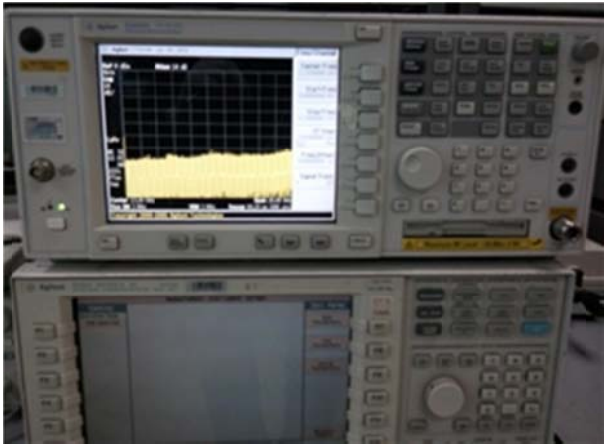






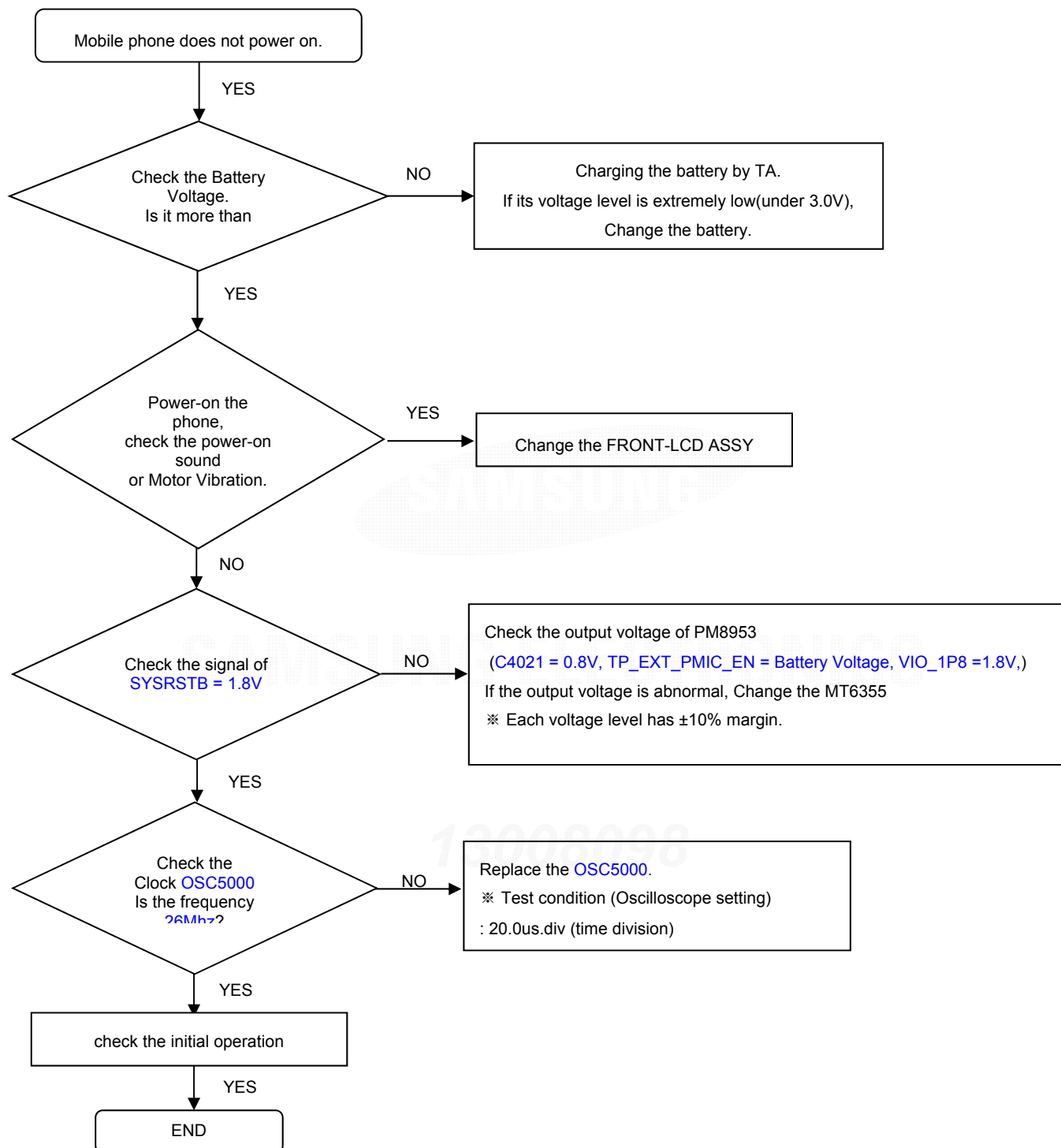
8. Level 3 Repair

8-3. Flow Chart of Troubleshooting

	
Oscilloscope	Digital Multimeter
	
Power Supply	+ driver, ESD Safe Tweezer
	
8960 & Spectrum Analyzer	Soldering iron

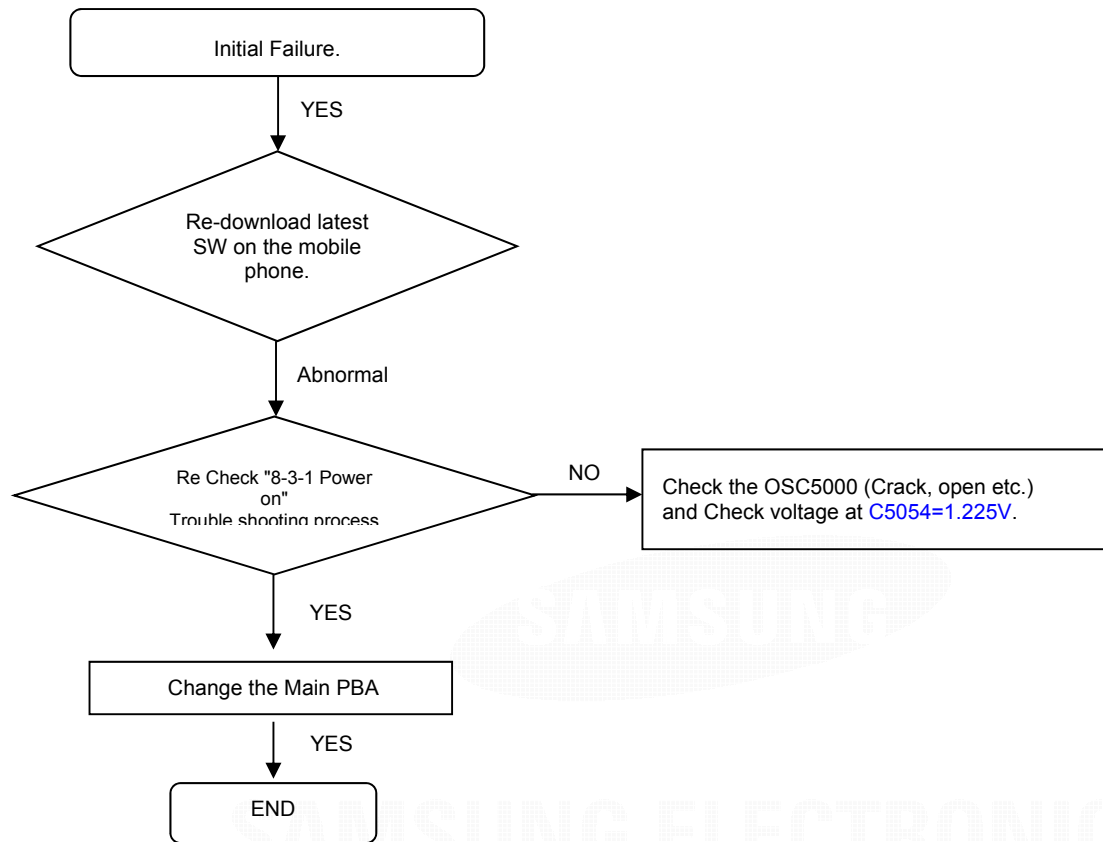
8. Level 3 Repair

8-4-1. Power On



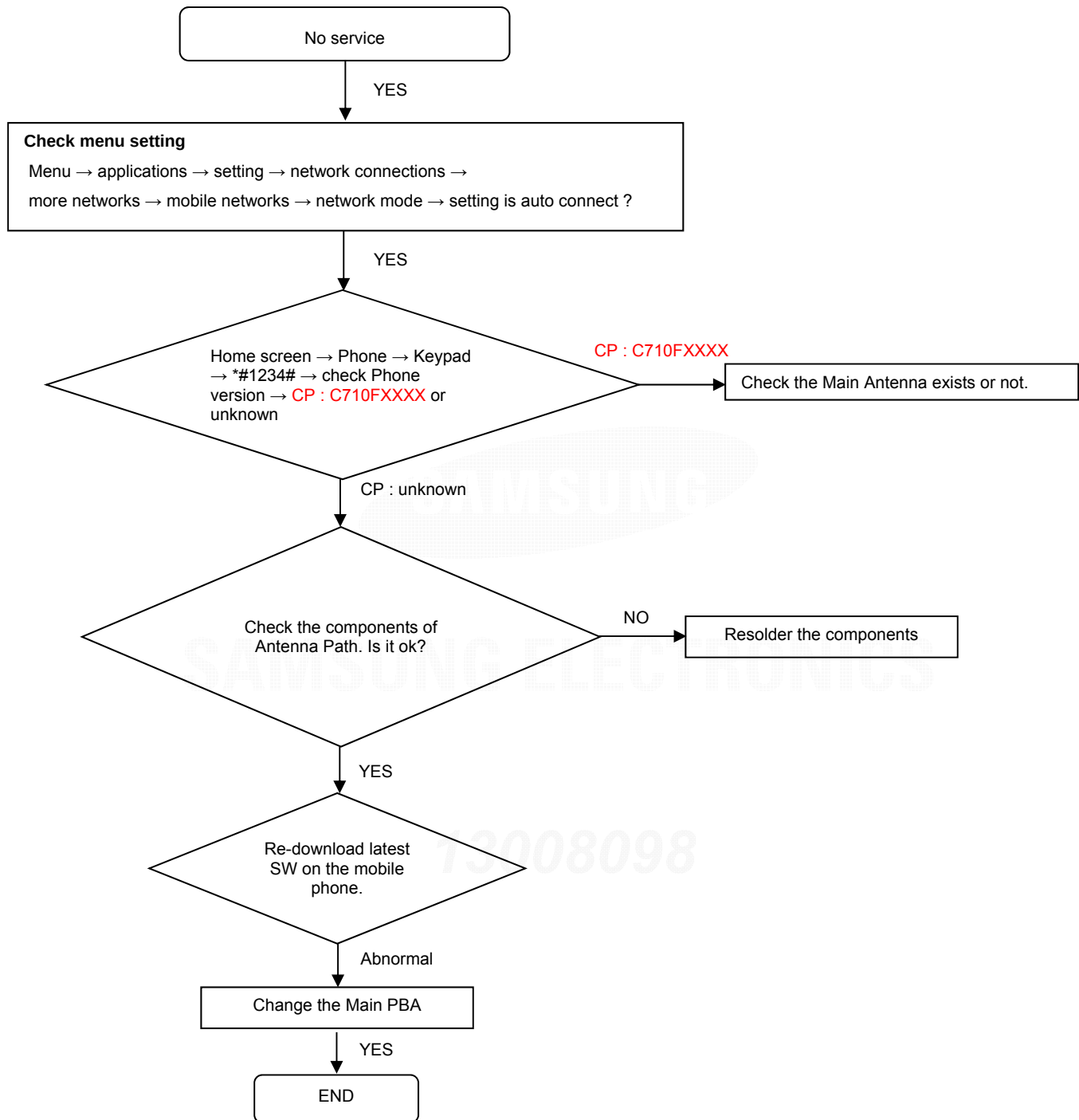
8. Level 3 Repair

8-4-2. Initial



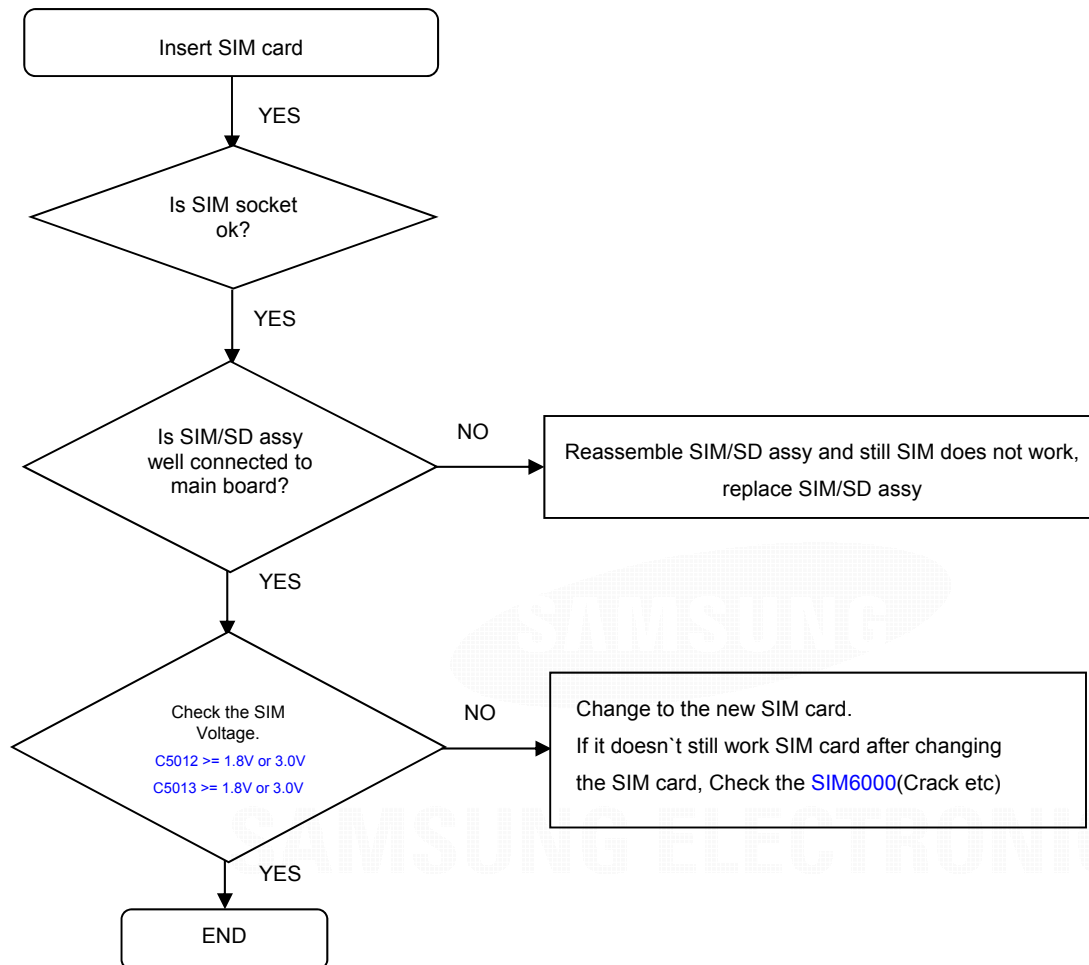
8. Level 3 Repair

8-4-3. No Service



8. Level 3 Repair

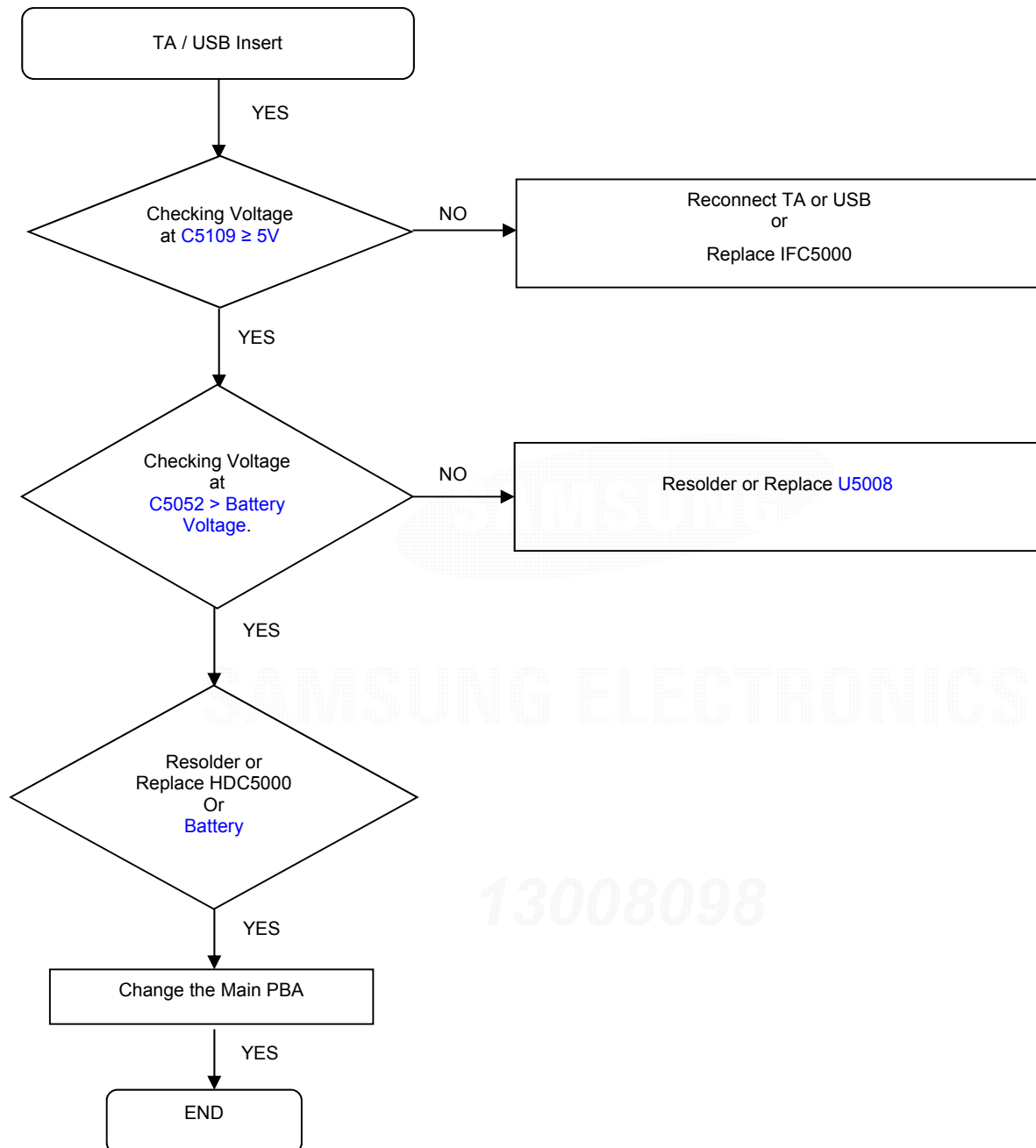
8-4-4. Sim Part



8. Level 3 Repair

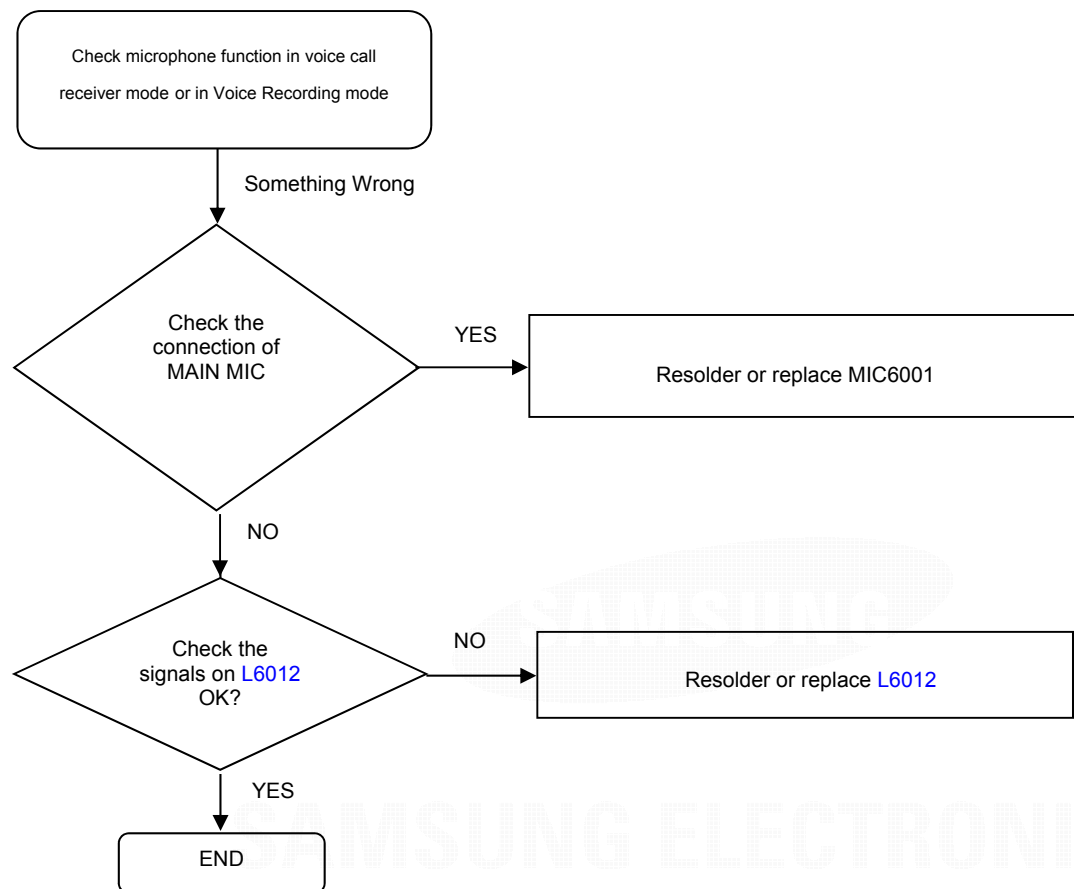
8-4-5. Charging Part

- First of all, Please check the battery status.



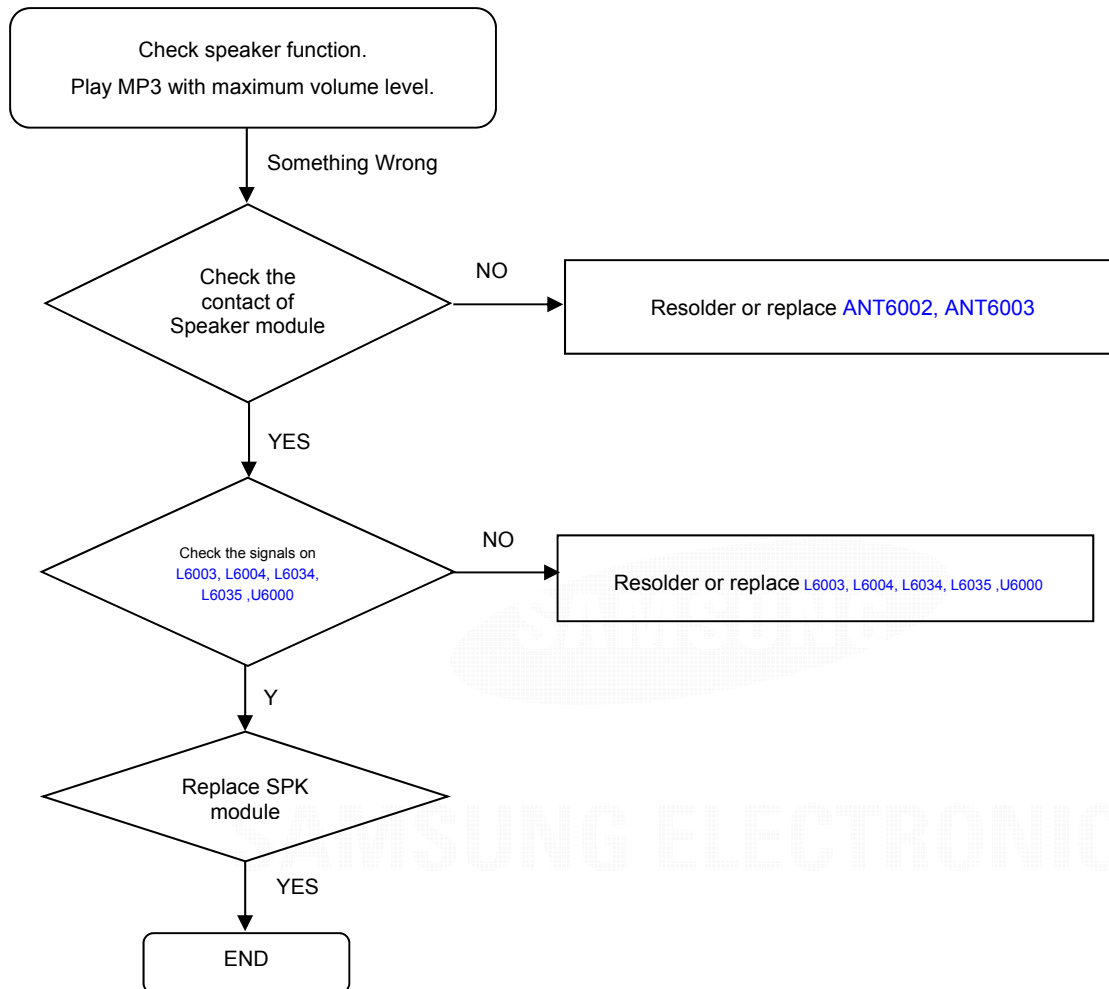
8. Level 3 Repair

8-4-6. Microphone Part



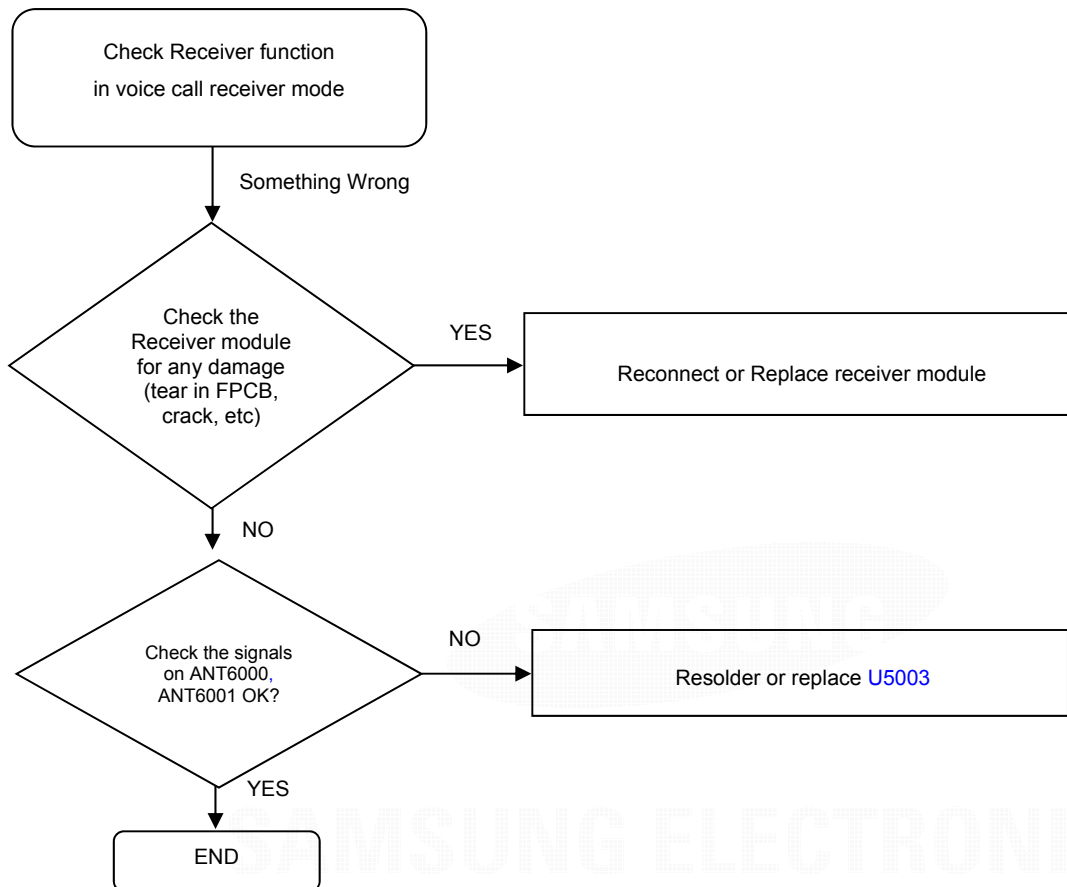
8. Level 3 Repair

8-4-7. Speaker Part



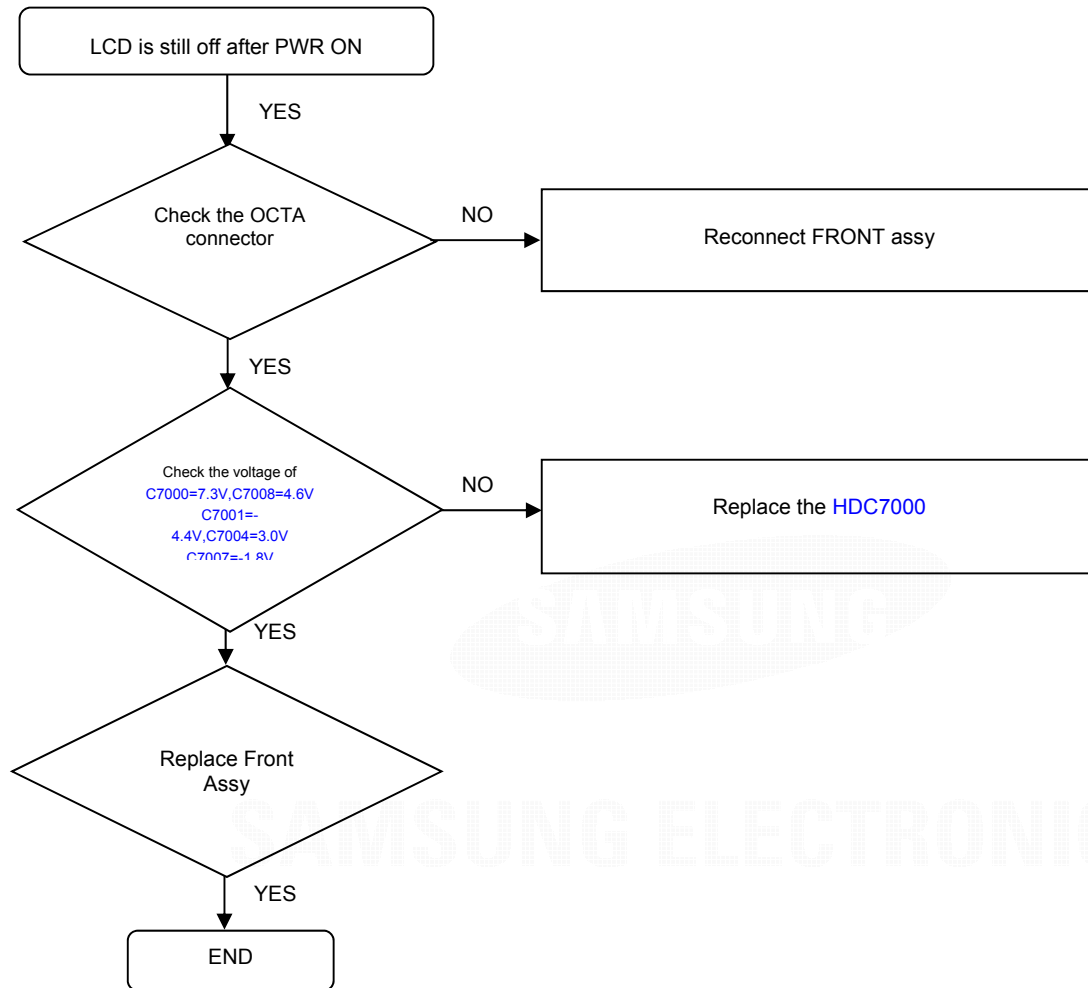
8. Level 3 Repair

8-4-8. Receiver Part



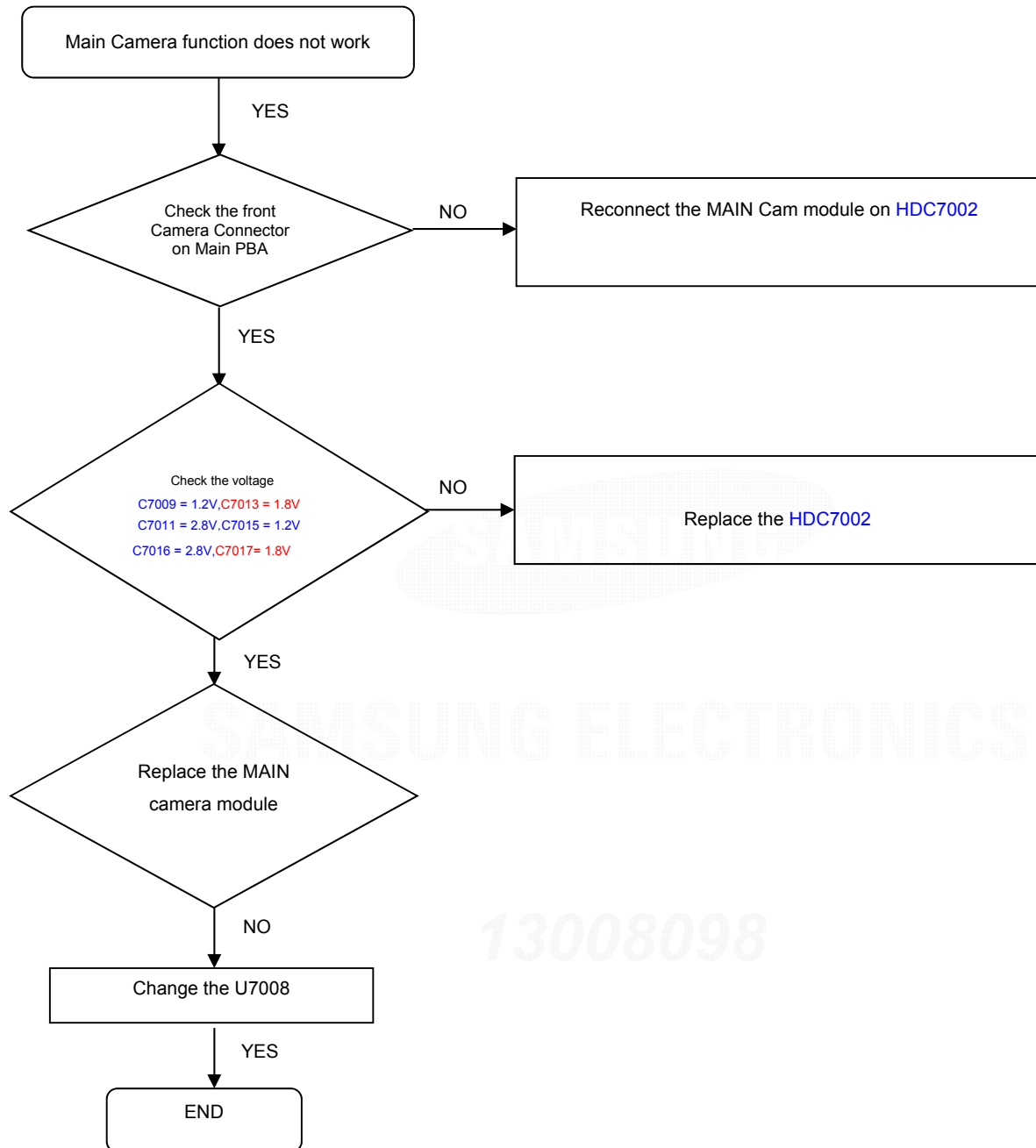
8. Level 3 Repair

8-4-9. OCTA



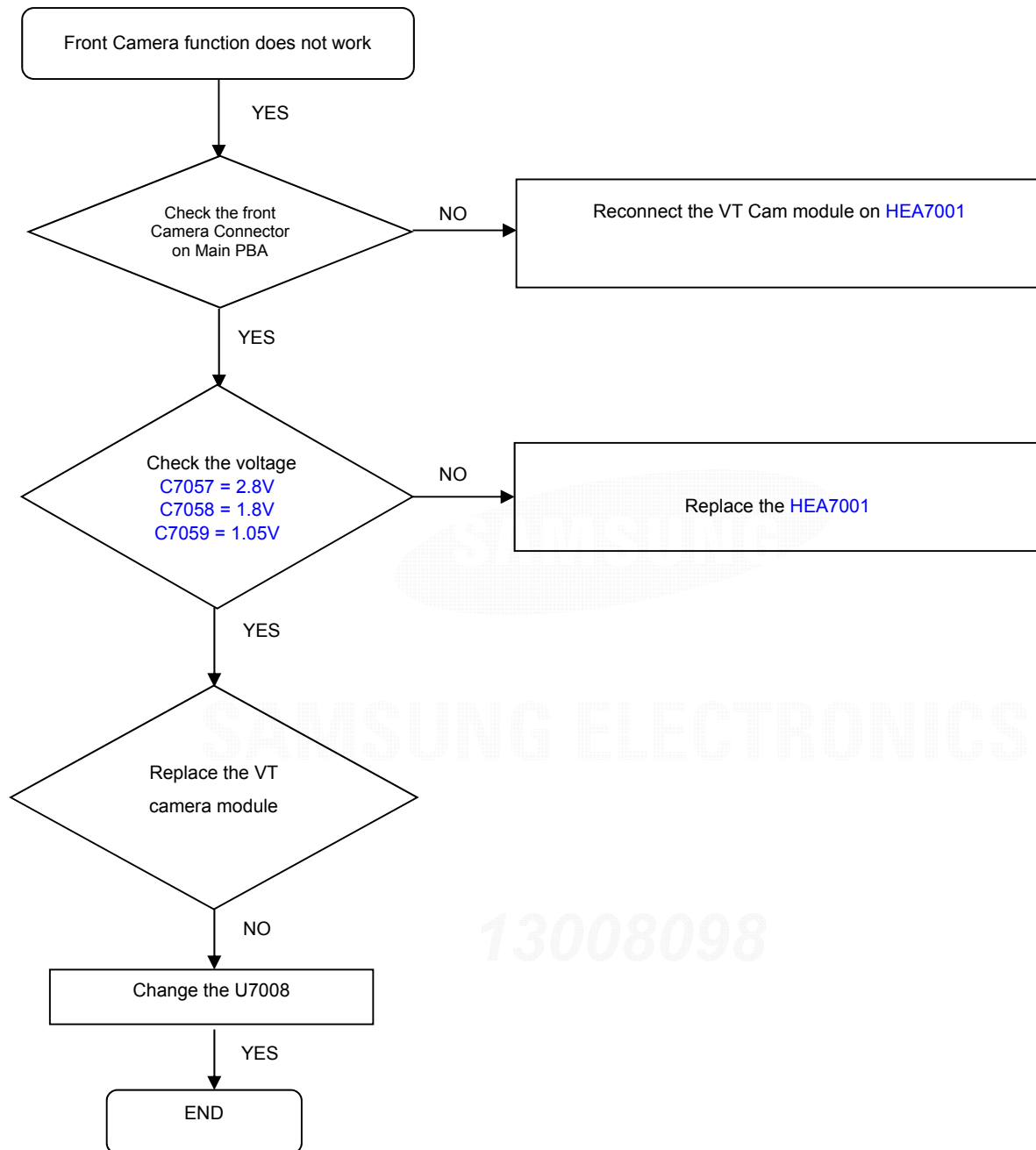
8. Level 3 Repair

8-4-10. MAIN CAMERA



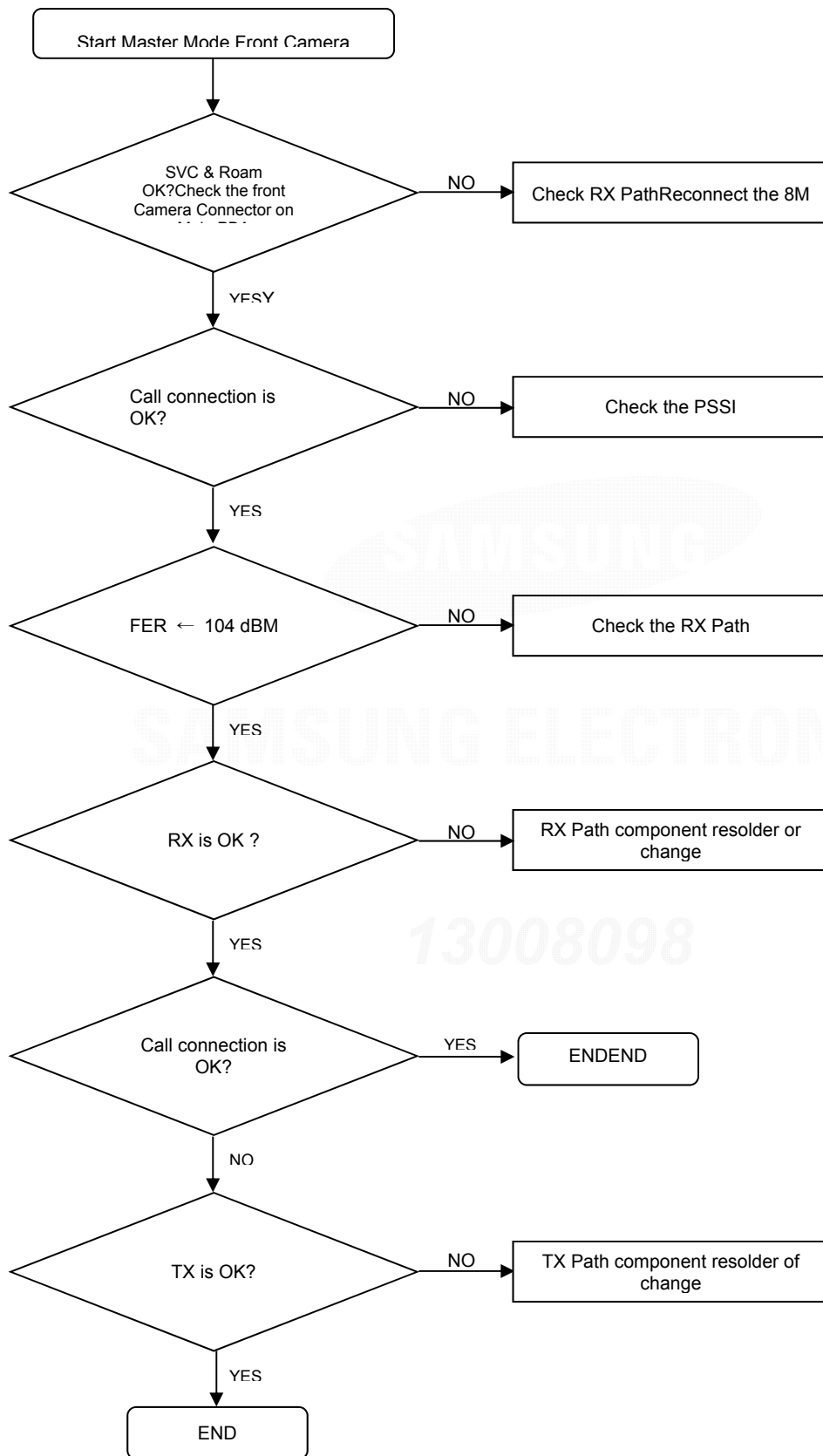
8. Level 3 Repair

8-4-11. VT CAMERA



8. Level 3 Repair

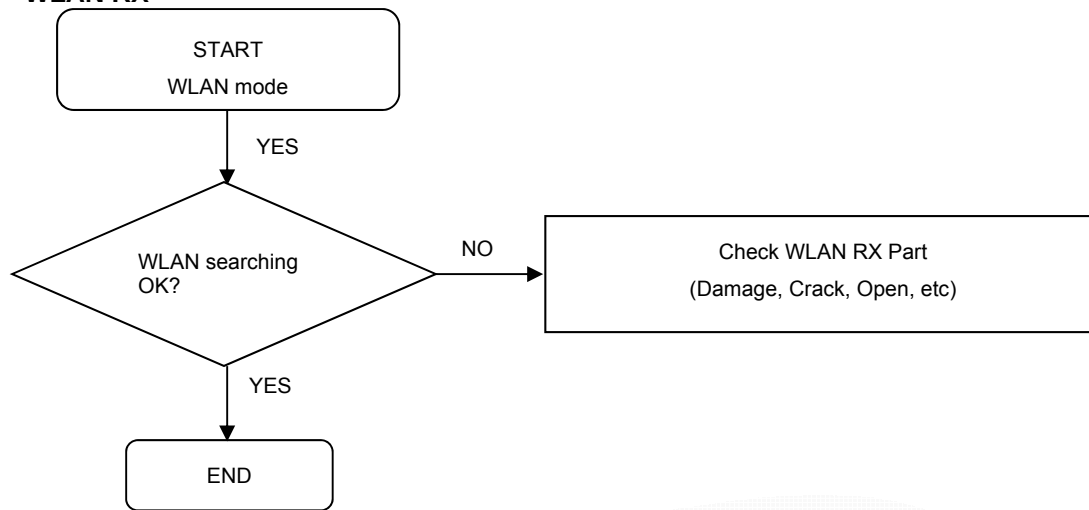
8-4-12. RF Section (LTE,WCDMA,TDSCDMA,CDMA,GSM)



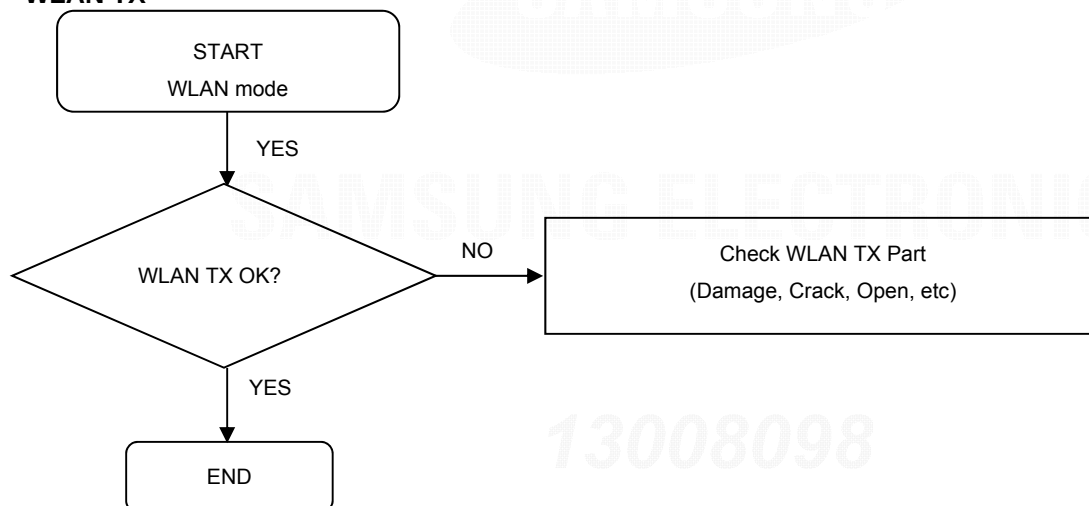
8. Level 3 Repair

8-4-13. RF Section (WLAN / BT)

- WLAN RX

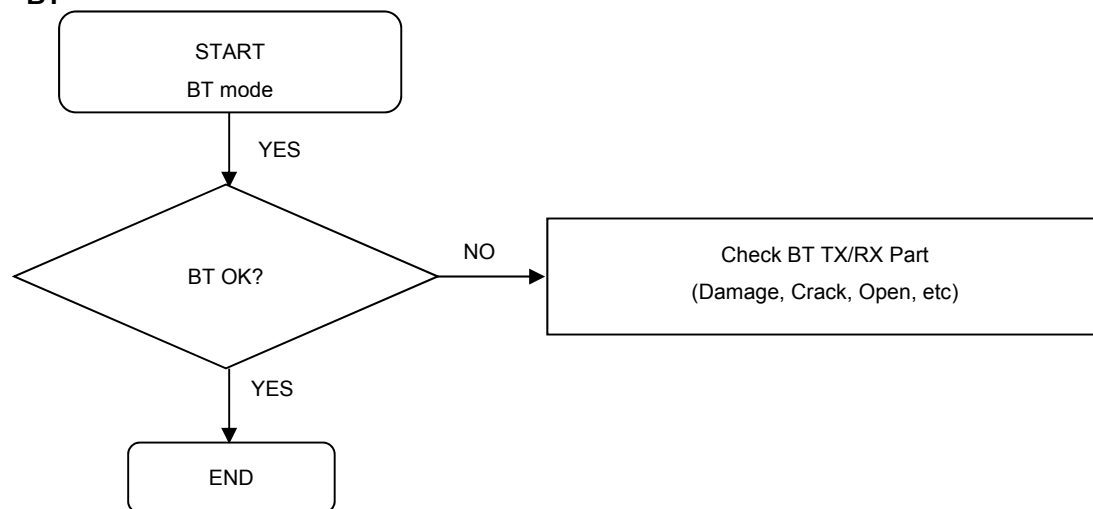


- WLAN TX

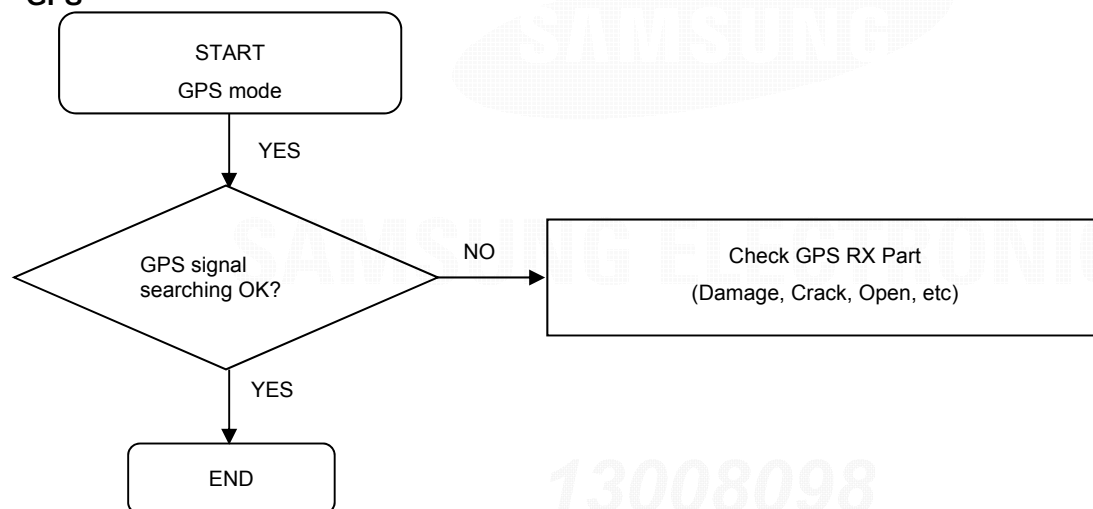


8. Level 3 Repair

- BT

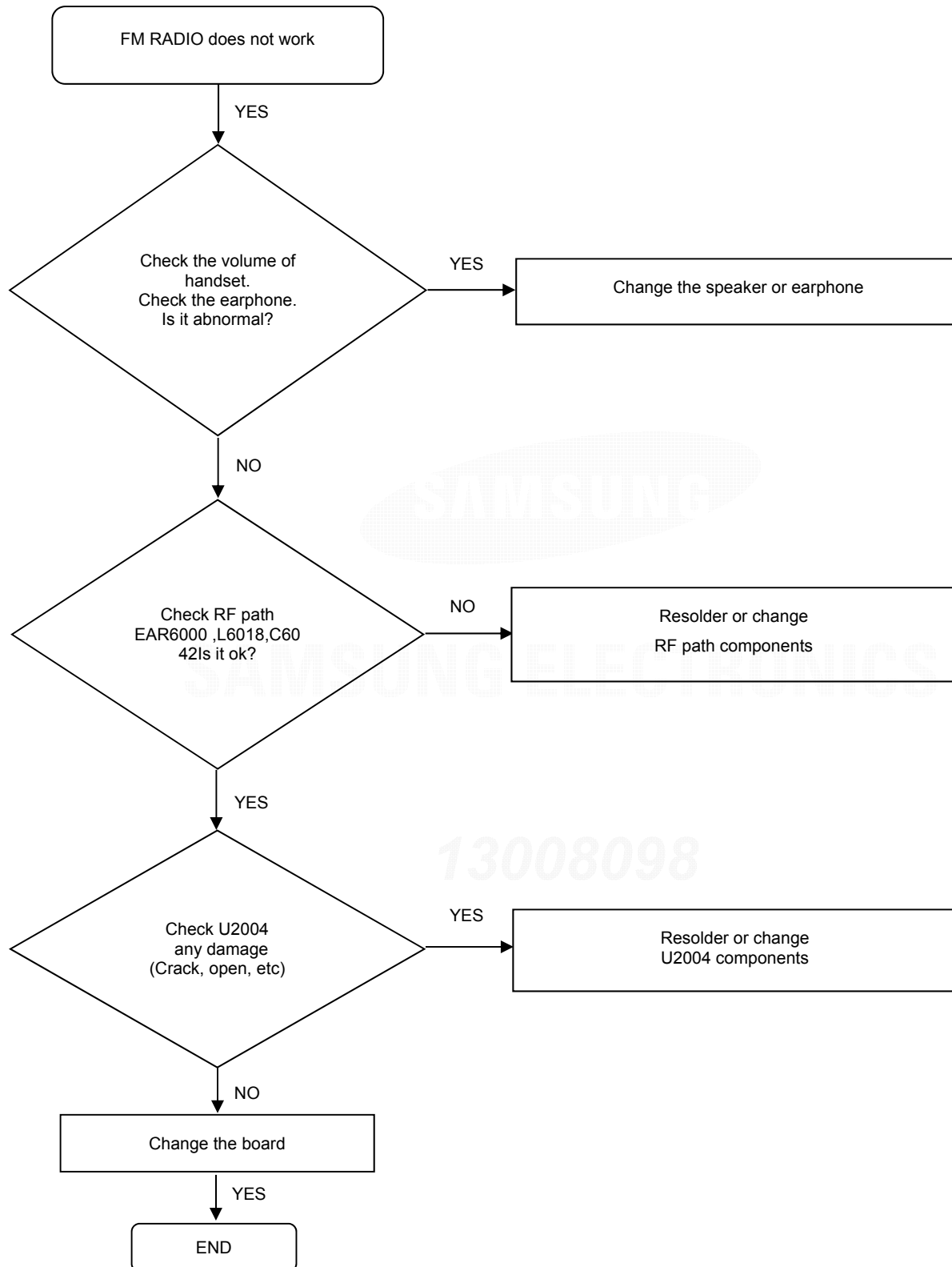


- GPS



8. Level 3 Repair

8-4-14. FM RADIO



8. Level 3 Repair

8-5. Service Schematics

- NC Point(Top View)

UCP3000(AP)

771	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
A			DVDD1 R_MSD	CAS	CAS					DQ18	DQ21		DQ17	DQ22		DQ1	DQ4		DQ7	DQ8		DQ25		DQ30		MDIO DATA		DVDD1 R_MSD	DVSS		A	
B		WB_CT R_WB	WB_CT R_WB	CA4	CA2		CA3	CT1 N		DVSS	DQ16	DQ19	DQ20	DQ0	DQ1	DVSS	DQ3	DQ6	DQ5	DQ11	DQ14	DQ27	DQ28	DQ31		MDIO DATA	MDIO DATA	MDIO DATA	MDIO DATA	MDIO DATA		B
C	WB_X0 P	WB_X1 P	WB_X2 P	WB_X3 P	WB_X4 P	WB_X5 P	WB_X6 P	WB_X7 P	WB_X8 P	WB_X9 P	WB_X10 P	WB_X11 P	WB_X12 P	WB_X13 P	WB_X14 P	WB_X15 P	WB_X16 P	WB_X17 P	WB_X18 P	WB_X19 P	WB_X20 P	WB_X21 P	WB_X22 P	WB_X23 P	WB_X24 P	WB_X25 P	WB_X26 P	WB_X27 P	WB_X28 P	WB_X29 P	WB_X30 P	C
D	WB_X31 P	WB_X32 P	WB_X33 P	WB_X34 P	WB_X35 P	WB_X36 P	WB_X37 P	WB_X38 P	WB_X39 P	WB_X40 P	WB_X41 P	WB_X42 P	WB_X43 P	WB_X44 P	WB_X45 P	WB_X46 P	WB_X47 P	WB_X48 P	WB_X49 P	WB_X50 P	WB_X51 P	WB_X52 P	WB_X53 P	WB_X54 P	WB_X55 P	WB_X56 P	WB_X57 P	WB_X58 P	WB_X59 P	WB_X60 P	WB_X61 P	D
E	WB_X62 P	WB_X63 P	WB_X64 P	WB_X65 P	WB_X66 P	WB_X67 P	WB_X68 P	WB_X69 P	WB_X70 P	WB_X71 P	WB_X72 P	WB_X73 P	WB_X74 P	WB_X75 P	WB_X76 P	WB_X77 P	WB_X78 P	WB_X79 P	WB_X80 P	WB_X81 P	WB_X82 P	WB_X83 P	WB_X84 P	WB_X85 P	WB_X86 P	WB_X87 P	WB_X88 P	WB_X89 P	WB_X90 P	WB_X91 P	WB_X92 P	E
F	WB_X93 P	WB_X94 P	WB_X95 P	WB_X96 P	WB_X97 P	WB_X98 P	WB_X99 P	WB_X100 P	WB_X101 P	WB_X102 P	WB_X103 P	WB_X104 P	WB_X105 P	WB_X106 P	WB_X107 P	WB_X108 P	WB_X109 P	WB_X110 P	WB_X111 P	WB_X112 P	WB_X113 P	WB_X114 P	WB_X115 P	WB_X116 P	WB_X117 P	WB_X118 P	WB_X119 P	WB_X120 P	WB_X121 P	WB_X122 P	WB_X123 P	F
G	WB_X124 P	WB_X125 P	WB_X126 P	WB_X127 P	WB_X128 P	WB_X129 P	WB_X130 P	WB_X131 P	WB_X132 P	WB_X133 P	WB_X134 P	WB_X135 P	WB_X136 P	WB_X137 P	WB_X138 P	WB_X139 P	WB_X140 P	WB_X141 P	WB_X142 P	WB_X143 P	WB_X144 P	WB_X145 P	WB_X146 P	WB_X147 P	WB_X148 P	WB_X149 P	WB_X150 P	WB_X151 P	WB_X152 P	WB_X153 P	WB_X154 P	G
H	WB_X155 P	WB_X156 P	WB_X157 P	WB_X158 P	WB_X159 P	WB_X160 P	WB_X161 P	WB_X162 P	WB_X163 P	WB_X164 P	WB_X165 P	WB_X166 P	WB_X167 P	WB_X168 P	WB_X169 P	WB_X170 P	WB_X171 P	WB_X172 P	WB_X173 P	WB_X174 P	WB_X175 P	WB_X176 P	WB_X177 P	WB_X178 P	WB_X179 P	WB_X180 P	WB_X181 P	WB_X182 P	WB_X183 P	WB_X184 P	WB_X185 P	H
I	WB_X186 P	WB_X187 P	WB_X188 P	WB_X189 P	WB_X190 P	WB_X191 P	WB_X192 P	WB_X193 P	WB_X194 P	WB_X195 P	WB_X196 P	WB_X197 P	WB_X198 P	WB_X199 P	WB_X200 P	WB_X201 P	WB_X202 P	WB_X203 P	WB_X204 P	WB_X205 P	WB_X206 P	WB_X207 P	WB_X208 P	WB_X209 P	WB_X210 P	WB_X211 P	WB_X212 P	WB_X213 P	WB_X214 P	WB_X215 P	WB_X216 P	I
J	WB_X217 P	WB_X218 P	WB_X219 P	WB_X220 P	WB_X221 P	WB_X222 P	WB_X223 P	WB_X224 P	WB_X225 P	WB_X226 P	WB_X227 P	WB_X228 P	WB_X229 P	WB_X230 P	WB_X231 P	WB_X232 P	WB_X233 P	WB_X234 P	WB_X235 P	WB_X236 P	WB_X237 P	WB_X238 P	WB_X239 P	WB_X240 P	WB_X241 P	WB_X242 P	WB_X243 P	WB_X244 P	WB_X245 P	WB_X246 P	WB_X247 P	J
K	WB_X248 P	WB_X249 P	WB_X250 P	WB_X251 P	WB_X252 P	WB_X253 P	WB_X254 P	WB_X255 P	WB_X256 P	WB_X257 P	WB_X258 P	WB_X259 P	WB_X260 P	WB_X261 P	WB_X262 P	WB_X263 P	WB_X264 P	WB_X265 P	WB_X266 P	WB_X267 P	WB_X268 P	WB_X269 P	WB_X270 P	WB_X271 P	WB_X272 P	WB_X273 P	WB_X274 P	WB_X275 P	WB_X276 P	WB_X277 P	WB_X278 P	K
L	WB_X279 P	WB_X280 P	WB_X281 P	WB_X282 P	WB_X283 P	WB_X284 P	WB_X285 P	WB_X286 P	WB_X287 P	WB_X288 P	WB_X289 P	WB_X290 P	WB_X291 P	WB_X292 P	WB_X293 P	WB_X294 P	WB_X295 P	WB_X296 P	WB_X297 P	WB_X298 P	WB_X299 P	WB_X300 P	WB_X301 P	WB_X302 P	WB_X303 P	WB_X304 P	WB_X305 P	WB_X306 P	WB_X307 P	WB_X308 P	WB_X309 P	L
M	WB_X310 P	WB_X311 P	WB_X312 P	WB_X313 P	WB_X314 P	WB_X315 P	WB_X316 P	WB_X317 P	WB_X318 P	WB_X319 P	WB_X320 P	WB_X321 P	WB_X322 P	WB_X323 P	WB_X324 P	WB_X325 P	WB_X326 P	WB_X327 P	WB_X328 P	WB_X329 P	WB_X330 P	WB_X331 P	WB_X332 P	WB_X333 P	WB_X334 P	WB_X335 P	WB_X336 P	WB_X337 P	WB_X338 P	WB_X339 P	WB_X340 P	M
N	WB_X341 P	WB_X342 P	WB_X343 P	WB_X344 P	WB_X345 P	WB_X346 P	WB_X347 P	WB_X348 P	WB_X349 P	WB_X350 P	WB_X351 P	WB_X352 P	WB_X353 P	WB_X354 P	WB_X355 P	WB_X356 P	WB_X357 P	WB_X358 P	WB_X359 P	WB_X360 P	WB_X361 P	WB_X362 P	WB_X363 P	WB_X364 P	WB_X365 P	WB_X366 P	WB_X367 P	WB_X368 P	WB_X369 P	WB_X370 P	WB_X371 P	N
P	WB_X372 P	WB_X373 P	WB_X374 P	WB_X375 P	WB_X376 P	WB_X377 P	WB_X378 P	WB_X379 P	WB_X380 P	WB_X381 P	WB_X382 P	WB_X383 P	WB_X384 P	WB_X385 P	WB_X386 P	WB_X387 P	WB_X388 P	WB_X389 P	WB_X390 P	WB_X391 P	WB_X392 P	WB_X393 P	WB_X394 P	WB_X395 P	WB_X396 P	WB_X397 P	WB_X398 P	WB_X399 P	WB_X400 P	WB_X401 P	WB_X402 P	P
R	WB_X403 P	WB_X404 P	WB_X405 P	WB_X406 P	WB_X407 P	WB_X408 P	WB_X409 P	WB_X410 P	WB_X411 P	WB_X412 P	WB_X413 P	WB_X414 P	WB_X415 P	WB_X416 P	WB_X417 P	WB_X418 P	WB_X419 P	WB_X420 P	WB_X421 P	WB_X422 P	WB_X423 P	WB_X424 P	WB_X425 P	WB_X426 P	WB_X427 P	WB_X428 P	WB_X429 P	WB_X430 P	WB_X431 P	WB_X432 P	WB_X433 P	R
T	WB_X434 P	WB_X435 P	WB_X436 P	WB_X437 P	WB_X438 P	WB_X439 P	WB_X440 P	WB_X441 P	WB_X442 P	WB_X443 P	WB_X444 P	WB_X445 P	WB_X446 P	WB_X447 P	WB_X448 P	WB_X449 P	WB_X450 P	WB_X451 P	WB_X452 P	WB_X453 P	WB_X454 P	WB_X455 P	WB_X456 P	WB_X457 P	WB_X458 P	WB_X459 P	WB_X460 P	WB_X461 P	WB_X462 P	WB_X463 P	WB_X464 P	T
U	WB_X465 P	WB_X466 P	WB_X467 P	WB_X468 P	WB_X469 P	WB_X470 P	WB_X471 P	WB_X472 P	WB_X473 P	WB_X474 P	WB_X475 P	WB_X476 P	WB_X477 P	WB_X478 P	WB_X479 P	WB_X480 P	WB_X481 P	WB_X482 P	WB_X483 P	WB_X484 P	WB_X485 P	WB_X486 P	WB_X487 P	WB_X488 P	WB_X489 P	WB_X490 P	WB_X491 P	WB_X492 P	WB_X493 P	WB_X494 P	WB_X495 P	U
V	WB_X496 P	WB_X497 P	WB_X498 P	WB_X499 P	WB_X500 P	WB_X501 P	WB_X502 P	WB_X503 P	WB_X504 P	WB_X505 P	WB_X506 P	WB_X507 P	WB_X508 P	WB_X509 P	WB_X510 P	WB_X511 P	WB_X512 P	WB_X513 P	WB_X514 P	WB_X515 P	WB_X516 P	WB_X517 P	WB_X518 P	WB_X519 P	WB_X520 P	WB_X521 P	WB_X522 P	WB_X523 P	WB_X524 P	WB_X525 P	WB_X526 P	V
W	WB_X527 P	WB_X528 P	WB_X529 P	WB_X530 P	WB_X531 P	WB_X532 P	WB_X533 P	WB_X534 P	WB_X535 P	WB_X536 P	WB_X537 P	WB_X538 P	WB_X539 P	WB_X540 P	WB_X541 P	WB_X542 P	WB_X543 P	WB_X544 P	WB_X545 P	WB_X546 P	WB_X547 P	WB_X548 P	WB_X549 P	WB_X550 P	WB_X551 P	WB_X552 P	WB_X553 P	WB_X554 P	WB_X555 P	WB_X556 P	WB_X557 P	W
Y	WB_X558 P	WB_X559 P	WB_X560 P	WB_X561 P	WB_X562 P	WB_X563 P	WB_X564 P	WB_X565 P	WB_X566 P	WB_X567 P	WB_X568 P	WB_X569 P	WB_X570 P	WB_X571 P	WB_X572 P	WB_X573 P	WB_X574 P	WB_X575 P	WB_X576 P	WB_X577 P	WB_X578 P	WB_X579 P	WB_X580 P	WB_X581 P	WB_X582 P	WB_X583 P	WB_X584 P	WB_X585 P	WB_X586 P	WB_X587 P	WB_X588 P	Y
AA	WB_X589 P	WB_X590 P	WB_X591 P	WB_X592 P	WB_X593 P	WB_X594 P	WB_X595 P	WB_X596 P	WB_X597 P	WB_X598 P	WB_X599 P	WB_X600 P	WB_X601 P	WB_X602 P	WB_X603 P	WB_X604 P	WB_X605 P	WB_X606 P	WB_X607 P	WB_X608 P	WB_X609 P	WB_X610 P	WB_X611 P	WB_X612 P	WB_X613 P	WB_X614 P	WB_X615 P	WB_X616 P	WB_X617 P	WB_X618 P	WB_X619 P	AA
AB	WB_X620 P	WB_X621 P	WB_X622 P	WB_X623 P	WB_X624 P	WB_X625 P	WB_X626 P	WB_X627 P	WB_X628 P	WB_X629 P	WB_X630 P	WB_X631 P	WB_X632 P	WB_X633 P	WB_X634 P	WB_X635 P	WB_X636 P	WB_X637 P	WB_X638 P	WB_X639 P	WB_X640 P	WB_X641 P	WB_X642 P	WB_X643 P	WB_X644 P	WB_X645 P	WB_X646 P	WB_X647 P	WB_X648 P	WB_X649 P	WB_X650 P	AB
AC	WB_X651 P	WB_X652 P	WB_X653 P	WB_X654 P	WB_X655 P	WB_X656 P	WB_X657 P	WB_X658 P	WB_X659 P	WB_X660 P	WB_X661 P	WB_X662 P	WB_X663 P	WB_X664 P	WB_X665 P	WB_X666 P	WB_X667 P	WB_X668 P	WB_X669 P	WB_X670 P	WB_X671 P	WB_X672 P	WB_X673 P	WB_X674 P	WB_X675 P	WB_X676 P	WB_X677 P	WB_X678 P	WB_X679 P	WB_X680 P	WB_X681 P	AC
AD	WB_X682 P	WB_X683 P	WB_X684 P	WB_X685 P	WB_X686 P	WB_X687 P	WB_X688 P	WB_X689 P	WB_X690 P	WB_X691 P	WB_X692 P	WB_X693 P	WB_X694 P	WB_X695 P	WB_X696 P	WB_X697 P	WB_X698 P	WB_X699 P	WB_X700 P	WB_X701 P	WB_X702 P	WB_X703 P	WB_X704 P	WB_X705 P	WB_X706 P	WB_X707 P	WB_X708 P	WB_X709 P	WB_X710 P	WB_X711 P	WB_X712 P	AD
AE	WB_X713 P	WB_X714 P	WB_X715 P	WB_X716 P	WB_X717 P	WB_X718 P	WB_X719 P	WB_X720 P	WB_X721 P	WB_X722 P	WB_X723 P	WB_X724 P	WB_X725 P	WB_X726 P	WB_X727 P	WB_X728 P	WB_X729 P	WB_X730 P	WB_X731 P	WB_X732 P	WB_X733 P	WB_X734 P	WB_X735 P	WB_X736 P	WB_X737 P	WB_X738 P	WB_X739 P	WB_X740 P	WB_X741 P	WB_X742 P	WB_X743 P	AE
AF	WB_X744 P	WB_X745 P	WB_X746 P	WB_X747 P	WB_X748 P	WB_X749 P	WB_X750 P	WB_X751 P	WB_X752 P	WB_X753 P	WB_X754 P	WB_X755 P	WB_X756 P	WB_X757 P	WB_X758 P	WB_X759 P	WB_X760 P	WB_X761 P	WB_X762 P	WB_X763 P	WB_X764 P	WB_X765 P	WB_X766 P	WB_X767 P	WB_X768 P	WB_X769 P	WB_X770 P	WB_X771 P	WB_X772 P	WB_X773 P	WB_X774 P	AF
AG	WB_X775 P	WB_X776 P	WB_X777 P	WB_X778 P	WB_X779 P	WB_X780 P	WB_X781 P	WB_X782 P	WB_X783 P	WB_X784 P	WB_X785 P	WB_X786 P	WB_X787 P	WB_X788 P	WB_X789 P	WB_X790 P	WB_X791 P	WB_X792 P	WB_X793 P	WB_X794 P	WB_X795 P	WB_X796 P	WB_X797 P	WB_X798 P	WB_X799 P	WB_X800 P	WB_X801 P	WB_X802 P	WB_X803 P	WB_X804 P	WB_X805 P	AG
AH	WB_X806 P	WB_X807 P	WB_X808 P	WB_X809 P	WB_X810 P	WB_X811 P	WB_X812 P	WB_X813 P	WB_X814 P	WB_X815 P	WB_X816 P	WB_X817 P	WB_X818 P	WB_X819 P	WB_X820 P	WB_X821 P	WB_X822 P	WB_X823 P	WB_X824 P	WB_X825 P	WB_X826 P	WB_X827 P	WB_X828 P	WB_X829 P	WB_X830 P	WB_X831 P	WB_X832 P	WB_X833 P	WB_X834 P	WB_X835 P	WB_X836 P	AH
AJ	WB_X837 P	WB_X838 P	WB_X839 P	WB_X840 P	WB_X841 P	WB_X842 P	WB_X843 P	WB_X844 P	WB_X845 P	WB_X846 P	WB_X847 P	WB_X848 P	WB_X849 P	WB_X850 P	WB_X851 P	WB_X852 P	WB_X853 P	WB_X854 P	WB_X855 P	WB_X856 P	WB_X857 P	WB_X858 P	WB_X859 P	WB_X860 P	WB_X861 P	WB_X862 P						

9. Reference Abbreviate

Reference Abbreviate

- **AAC**: Advanced Audio Coding.
- **AVC** : Advanced Video Coding.
- **BER** : Bit Error Rate
- **BPSK**: Binary Phase Shift Keying
- **CA** : Conditional Access
- **CDM** : Code Division Multiplexing
- **C/I** : Carrier to Interference
- **DMB** : Digital Multimedia Broadcasting
- **EN** : European Standard
- **ES** : Elementary Stream
- **ETSI**: European Telecommunications Standards Institute
- **MPEG**: Moving Picture Experts Group
- **PN** : Pseudo-random Noise
- **PS** : Pilot Symbol
- **QPSK**: Quadrature Phase Shift Keying
- **RS** : Reed-Solomon
- **SI** : Service Information
- **TDM** : Time Division Multiplexing
- **TS** : Transport Stream

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